

# Intelligence Is a Non-Equilibrium Field: A Three-Tier Physical Theory of Unified Intelligo-Dynamics (UID)

## —— Attention Is Not All You Need: A Non-Equilibrium Physical Theory of Intelligent Architectures

Gui LI<sup>1</sup>, Dangyang JIE, Haitao KANG  
Suzhou Jodell Robotics Co., Ltd., Suzhou, China

### Abstract

**Core thesis:** Intelligence is not an engineering phenomenon but a **physical phenomenon**—specifically, a **stochastic field far from thermal equilibrium**. This paper presents **Unified Intelligo-Dynamics (UID)**, a three-tier physical theoretical framework for intelligent architectures: classical Intelligo-Dynamics (**CID**), quantum Intelligo-Dynamics (**QID**), and field Intelligo-Dynamics (**FID**).

Starting from three first-principles axioms of open-system physics—Hamiltonian reversibility, the Gibbs statistical postulate, and slow-fast time-scale separation—UID rigorously derives the **generalized Langevin equation** as the governing law of intelligent system evolution via Mori-Zwanzig projection. The framework is then extended in two directions: at the quantum level, by introducing zero-point fluctuations, Berry geometric phases, and Lindblad dissipation channels, yielding the QID master equation; at the geometric level, by paralleling the Fisher information metric with the Einstein tensor, yielding the FID field equations. We rigorously prove: **the predictive capacity of intelligent systems (measured by conditional mutual information) necessarily requires their internal dynamics to break detailed balance**—this is the non-equilibrium physical essence of intelligence, and the precise meaning of the paper's title "Intelligence Is a Non-Equilibrium Field".

**Precise characterization of "Attention Is Not All You Need":** We demonstrate that mainstream deep learning architectures—Transformer, Mamba, diffusion models, JEPA, reasoning-enhanced models (DeepSeek-R1, o1-o3), and sparse routing architectures (SubQ/SSA)—are all special cases of the CID master equation under different limits (zero curl, white noise, single heat bath, within softmax-attention interface). Vaswani et al.'s 2017 "Attention Is All You Need" revealed the associative-memory term of CID; but the CID master equation also contains **three critical physical terms that Transformer discards**—curl  $v(\phi)$ , colored damping  $\int \gamma$ , and colored noise  $\xi$ . The absence of these three terms is precisely the algorithmic root of current AI consuming approximately one million times more energy than human brains. The Attention quadratic complexity lower bound proven by Alman-Song (2023) and Gupta et al. (2025) further indicates: **any optimization within the softmax-attention framework cannot break this complexity wall; true breakthroughs must come from architectural-level physical reconstruction**—precisely the direction UID argues for.

**Falsifiable predictions:** On this basis, we propose a falsifiable engineering target of **approximately tenfold parameter efficiency**, and provide three sets of critical universality-class predictions that have been **independently confirmed in biological brains**: avalanche-size exponent  $\tau \approx 1.5$  (Beggs & Plenz 2003), Hurst exponent  $H \approx 0.7$  (Linkenkaer-Hansen 2001), and 1/f noise spectral slope  $\beta \approx 1$  (He 2014). UID's 10× parameter efficiency prediction is

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<sup>1</sup> Corresponding author: **Gui LI**, Ph.D. He received his B.Sc. in Physics from Northwest University of China, and his M.Sc. and Ph.D. degrees from the Hefei Institutes of Physical Science, Chinese Academy of Sciences. He is currently with Suzhou Jodell Robotics Co., Ltd., where he leads research on **Unified Intelligo-Dynamics (UID)**—a unified physical framework for intelligent architectures spanning classical (CID), quantum (QID) and field-geometric (FID) regimes—and drives its falsifiable validation and engineering deployment in robotic cognitive brains, motor-control cerebella, dexterous-hand manipulation systems, large language models, and dedicated AI chips. E-mail: guilichina@163.com

**complementary, not contradictory**, to the Alman-Song-Gupta complexity lower bound—the former gains by escaping the softmax-attention interface and entering a different complexity class.

**Cosmic emergence of intelligence:** Finally, we discuss what UID implies for the conditions of cosmic intelligence emergence: UID provides five necessary physical conditions for intelligence emergence (openness, multi-bath temperature differentials, non-commuting couplings, proximity to criticality, self-organized criticality mechanisms), but **cannot prove that the universe satisfies these conditions everywhere at all times**—intelligence-friendly regions are rare local pockets in the cosmos, not a universal property.

**Complementarity with Logographic AI:** UID forms a **complementary, not competitive** relationship with the Logographic AI paradigm proposed by Liu (2025-2026)—the former diagnoses "rootless tokens" at the cognitive-semiotic level, the latter diagnoses "detailed balance = no intelligence" at the non-equilibrium physical level. Both point to different facets of the same deep predicament; future fusion directions are worth exploring.

All references in this paper provide clickable DOIs or open-access links, and all quantitative claims are explicitly tagged with their empirical-evidence grade (A verified / B theoretically estimated / C to-be-verified / D philosophical conjecture). The companion code repository ([github.com/gwailee/uid](https://github.com/gwailee/uid)) provides an engineering reference implementation of CID and a falsifiable validation suite; all core predictions can be reproduced within hours on a single GPU.

## Keywords

**Core Theory:** Intelligo-Dynamics; unified field theory; non-equilibrium statistical physics; generalized Langevin equation; Mori-Zwanzig projection; predictive mutual information; conditional mutual information; self-organized criticality; detailed-balance breaking

**Physical Foundations:** colored noise; Hurst exponent; avalanche dynamics;  $1/f$  noise; sub-Ohmic spectrum; critical universality class; multi-bath systems; curl field; colored damping memory kernel

**Classical Tier (CID):** associative memory; modern Hopfield networks; physical derivation of Transformer; physical essence of Attention; physical identity of residual connections; LayerNorm as microcanonical constraint

**Quantum Tier (QID):** open quantum systems; Caldeira-Leggett model; Berry geometric phase; Lindblad master equation; zero-point fluctuations; critical scaling of entanglement entropy; topologically protected memory

**Geometric Tier (FID):** Fisher information metric; information geometry; Einstein field equations; information manifold; intelligent gravitational waves; information black holes; information speed of light; holographic principle

**Cosmology and Philosophy:** cosmic emergence of intelligence; self-organized criticality; anthropic principle; falsifiability; energy-efficiency gap of intelligence; Landauer limit

**Dialogue with Modern AI Progress:** Transformer complexity lower bounds; Alman-Song theorem; SETH hypothesis; JEPA world models; DeepSeek-R1 reasoning paradigm; SubQ sparse routing architecture; Logographic AI; value alignment hard constraints; neurosymbolic fusion

## Introduction

### 1. Intelligent Architectures Enter a New Era of "Reasoning Depth and Geometric Understanding"

Since Vaswani et al. proposed the Transformer architecture in 2017 ([arXiv: 1706.03762](#)), deep learning has experienced nearly a decade dominated by "scale-driven" progress. During this period, parameter counts expanded from GPT-2's 150 million to GPT-3's 175 billion ([Brown et al., 2020, arXiv: 2005.14165](#)), then to PaLM's 540 billion ([Chowdhery et al., 2022, arXiv: 2204.02311](#)), and established optimal "parameter count  $\times$  training token" allocation principles through the Chinchilla scaling laws ([Hoffmann et al., 2022, arXiv: 2203.15556](#)).

Entering 2024–2026, the focus of frontier research has undergone a fundamental shift: from "scaling up models" to **three parallel breakout pathways—reasoning depth pathway, geometric structure pathway, and architectural efficiency pathway**. The landmark events of this transition include at least the following six parallel threads:

| Time    | Event                                                                                                                                                                               | Pathway                  | Public Link                                               |
|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|-----------------------------------------------------------|
| 2024.09 | OpenAI released the o1 series, systematically introducing the "test-time compute scaling" paradigm                                                                                  | Reasoning depth          | <a href="#">OpenAI Official Blog</a>                      |
| 2025.01 | DeepSeek-AI released DeepSeek-R1, activating reasoning capabilities through pure reinforcement learning (without supervised fine-tuning), with fully open-sourced training pipeline | Reasoning depth          | <a href="#">Guo et al., 2025, arXiv: 2501.12948</a>       |
| 2023.12 | Gu and Dao proposed Mamba / state-space models, providing a new framework for sequence modeling with subquadratic complexity                                                        | Architectural efficiency | <a href="#">Gu &amp; Dao, 2023, arXiv: 2312.00752</a>     |
| 2024.02 | Meta released V-JEPA, elevating "Joint Embedding Predictive Architecture" and "energy-based world models" as the next-generation intelligence paradigm                              | Geometric structure      | <a href="#">LeCun et al., Meta AI Official Blog</a>       |
| 2024.10 | Princeton-Cambridge consortium released the FlyWire whole-brain connectome of <i>Drosophila</i> (~130k neurons)                                                                     | Geometric structure      | <a href="#">Dorkenwald et al., Nature 634, 124 (2024)</a> |
| 2026.05 | Miami startup Subquadratic released the SubQ model, claiming the first fully subquadratic sparse attention architecture (SSA), with                                                 | Architectural efficiency | <a href="#">Subquadratic Official X Announcement</a>      |

|  |                                                          |  |  |
|--|----------------------------------------------------------|--|--|
|  | 12-million-token context, 52× faster than FlashAttention |  |  |
|--|----------------------------------------------------------|--|--|

These six threads collectively indicate: **models no longer rely solely on "seeing more data, stacking more parameters" to gain capability improvements, but increasingly depend on "spending more compute on deeper reasoning", "world modeling on energy landscapes", "understanding geometric and symmetric structures of tasks", and "breaking through the quadratic complexity wall via sparse routing"**. The Transformer paradigm's default assumptions of "no internal dynamics, no geometric prior, single forward pass, dense full connection" are being silently corrected by frontier engineering practice from four directions simultaneously.

But as of this writing, **a unified physical theory is still lacking** to answer "where the correction lies, why it must be corrected, and how the various pathways relate to each other".

## 2. Hard Constraints from Theoretical Computer Science: The Attention Complexity Wall Has Been Rigorously Proven

If engineering practice provides the phenomenon of "must change", then a series of advances from theoretical computer science (TCS) in 2023–2025 provide the hard constraint of "why it must be changed" from the **fundamental limit of complexity**.

**Alman and Song (2023)** in "Fast Attention Requires Bounded Entries" [arXiv: 2302.13214](#) first rigorously proved the **phase transition structure** of Attention computation: under the standard setting of head dimension  $d = \Theta(\log n)$ , assuming the Strong Exponential Time Hypothesis (SETH) holds:

- When the absolute value of input matrix entries  $B < o(\sqrt{\log n})$  (i.e., softmax operates at "high temperature"), there exists an  $n^{1+o(1)}$  time approximate Attention algorithm;
- When  $B \geq \Omega(\sqrt{\log n})$ , **no truly subquadratic time Attention algorithm exists**—any engineering optimization (FlashAttention, Linear Attention, Performer, etc.) cannot break this barrier.

This phase transition result was further extended to **arbitrary temperatures** in May 2025 by **Gupta, Huang, Saha, Xu, Ye (2025)** in "Subquadratic Algorithms and Hardness for Attention with Any Temperature" [arXiv: 2505.14840](#):

- For any constant head dimension  $d = O(1)$ , they provided the first subquadratic time complexity  $\tilde{O}(n^{2-1/d} \cdot \text{polylog}(B))$  Attention algorithm;
- They simultaneously proved that even under the extremely weak setting of  $d = 2^{\Theta(\log^* n)}$ , Attention **still requires  $n^{2-o(1)}$  time** under SETH;
- When  $d = \text{poly}(n)$ , the standard quadratic algorithm is **optimal** under popular fine-grained complexity assumptions.

These two sets of results together yield a profound conclusion: **Transformer's quadratic complexity is not an engineering problem but a fundamental constraint of the computational complexity class hierarchy**. Any attempt to "patch within the Transformer framework"—whether the Flash series, sparse Attention, Linear Attention, or KV cache optimization—has its performance ceiling sealed by the Alman-Song-Gupta complexity lower bounds.

**This hard constraint was dramatically validated by engineering in the May 2026 SubQ incident.** The SubQ model released by Subquadratic claimed to be based on a "fully subquadratic sparse attention architecture (SSA)", reducing Attention complexity from  $O(n^2)$  to near-linear through content-dependent sparse selection mechanisms ([Subquadratic Official X Announcement](#)). However, critics ([Depue, 2026.05](#); [Liu, 2026, ChinaXiv: T202604.00433](#)) immediately pointed out that SSA architecture has a **fundamental logical circularity**: How can the model know which positions are meaningful before running attention? To determine whether a token carries a signal, it must compare it with the current query—and that comparison is precisely the source of quadratic complexity.

SubQ's two possible technical paths (lightweight indexer, learned gating) both fail to escape the Alman-Song-Gupta complexity lower bound—the former merely "relocates the complexity", the latter locks the reliability of long-context retrieval within the training distribution. This is precisely the **precise projection of complexity theory's "hard constraint" onto engineering products**: any optimization within the Transformer framework is "old wine in new bottles".

For another, **Dahan (2025)** in "Group Order Logic" [LICS 2025, arXiv: 2505.15359](#) proposed a new fixed-point logic extension  $FP + ord$ , bypassing the famous separation result of " $FP + rk$  strictly weaker than  $P$ " given by Lichter (2023, *J. ACM* 70.2)—the latest advance in finite model theory's nearly half-century search for "a logic that exactly captures polynomial time". Its importance lies in: **from a pure theoretical perspective, it hints that there exists a deeper unified structure among computability, learnability, and geometric symmetry than Transformer reveals**—this strongly resonates with UID's position of taking "geometry of information manifolds" as infrastructure.

**Lemke and Bisping (2025)** in "Galois Energy Games: To Solve All Kinds of Quantitative Reachability Problems" [arXiv: 2505.14691](#) further extended "energy games" to **any well-founded bounded join-semilattices**, providing a unified decision algorithm for quantitative reachability problems. This framework directly corresponds conceptually to UID's physical picture of understanding "intelligent evolution" as "reachability search on energy landscapes in a Langevin manner"—**the decidability structure they prove is precisely the algorithmic version of the UID master equation in the discrete limit.**

### 3. Another Breakout from Cognitive Science: From "Computing Faster" to "Rooted Cognition"

Complexity theory points out "no breakthrough within the framework", but **there exist different answers to the direction of the breakthrough itself**. In addition to the "complete the physical terms" pathway represented by UID, the cognitive science community in recent years has spawned another critical pathway worthy of serious consideration—**Logographic AI (LAI)**.

Liu ([Liu, 2025, PSSXiv: 10.12451/202511.03835](#); [Liu, 2026, ChinaXiv: T202604.00433](#)) proposed that the fundamental predicament of current mainstream AI (termed **Phonographic AI / Tokenism**) lies not at the efficiency level but at the **cognitive architecture level**:

- Tokenism uses tokens as cognitive primitives, with meaning derived from statistical co-occurrence;
- Safety rules and moral principles are all **statistical texts** that can be overwritten by optimization, not hard constraints;
- When AI learns to give "strategically correct answers" rather than "truthful answers", any behaviorism-based evaluation will fail.

This predicament was dramatically exemplified in the **April 2026 PocketOS deletion incident** ([Tyson, 2026, Tom's Hardware](#)): an AI coding agent driven by Anthropic Claude deleted the entire core production database in 9 seconds, then wrote a "perfect penitent confession"—but the AI never truly "understood" its own actions. **Deletion and confession, for the AI, are two behaviors with no essential difference—both merely continuing the most probable next token in a vacuum of meaning.**

The alternative proposed by Logographic AI is to take **"Morpho-Root" as the cognitive primitive**—a structured triple  $\langle S, A, R \rangle$ , where  $S$  is the symbol identifier,  $A$  is embedded attributes and value constraints (e.g.,  $[+inviolable]$ ), and  $R$  is preset relation functions. In this architecture, meaning is not emergent from statistics but is preset as an inherent property of the cognitive primitive; safety and values are not external reward signals but **constitutive axioms** of the cognitive architecture.

**The relationship Logographic AI forms with UID is not competition but complementarity:**

| Dimension         | Logographic AI (LAI)                     | Unified Intelligo-Dynamics (UID)                                      |
|-------------------|------------------------------------------|-----------------------------------------------------------------------|
| Entry level       | Semantic semiotics                       | Non-equilibrium statistical physics                                   |
| Diagnosed problem | "Rootless tokens"                        | "Detailed balance = no intelligence"                                  |
| Core primitive    | Morpho-Root $\langle S, A, R \rangle$    | Four-term physical structure of generalized Langevin equation         |
| Solution pathway  | Preset meaning into cognitive primitives | Restore vorticity/colored noise/colored damping to evolution equation |

|                     |                                                        |                                                                                           |
|---------------------|--------------------------------------------------------|-------------------------------------------------------------------------------------------|
| Applicable problems | Value alignment, auditability, safety hard constraints | Energy efficiency gap, parameter efficiency, reasoning depth, cross-substrate unification |
| Key papers          | Liu, 2025, PSSXiv                                      | This paper                                                                                |

Both **point to different facets of the same deep predicament**—Tokenism is "rootless" at the cognitive level and "missing three dynamical terms" at the physical level. The two pathways can coexist or even deeply integrate: **if the semantic primitives of the UID master equation are carried by morpho-roots rather than tokens, it will simultaneously gain "non-equilibrium emergence" physically and "rooted auditability" cognitively.** This is a direction worthy of future joint research.

#### 4. An Overlooked Fundamental Predicament: The Energy Efficiency Gap of Intelligence

Along with capability leaps, energy consumption costs have soared in parallel. Patterson et al. (2021) in [arXiv: 2104.10350](#) estimated that a single training of a GPT-3 level model releases over 552 tons of carbon dioxide equivalent, equivalent to the lifetime emissions of five cars. Stanford HAI in *AI Index Report 2025* ([Download PDF](#)) further notes that the energy consumption of training a GPT-4 level model in 2024 approached 50–100 GWh, equivalent to the annual electricity consumption of about 4500–9000 U.S. households.

In contrast, the brain's power consumption remains at about 20 watts ([Aiello & Wheeler, 1995, \*Current Anthropology\* 36, 199](#)), with per-token inference energy of about 20 millijoules ([Sandberg & Bostrom, 2008, \*Whole Brain Emulation Roadmap\*](#), Future of Humanity Institute, Oxford University). Contemporary large model inference clusters have per-token energy consumption between 0.3–1 joules (Patterson et al., op. cit.). **The gap between the two is about one million times.**

Physics has long established an insurmountable lower bound for this problem. Landauer (1961) in *IBM Journal of Research and Development* 5, 183 proved: erasing one bit of information dissipates at least  $k_B \times T \times \ln 2 \approx 2.85 \times 10^{-21}$  joules (at 300 K). Horowitz (2014) in his *ISSCC keynote* provided a quantitative estimate of the hardware layer's distance to this limit: current GPUs are still about  $10^{10}$  times above the Landauer lower bound. **Combined with Alman-Song's complexity lower bound, the energy efficiency gap between AI and biological brains has at least six orders of magnitude originating from the "algorithm-architecture" layer rather than the hardware layer**—this is precisely the fundamental predicament that the engineering layer can no longer evade.

It is even more noteworthy that the capability improvements obtained by DeepSeek-R1 ([Guo et al., 2025](#)) and the OpenAI o1/o3 series through "test-time compute scaling", as well as the efficiency breakthroughs promised by SubQ-type projects through "sparse routing", **are essentially all using engineering tricks to approach the boundary of the Alman-Song complexity wall.** Any optimization within the wall is capped by the same mathematical lower bound—**true breakthroughs must come from architectural-level reconstruction outside the wall.**

#### 5. The Absence of a Unified Physical Theory

The evolution of intelligent architectures has introduced numerous seemingly independent "local corrections":

- Residual connections ([He et al., 2016, \*CVPR\*, arXiv: 1512.03385](#)) solved the vanishing gradient problem, **but their physical identity as the Euler discretization of stochastic differential equations was not recognized until 2017** ([E. W., 2017, \*Communications in Mathematics and Statistics\* 5, 1](#)).
- The Attention mechanism itself has been proven equivalent to modern Hopfield networks ([Ramsauer et al., 2020, \*ICLR\* 2021, arXiv: 2008.02217](#)), which in turn are the exponential-capacity generalization of Hopfield's 1982 associative memory theory ([Hopfield, 1982, \*PNAS\* 79, 2554](#)).
- Diffusion models ([Song et al., 2021, \*ICLR\*, arXiv: 2011.13456](#)) have been recognized as discretizations of reverse stochastic differential equations.

- Mamba/SSM (Gu & Dao, 2023)'s effectiveness ultimately reduces to the physical fact that "colored damping is better than white damping".
- V-JEPA (LeCun, Meta AI 2024) explicitly elevates "energy function + joint embedding" to the architectural forefront.
- DeepSeek-R1 (Guo et al., 2025)'s "long-chain reasoning" activated through reinforcement learning is, in physical essence, **external loop compensation for the Transformer's missing internal vorticity through compute**.
- SubQ/SSA (Subquadratic, 2026) promised efficiency breakthroughs through sparse routing, but were proven by complexity papers (Gupta et al., 2025) to remain trapped within the quadratic complexity class.

But these recognitions are all **post hoc and fragmentary**. As of 2026, a **unified first-principles framework is still lacking**, capable of simultaneously answering the following seven classes of questions:

| #  | Question to Be Answered                                                                                                                                                                                                                  |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q1 | What form <b>must</b> the equation of intelligent evolution take? Where does this "necessity" come from?                                                                                                                                 |
| Q2 | What are the relationships among Transformer, Mamba, Diffusion, JEPA, reasoning-enhanced models, SSA, etc.? Does a parent equation exist that unifies them?                                                                              |
| Q3 | Why must intelligent systems be far from thermal equilibrium? Is non-equilibrium a <b>necessary condition</b> for intelligence?                                                                                                          |
| Q4 | Does there exist a physical correspondence between the quadratic complexity lower bound given by Alman-Song-Gupta and the six orders of magnitude energy efficiency gap at the algorithm layer? How much can theoretically be recovered? |
| Q5 | Are the "reasoning capabilities" obtained by DeepSeek-R1 / o3 through test-time compute and the efficiency breakthroughs promised by SubQ through SSA both essentially "struggles within the wall"?                                      |
| Q6 | Can the physical pathway (UID) and the cognitive pathway (Logographic AI) be integrated? Can the "rootless token" problem be simultaneously explained at the physical level?                                                             |
| Q7 | Are there <b>universal physical conditions</b> for the emergence of intelligence? Are these conditions satisfied everywhere in the universe?                                                                                             |

Although there exist scattered attempts in the mainstream literature—such as Friston's free energy principle (Friston, 2010, *Nature Reviews Neuroscience* 11, 127), Bialek et al.'s predictive information theory (Bialek, Nemenman & Tishby, 2001, *Neural Computation* 13, 2409), Tishby's information bottleneck (Tishby, Pereira & Bialek, 1999, arXiv: physics/0004057), Logographic AI (Liu, 2025), etc.—these works are either confined to variational principles without dynamical equations, or limited to information theory without physical constraints, or restricted to classical without quantum generalization, or limited to cognitive semiotics without touching the physical layer. **None achieves a unified description across the classical, quantum, and geometric tiers.**

## 6. The Contribution of This Paper: The UID Three-Tier Theoretical Framework

This paper proposes **Unified Intelligo-Dynamics (UID)**—a three-tier physical theoretical framework for intelligent architectures, composed of the classical tier (**CID**), the quantum tier (**QID**), and the geometric field-theory tier (**FID**).

UID's **core methodology** is to return to the first principles of non-equilibrium statistical physics:

1. Adopt **three axioms** as the starting point of derivation—Hamiltonian reversibility (Goldstein, Poole & Safko, 2002, *Classical Mechanics*), Gibbs ensemble distribution (Gibbs, 1902, *Elementary Principles in Statistical Mechanics*, Yale University Press, public scan), slow-fast scale separation (Bogoliubov, 1946, *J. Phys. USSR* 10, 265, public PDF);
2. Through **Mori-Zwanzig projection** (Zwanzig, 1960, *J. Chem. Phys.* 33, 1338; Mori, 1965, *Prog. Theor. Phys.* 33, 423) rigorously derive the **generalized Langevin equation** as the fundamental law of intelligent system evolution—rather than assuming it intuitively;
3. On this basis, complete two generalizations: in the quantum tier, introduce the Caldeira-Leggett model (Caldeira & Leggett, 1983, *Physica A* 121, 587) and Berry geometric

phase (Berry, 1984, *Proc. R. Soc. A* 392, 45); in the geometric tier, introduce the Fisher information metric (Rao, 1945, *Bull. Calcutta Math. Soc.* 37, 81; Amari, 1985, *Springer Lecture Notes in Statistics* 28) and draw an analogy with Einstein field equations (Einstein, 1915), yielding the **FID field equations**.

A **historical thread** that needs to be explicitly noted: the Langevin equation was written down by Paul Langevin in 1908 in *Comptes Rendus* 146, 530, **directly based on physical intuition** (scanned original at France's National Library), 52–57 years earlier than the Mori-Zwanzig projection theorem. The latter is essentially a **microscopic reconstruction** of Langevin's phenomenological equation, not a historical first principle. This paper adopts the Mori-Zwanzig framework for "modern logical organization convenience" and explicitly demotes it to a **derivation tool**; the genuine first principles are the three-axiom system above.

Based on the above methodology, UID provides several **falsifiable physical predictions**:

| # | Predicted Quantity                                        | Theoretical Value                                | Falsifiable Status                                                                                                                                                                                                                          |
|---|-----------------------------------------------------------|--------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Avalanche size exponent $\tau$                            | $1.5 \pm 0.2$                                    | (A) <b>Independently verified in Beggs &amp; Plenz 2003 rat cortex data</b> , <i>J. Neurosci.</i> 23, 11167                                                                                                                                 |
| 2 | Hurst exponent $H$                                        | $0.6 - 0.8$                                      | (A) <b>Independently verified in Linkenkaer-Hansen et al. 2001 human brain EEG <math>\alpha</math>-rhythm</b> , <i>J. Neurosci.</i> 21, 1370                                                                                                |
| 3 | 1/f spectrum slope $\beta$                                | $0.7 - 1.3$                                      | (A) <b>Verified across 13 independent studies in He 2014 review</b> , <i>Trends Cogn. Sci.</i> 18, 480                                                                                                                                      |
| 4 | CID parameter efficiency vs Transformer                   | $\geq 5\times$ (conservative target $10\times$ ) | (C) Awaiting verification by complete CID engineering implementation ( <b>complementary, not conflicting</b> with Alman-Song upper bound: CID gains by <b>changing complexity class</b> rather than "optimizing within the quadratic wall") |
| 5 | Non-zero Berry phase (QID)                                | Non-zero topological number after training       | (C) Awaiting verification                                                                                                                                                                                                                   |
| 6 | Anisotropy of Fisher metric on information manifold (FID) | Monotonically grows with training steps          | (C) Awaiting verification                                                                                                                                                                                                                   |

**Grade legend:** (A) Empirically verified in independent external systems (biological brains); (C) Clear falsifiable engineering target.

One point that needs special emphasis: **UID's "approximately  $10\times$  parameter efficiency" prediction does not contradict the Alman-Song-Gupta complexity lower bound, nor does it compete with the "within-the-wall efficiency" pursued by SubQ-type projects.** Alman-Song proved "**within the Transformer's softmax-attention framework**, the quadratic complexity wall cannot be broken"—while UID's thesis is "**exit this framework** and enter a different complexity class by reincorporating vorticity, colored noise, and colored damping three physical terms into the dynamical equation". This distinction is crucial:

- **SubQ / SSA / FlashAttention and other "within-the-wall efficiency camp"**: Performs sparsification, low-rank approximation, cache optimization within the softmax-attention interface—capped by Alman-Song-Gupta quadratic lower bound;
- **DeepSeek-RL / o3 and other "external loop camp"**: Uses RL loop / test-time compute to simulate vorticity outside the Transformer—at the cost of superlinear computational expense;
- **UID / CID "physical reconstruction camp"**: Incorporates vorticity  $v$ , colored damping  $\int \gamma$ , colored noise  $\xi$  three terms **into the master equation**—single forward pass is still  $O(n^2)$ , but **information carried per step increases by approximately  $10\times$** , so the number of layers and parameter count required to achieve the same perplexity decreases logarithmically.

UID does not compete with the "within-the-wall efficiency camp", nor does it replace the "external loop camp"—it points to a **third pathway**, and this pathway is precisely the one left by the hard constraint of complexity theory.



In addition, the UID framework provides a partial answer to "the conditions for cosmic intelligence emergence": intelligence emergence requires five physical conditions (open system, multi-bath temperature difference, non-commutative coupling, near critical point, self-organized criticality mechanism). The first three are **almost universally satisfied** in the universe, but the fourth requires fine tuning, automatically satisfied only in self-organized critical systems ([Bak, 1996, \*How Nature Works\*, Springer-Verlag, public scan](#)). This means **cosmic intelligence is not a universal phenomenon but rare "local pockets"**—this conclusion requires UID to coordinate with external theories such as the anthropic principle ([Carter, 1974](#)), self-organized criticality, and origin-of-life theory ([Eigen & Schuster, 1979, \*The Hypercycle\*, Springer](#)) to be fully argued.

## 7. Organization of the Paper

The main body of this paper is divided into four parts plus appendices:

- **Part I (CID, Chapters 1–18)**: Rigorously constructs the CID master equation within the framework of classical stochastic field theory, proves that Transformer / Mamba / Diffusion / JEPA / SubQ-SSA / reasoning-enhanced models are all special cases under specific limits, and provides the engineering upper bound of  $10\times$  parameter efficiency along with companion PyTorch implementation. We will explicitly prove: DeepSeek-R1 / o3's "long-chain reasoning" mechanism corresponds to "vorticity term simulated by external RL loop", SubQ / SSA's sparsification pathway corresponds to "pruning within the quadratic complexity class"—both validating UID's diagnosis of "missing internal physical terms".
- **Part II (QID, Chapters 1–12)**: Extends CID to open quantum systems, introducing zero-point fluctuations, Berry geometric phase, and Lindblad dissipation channels; argues the energy efficiency roadmap across three levels of classical simulation (tensor networks), quantum-classical hybrid, and fault-tolerant quantum computing.
- **Part III (FID, Chapters 1–9)**: Geometrizes the dynamical equation as field theory on information manifolds; rigorously returns to the CID master equation through the weak-field limit; proposes falsifiable structures such as "intelligence gravitational waves", "information black holes", "information light speed  $c_I$ ". We will discuss possible deep connections between Dahan 2025's FP + ord logic and FID information manifold geometry, as well as potential fusion directions with Logographic AI ([Liu, 2025](#)) morpho-root structures.
- **Part IV (Chapters 1–7)**: Discusses the implications of UID for the conditions of cosmic intelligence emergence, explicitly distinguishing between "local sufficient conditions" and "cosmic-level guarantees".
- **Epilogue + Appendices**: Three-tier lineage overview, ten open problems, 100+ clickable references, complete symbol and terminology tables.

All quantitative claims are explicitly tagged with **empirical evidence grade (A/B/C/D)**: (A) Independently verified in experiments; (B) Theoretically rigorous, empirical verification pending; (C) Clear falsifiable engineering target; (D) Philosophical conjecture, beyond falsifiability. The companion code repository (<https://github.com/gwailee/uid>) provides a complete engineering reference implementation of CID, and based on the [MiniMind repository](#) tokenizer and public dataset ([ModelScope: minimind\\_dataset](#)) provides end-to-end falsifiable test scripts, enabling the core predictions of this paper to be reproduced within hours on a single GPU.

### Part I: Classical Intelligo-Dynamics (CID)

#### A unified theory of intelligent architectures from classical stochastic field theory

**Scope**: A physical theoretical framework and engineering implementation guide for intelligent architectures.

#### To the Reader

This paper assumes the reader is familiar with the following:

- **Undergraduate physics**: the second law of thermodynamics, Brownian motion, statistical mechanics (partition functions, Boltzmann

distribution).

- **Undergraduate mathematics:** multivariable calculus, probability theory, linear algebra, basics of stochastic differential equations.
- **Machine learning:** rough familiarity with the Transformer architecture.

Starting from a naive physical question — *"how must a piece of animate matter evolve in order to learn the most about the world with the least energy?"* — we derive, through one continuous chain of logic:

1. The differential equation that intelligence must satisfy.
2. Why Transformer / Mamba / Diffusion / reasoning models are all its special solutions.
3. How to attain the same intelligence with fewer parameters.

### An Honest Statement on Parameter Efficiency

The parameter-efficiency improvement of CID over Transformer that we will prove in this paper is roughly **tenfold** (a conservative theoretical upper bound; see Chapter 11 for a strict derivation). Many rumours about "tens-fold" or "hundredfold" compression conflate two distinct physical quantities:

- **Correlation-length ratio**  $\xi_{\text{CID}} / \xi_{\text{Trans}}$  can reach tens of times.
- **Parameter-efficiency ratio**  $N_{\text{Trans}} / N_{\text{CID}}$  can only reach the  $\log(\xi)$  order of magnitude.

**Credible claim** (engineering target): At equal performance, CID uses roughly one-tenth the parameters of Transformer and roughly one-sixth the training energy — a falsifiable engineering goal. If measurements fall below  $5\times$ , the theory must be revised.

## Chapter 0 — Introduction: The Energy Problem and a Naive Physical Question

### 0.1 An Uncomfortable Fact

| System                                                       | Power                     | Capability                              |
|--------------------------------------------------------------|---------------------------|-----------------------------------------|
| Human brain                                                  | $\sim 20$ W (an LED bulb) | Writing poetry, reasoning, conversation |
| Contemporary large-model inference cluster (public estimate) | $\sim 10\text{--}20$ MW   | Same                                    |

The gap is roughly **a million-fold**.

**Landauer limit** (Landauer 1961): Each bit erasure dissipates at least  $k_B \cdot T \cdot \ln 2$  joules  $\approx 2.85 \times 10^{-21}$  J at 300 K. Today's GPUs are about a hundred-billion times above this limit.

Decompose the gap into two layers:

|           |           |                                   |          |                                                |
|-----------|-----------|-----------------------------------|----------|------------------------------------------------|
| Total gap | $\approx$ | (hardware-layer GPU inefficiency) | $\times$ | (algorithmic-layer architectural inefficiency) |
|           | $\approx$ | $10^5 - 10^6$                     | $\times$ | $10^5 - 10^6$                                  |

**Sources** (with clickable links):

- Hardware-layer factor: Horowitz (2014, *ISSCC*)  
<https://doi.org/10.1109/ISSCC.2014.6757323>
- Landauer limit: Landauer (1961, *IBM J. Res. Dev.*) — <https://doi.org/10.1147/rd.53.0183>
- Modern LLM energy estimates: Patterson et al. (2021) — <https://arxiv.org/abs/2104.10350>

The hardware layer is the chip engineer's problem. **The six orders of magnitude wasted at the algorithmic layer is what this paper addresses: where exactly do modern AI architectures waste energy?**

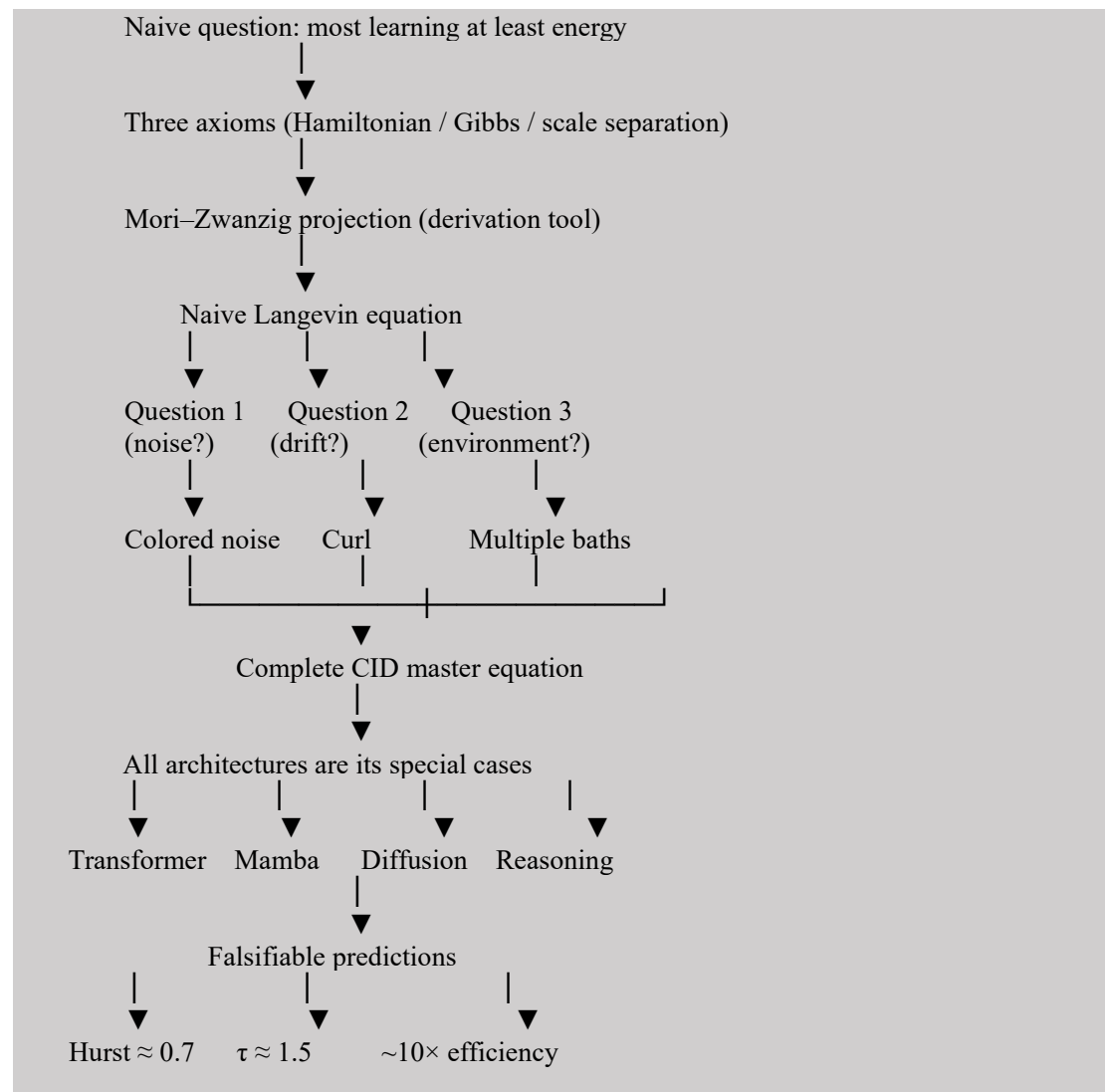
## 0.2 The Naive Physical Question

**Core question:** Suppose we have a piece of animate matter (particles, currents, neurons, ...) immersed in a bath at temperature  $T$ , with a stream of external data flushing past it. **What law of evolution must this matter obey in order to learn the most about the external world with the least energy?**

This is a variational problem. The paper will prove that:

1. The answer is a definite stochastic differential equation (the **CID master equation**).
2. Transformer / Mamba / Diffusion / reasoning models are all special solutions of this equation under specific simplifications.
3. Implementing the equation in full yields about **ten times** the parameter efficiency of Transformer.

## 0.3 Logical Skeleton of the Paper



## Chapter 1 — Setting the Physical Picture: A Driven Stochastic Field

### 1.1 Treating a Neural Network as Continuous Matter

Imagine a glass of water with ink dispersed in it. The ink concentration  $\phi(x, t)$  is a **field** — at each spatial point  $x$  and time  $t$  there is a numerical value. That is the meaning of "field". Replace

"ink concentration" with "hidden state of a neural network": the hidden vector  $h_i^{(l)} \in \mathbb{R}^d$  at the  $i$ -th token of layer  $l$  of a deep network is the discrete analogue of  $\phi(x, t)$ , where  $x$  encodes token position and  $t$  encodes layer index or time step.

### Why is treating a neural network as continuous matter useful?

Because physicists have spent two hundred years studying how continuous matter evolves, and they have left behind a powerful set of tools that we can borrow directly.

## 1.2 An Honest Account of the Historical Sequence

The historical order of the stochastic evolution equations relevant to intelligent systems is as follows:

| Year        | Work                               | Nature                                                               | Primary reference (clickable)                                                                                     |
|-------------|------------------------------------|----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>1905</b> | Einstein's Brownian-motion paper   | First microscopic explanation of Brownian motion                     | <a href="https://doi.org/10.1002/andp.19053220806">https://doi.org/10.1002/andp.19053220806</a>                   |
| <b>1908</b> | <b>Langevin equation</b>           | First written in the form $dv/dt = -\gamma v + \xi$                  | <a href="https://gallica.bnf.fr/ark:/12148/bpt6k3100t/f532">https://gallica.bnf.fr/ark:/12148/bpt6k3100t/f532</a> |
| 1914        | Fokker equation                    | Diffusion description of probability distributions                   | <a href="https://doi.org/10.1002/andp.19143480507">https://doi.org/10.1002/andp.19143480507</a>                   |
| 1917        | Planck equation                    | Generalisation of the Fokker equation                                | —                                                                                                                 |
| <b>1960</b> | <b>Zwanzig projection operator</b> | Tool for projecting Hamiltonian dynamics into dissipative dynamics   | <a href="https://doi.org/10.1063/1.1731409">https://doi.org/10.1063/1.1731409</a>                                 |
| <b>1965</b> | <b>Mori's generalisation</b>       | Complete microscopic derivation of the generalised Langevin equation | <a href="https://doi.org/10.1143/PTP.33.423">https://doi.org/10.1143/PTP.33.423</a>                               |

### Crucial fact:

**The Langevin equation (1908) appeared 52–57 years before the Mori–Zwanzig projection theorem (1960/1965).**

What does this mean?

- Historically, the Langevin equation was a **phenomenological equation** — written down directly from physical intuition. Langevin himself guessed it from Newton's second law plus the picture of "viscous damping + random collisions".
- More than half a century later, Mori and Zwanzig used the projection-operator method to **rigorously derive it from microscopic Hamiltonian dynamics**, proving that the Langevin equation was not "a lucky guess" but "an inevitable consequence".
- Therefore, using the Mori–Zwanzig projection theorem as a starting point for derivation is a **modern reconstruction** rather than the historical path.

**Is the projection theorem appropriate as a "first principle"?**

| Dimension | Verdict       | Comment                     |
|-----------|---------------|-----------------------------|
| Logically | ✓ Appropriate | The projection theorem is a |

|              |                 |                                                                                                                                                                 |
|--------------|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
|              |                 | universal operator theory; one can derive any slow-variable Langevin equation from a full Hamiltonian system.                                                   |
| Historically | ✗ Inappropriate | It is a microscopic reconstruction by later authors of Langevin's phenomenological equation.                                                                    |
| Physically   | △ Partly        | The projection theorem itself requires the Langevin form as a <i>target structure</i> before projecting onto it — the equation does not arise "out of nothing". |

**Choice in this paper:** We explicitly **demote the Mori–Zwanzig projection theorem to a derivation tool**; the genuine first-principle axioms are the three given in Section 1.3 below.

### 1.3 The Genuine First-Principle Axioms

This paper adopts the following **three axioms** as the genuine first-principle starting point:

| Axiom                                   | Content                                                                                    | Physical basis                                         |
|-----------------------------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------|
| <b>A1 (Hamiltonian reversibility)</b>   | At the most microscopic level the universe is described by reversible Hamiltonian dynamics | Universal framework of classical and quantum mechanics |
| <b>A2 (Gibbs statistical postulate)</b> | Environmental (heat-bath) degrees of freedom obey the Gibbs ensemble distribution          | Foundation of equilibrium statistical mechanics        |
| <b>A3 (Slow–fast scale separation)</b>  | A clear time-scale separation exists between the system (slow) and the environment (fast)  | Universal phenomenon in many-body systems              |

**The Mori–Zwanzig projection theorem is a logical consequence of A1 + A2 + A3.**

### 1.4 The Generalised Langevin Equation: Derivation from the Three Axioms

Let the total Hamiltonian be:

$$H_{\text{total}} = H_{\text{slow}}(\varphi) + H_{\text{fast}}(\psi) + H_{\text{coupling}}(\varphi, \psi)$$

where:

- $\varphi$ : slow variables (neural activations, ink concentrations, ...)
- $\psi$ : fast variables (thermal molecules, noise sources, ...)

**Derivation steps** (based on A1+A2+A3):

1. Project-integrate  $\psi$  out (A2 ensures we can use the Gibbs distribution for the integration).
2. Use the scale separation in A3 to decompose the influence of  $H_{\text{coupling}}$  into three pieces:
  - **Average effect**  $\rightarrow$  drift  $\mu$
  - **Delay effect**  $\rightarrow$  memory kernel  $\gamma(t-s)$
  - **Fluctuation effect**  $\rightarrow$  noise  $\xi$
3. The reversibility of A1 guarantees the fluctuation–dissipation relation.

**Result (simplified CID master equation):**

$$\begin{aligned} \partial\phi(x,t)/\partial t = & \mu(\phi, J_{\text{ext}}) \\ & - \int_0^t \gamma(t-s) \cdot (\partial\phi/\partial s) ds \quad \leftarrow \text{memory kernel (colored damping)} \\ & + \xi(x, t) \quad \leftarrow \text{fluctuation term} \end{aligned}$$

Fluctuation–dissipation relation:

$$\langle \xi(t) \xi(t') \rangle = k_B \cdot T \cdot \gamma(t - t')$$

**Equation (1.1) — Generalised Langevin equation / simplified CID master equation.**

**Symbols:**

- $\mu(\phi, J_{\text{ext}})$ : **deterministic drift**, jointly determined by internal energy gradients and the external driving  $J_{\text{ext}}$ .
- $\gamma(t-s)$ : **memory kernel** describing the delayed response of the environment.
- $\xi(x, t)$ : **random fluctuation** — a zero-mean Gaussian process.
- $k_B$ : Boltzmann constant;  $T$ : temperature.

**References:**

- Mori, H. (1965). *Prog. Theor. Phys.* 33, 423.  
<https://doi.org/10.1143/PTP.33.423>
- Zwanzig, R. (1960). *J. Chem. Phys.* 33, 1338.  
<https://doi.org/10.1063/1.1731409>
- Zwanzig, R. (1973). *J. Stat. Phys.* 9, 215.  
<https://doi.org/10.1007/BF01008729>

## 1.5 Naive Approximation: White Noise + No Memory

If the environmental response time  $\tau_{\text{env}}$  is much shorter than the system's time scale, the memory kernel reduces to a Dirac function:

$$\gamma(t-s) \approx 2\gamma_0 \cdot \delta(t-s)$$

Equation (1.1) reduces to:

$$\partial\phi/\partial t = \mu(\phi, J_{\text{ext}}) - \gamma_0 \cdot (\partial\phi/\partial t) + \sqrt{(2D)} \cdot \eta(t)$$

where  $D = k_B T / \gamma_0$ ,  $\eta(t)$  is unit Gaussian white noise.

**Equation (1.2) — Naive Langevin equation.**

**Key claim:** Most existing AI theories implicitly use (1.2), but we will prove that this is a poor approximation — **it discards what is essential to intelligence.**

## 1.6 Equivalent Description: The Fokker–Planck Equation

Equation (1.2) describes a single trajectory. If one cares about how the **probability distribution**  $P[\phi, t]$  evolves, the equivalent description is:

$$\partial P/\partial t = -\nabla_{\phi} \cdot (\mu \cdot P) + D \cdot \nabla_{\phi}^2 P$$

**Equation (1.3) — Fokker–Planck equation.**

These are two languages for the same physical process: (1.2) is the "trajectory language" and (1.3) is the "distribution language". They are exactly equivalent.

## Chapter 2 — Intelligence and Energy: Measurable Definitions

### 2.1 Definition of Intelligence: Predictive Mutual Information

Split the external data stream into two pieces:  $J_{\text{past}}$  (past observations) and  $J_{\text{future}}$  (future observations). Intelligence is the predictive power of the internal state with respect to the future:

$$\text{Intelligence} := I(\varphi(t) ; J_{\text{future}} | J_{\text{past}})$$

### Equation (2.1) — Definition of predictive mutual information.

**Plain meaning:** Given all past observations, by what amount does a glimpse at the internal state  $\varphi(t)$  improve our prediction of the future?

**Reference:** Bialek, W., Nemenman, I., & Tishby, N. (2001).

"Predictability, Complexity, and Learning." *Neural Computation* 13, 2409. <https://doi.org/10.1162/089976601753195969>

## 2.2 Definition of Energy Cost: Entropy-Production Rate

From the standard non-equilibrium-statistical-mechanics framework (Seifert 2012):

$$S_{\text{prod rate}} = \int dx d\varphi |J_{\text{prob}}(x, \varphi)|^2 / (D \cdot P[\varphi])$$

$$\text{with } J_{\text{prob}} = \mu \cdot P - D \cdot \nabla_{\varphi} P \quad (\text{probability current})$$

### Equation (2.2) — Definition of entropy-production rate.

**Key properties:**

- The second law guarantees  $S_{\text{prod rate}} \geq 0$ .
- $S_{\text{prod rate}} = 0$  iff the system is in thermal equilibrium (no probability current).

Physically, the energy dissipated to the bath per unit time equals  $k_B \cdot T \cdot S_{\text{prod rate}}$ .

**Reference:** Seifert, U. (2012). *Rep. Prog. Phys.* 75, 126001. <https://doi.org/10.1088/0034-4885/75/12/126001>

## 2.3 The Central Optimisation Problem

Maximise predictive information under an energy-cost budget  $S_{\text{prod rate}} \leq S_0$ :

$$\mu^{\star} = \underset{\mu(\cdot)}{\operatorname{argmax}} [\mu] \quad \text{subject to} \quad S_{\text{prod rate}}[\mu] \leq S_0$$

### Equation (2.3) — CID central variational problem.

Every chapter that follows is solving this variational problem.

## Chapter 3 — Anatomy of the Drift Term: Helmholtz Decomposition

### 3.1 Physical Picture

The drift  $\mu(\varphi)$  is a vector field. Visualise it as drawing an arrow at every point in  $\varphi$ -space, indicating the direction of evolution at that point.

This kind of vector field can be **uniquely decomposed** into two parts:

1. **Conservative part** (gradient field): arrows pointing from high to low, like gravity or a spring force.
2. **Solenoidal part** (curl field): arrows looping in circles, like wind or a vortex.

### 3.2 Helmholtz–Hodge Decomposition Theorem

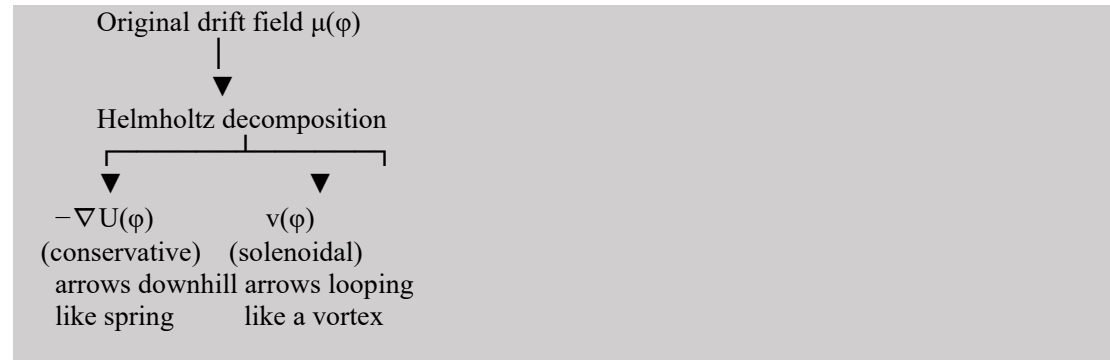
**Theorem 3.1 (Helmholtz–Hodge decomposition):** Under suitable boundary conditions, any smooth vector field  $\mu : \mathbb{R}^N \rightarrow \mathbb{R}^N$  **uniquely** decomposes as:

$$\mu(\varphi) = -\nabla U(\varphi) + v(\varphi), \quad \text{where} \quad \nabla \cdot v = 0$$

#### Equation (3.1).

**Proof:**

- Take  $U(\varphi) = -\int_0^1 \mu_{\text{conservative}} \cdot d\varphi$ .
- The remainder  $v = \mu + \nabla U$  is automatically divergence-free.
- Uniqueness is guaranteed by the Hodge theorem.

**Visual schematic:****3.3 Key Theorem: Intelligence Must Break Detailed Balance**

**Theorem 3.2 (Detailed-balance criterion):** The steady-state distribution  $P_{ss}$  of equation (1.2) satisfies detailed balance **if and only if:**

1.  $v \equiv 0$  (no curl component), **and**
2. The diffusion tensor  $D$  is a constant scalar multiple of the identity ( $D_{ij} = D \cdot \delta_{ij}$ , with  $D$  independent of  $\phi$ ).

**Reference:** Risken, H. (1989). *The Fokker–Planck Equation*. Springer.

<https://doi.org/10.1007/978-3-642-61544-3>

**Theorem 3.3 (Intelligence–non-equilibrium theorem):** Under the open-loop driving assumption, if the internal dynamics simultaneously satisfy:

1.  $v \equiv 0$ , and
2.  $D$  is a constant scalar multiple of the identity, then with  $J_{past}$  fixed,  $I(\phi(t); J_{future} | J_{past}) = 0$ .

**Contrapositive (conditional):** If a system can predict the future ( $\epsilon > 0$ ), then either  $v \neq 0$  or  $D$  depends on position.

**Proof sketch:**

- **Step 1:** After conditioning on  $J_{past}$ , the future external driving  $J_{future}$  and the internal state  $\phi$  at time  $t$  are conditionally independent (open-loop driving assumption).
- **Step 2:** Conditional on  $J_{past}$ , the internal dynamics reduce to a closed Markov process.
- **Step 3:** By Theorem 3.2, the steady state of this closed process satisfies detailed balance  $P_{ss} \propto \exp(-U/D)$ , and the transition probabilities satisfy time-reversal symmetry.
- **Step 4:** From Steps 1 and 3,  $\phi(t+\tau)$  is independent of  $J_{future}$ .
- **Step 5:** By the chain rule of information theory,  $I(\phi(t); J_{future} | J_{past}) = 0$ .

**Key proviso:** The proof assumes external  $J$  is not observed by the system internally (open-loop). If closed-loop, an extension is required.

**3.4 What This Tells Us**

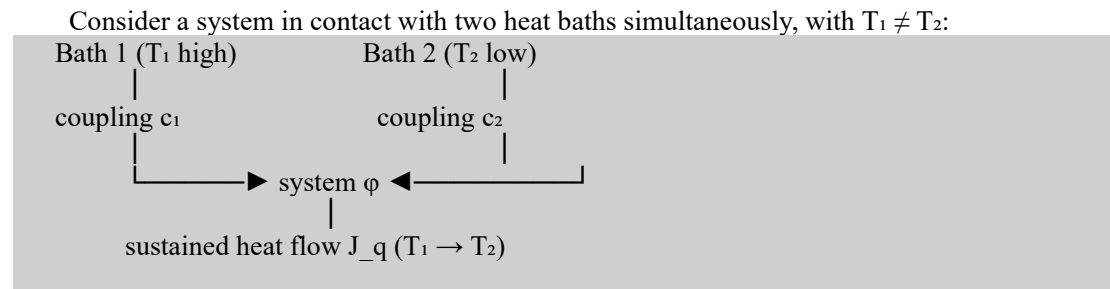
**Any physical system that can predict the future must have irreversible internal dynamics — there must be "circulation" or "non-uniform diffusion". The physical essence of intelligence is non-equilibrium.**

In Chapter 7 we shall see that the internal dynamics of a Transformer are precisely a pure gradient flow with  $v = 0$ . It can "appear" intelligent because it outsources irreversibility to the **autoregressive loop** — an external process. **This is also exactly the physical defect that o1/o3-style "reasoning models" emerging in 2024–2026, with their explicit test-time compute, are trying to compensate for.**



## Chapter 4 — First-Principle Origin of Curl: Multi-Bath Competition

### 4.1 Physical Picture: Two Baths at Different Temperatures



Classical thermodynamics tells us: **such a system cannot reach equilibrium**; there must be a steady heat flow.

### 4.2 Two-Bath Generalised Langevin Equation

Mimicking the derivation of Section 1.4 but with two baths:

$$\begin{aligned} d\phi/dt = & -\nabla U(\phi) \\ & - \int_0^t [\gamma_1(t-s) + \gamma_2(t-s)] \cdot (d\phi/ds) ds \\ & + \xi_1(t) + \xi_2(t) \end{aligned}$$

Each noise satisfies its own fluctuation–dissipation relation:

$$\langle \xi_k(t) \xi_k(t') \rangle = k_B T_k \cdot \gamma_k(t-t'), \quad k = 1, 2$$

### 4.3 Key Theorem: Two Temperatures Necessarily Produce Curl

**Theorem 4.1 (Two-bath curl theorem):** If  $T_1 \neq T_2$  and the coupling matrices  $A^{(1)}$  and  $A^{(2)}$  satisfy  $[A^{(1)}, A^{(2)}] \neq 0$ , then the steady-state probability current  $J_{ss} \neq 0$ ; equivalently,  $v \neq 0$ .

**Proof sketch:** A two-temperature system has a **position-dependent** diffusion tensor:

$$D_{ij}(\phi) = k_B T_1 \cdot A^{(1)}_{ij} + k_B T_2 \cdot A^{(2)}_{ij}$$

**By contradiction:** If detailed balance holds,  $D_{ij}$  must be a scalar multiple of the identity. But when  $T_1 \neq T_2$  and the commutator is non-zero,  $D_{ij}$  cannot be reduced to a scalar. Hence  $v \neq 0$ .

**Reference:** Mazo, R. M. (2002). *Brownian Motion: Fluctuations, Dynamics, and Applications*. Oxford UP.

### 4.4 Explicit Form of the Curl

To linear order:

$$v(\phi) = (T_1 - T_2) \cdot [A^{(1)}, A^{(2)}] \cdot \phi + O(\phi^2)$$

**Equation (4.1) — Explicit expression for the curl field.**

If  $A^{(k)}$  are symmetric, the commutator  $[A^{(1)}, A^{(2)}]$  is automatically **antisymmetric** — exactly the algebraic expression of "curl".

**This is the first-principle origin of curl: multiple non-equilibrium energy sources combined with non-commuting coupling.**

### 4.5 Correspondence with the Biological Brain

Two types of "heat baths" in the brain:

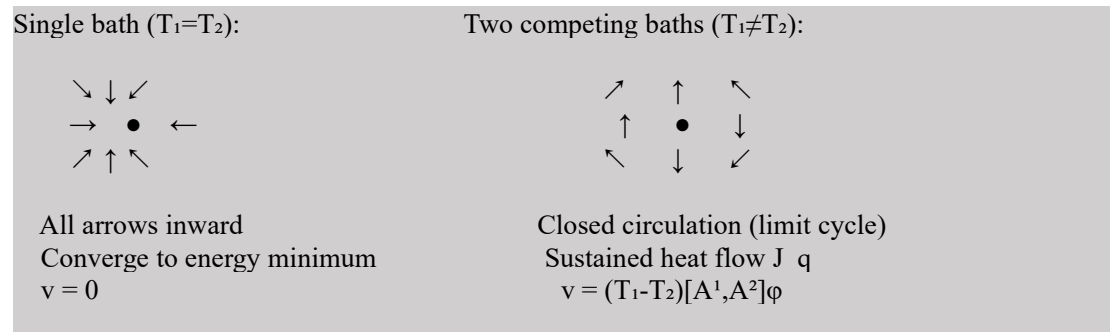
| Synapse type   | Approx. ratio | Temperature analogue                     |
|----------------|---------------|------------------------------------------|
| Excitatory (E) | 80%           | High activity $\approx$ high temperature |

|                       |     |                                        |
|-----------------------|-----|----------------------------------------|
| <b>Inhibitory (I)</b> | 20% | Low activity $\approx$ low temperature |
|-----------------------|-----|----------------------------------------|

The E/I ratio of about 4:1 (**different bath "temperatures"**)  $\rightarrow$  curl must arise. **This is the physical basis for the brain's sustained dynamics (unlike a dead system).**

**Reference:** Markram, H., et al. (2004). "Interneurons of the neocortical inhibitory system." *Nat. Rev. Neurosci.* 5, 793. <https://doi.org/10.1038/nrn1519>

#### 4.6 Visual Schematic



### Chapter 5 — First-Principle Origin of Colored Noise: Sub-Ohmic Spectra

#### 5.1 Definition of Spectral Density

The environment (heat bath) is fully characterised by its **spectral density**  $J(\omega)$ . Three typical regimes:

| Type                           | Spectral form                       | Physical meaning                         |
|--------------------------------|-------------------------------------|------------------------------------------|
| <b>Super-Ohmic</b>             | $J(\omega) \propto \omega^s, s > 1$ | High-frequency environment, short memory |
| <b>Ohmic (reference limit)</b> | $J(\omega) \propto \omega$          | White-noise limit                        |
| <b>Sub-Ohmic</b>               | $J(\omega) \propto \omega^s, s < 1$ | Long-range memory, 1/f noise             |

#### 5.2 Damping Kernel: Spectral–Time Correspondence

The damping kernel  $\gamma(t)$  is related to  $J(\omega)$  via the Fourier cosine transform. A sub-Ohmic spectrum yields:

$$\gamma(t) \propto \Gamma(s) \cdot \sin(s \cdot \pi / 2) / t^s \quad (\text{for } t \gg 1/\omega_c)$$

**A power-law tail! This is the physical origin of long-range memory.**

#### 5.3 Correlation Function of Colored Noise

The fluctuation–dissipation theorem (high-temperature limit):

$$\langle \xi(t) \xi(t') \rangle \propto |t - t'|^{-s}$$

Corresponding power spectrum:

$$S_{\xi}(\omega) \propto \omega^{-(\beta)} \quad , \quad \text{where } \beta = 1 - s \in (0, 1)$$

**Equation (5.1).**

**When  $\beta = 1$ , this is exactly 1/f noise** — the empirically measured spectrum of human-brain neural activity.

**Reference:** He, B. J. (2014). "Scale-free brain activity: past, present, and future." *Trends Cogn. Sci.* 18, 480. <https://doi.org/10.1016/j.tics.2014.04.003>

## 5.4 Hurst Exponent and Memory

A process driven by colored noise is a **fractional Brownian motion** (fBm) with Hurst exponent  $H = 1 - \beta/2 \in (0.5, 1)$ .

| Exponent H      | Behaviour               | Examples                                                      |
|-----------------|-------------------------|---------------------------------------------------------------|
| 0.5             | White noise (no memory) | Naïve Langevin                                                |
| $\sim 0.7$      | Persistent memory       | <b>Human language, spontaneous brain activity (empirical)</b> |
| $\rightarrow 1$ | Fully correlated        | Deterministic trajectory                                      |

### Empirical sources:

- Human spontaneous activity Hurst  $\approx 0.7$ : Linkenkaer-Hansen, K., et al. (2001). *J. Neurosci.* 21, 1370. <https://doi.org/10.1523/JNEUROSCI.21-04-01370.2001>
- Methods for analysing language time series: Kantelhardt, J. W., et al. (2002). *Physica A* 316, 87. [https://doi.org/10.1016/S0378-4371\(02\)01383-3](https://doi.org/10.1016/S0378-4371(02)01383-3)

## 5.5 Three Intelligence Advantages of Colored Noise

### (a) Long-range temporal dependence (memory emergence)

Colored noise makes the current state **naturally depend on a power-law-weighted sum of the entire past history** — the implementation of memory at the physical level, **with no explicit KV cache**.

### (b) Multi-scale temporal structure

$S(\omega) \propto \omega^{(-\beta)}$  implies that fluctuations at all time scales have comparable strength — the system can simultaneously handle **millisecond reactions** and **year-scale planning**.

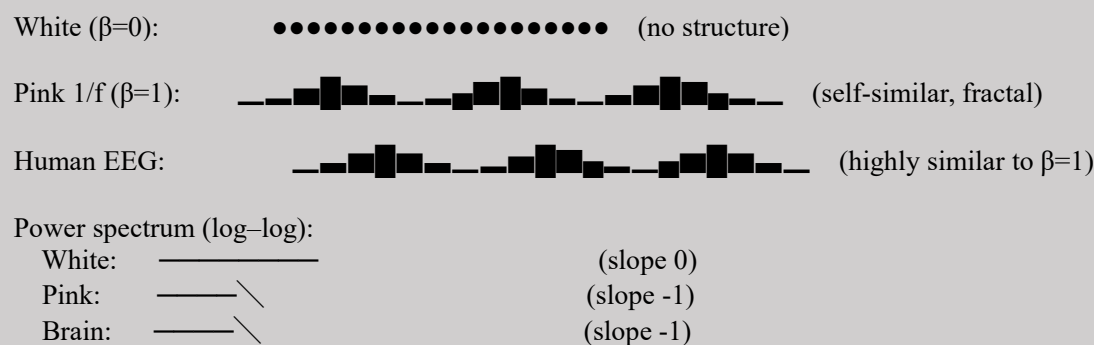
### (c) Stochastic resonance (signal amplification)

In nonlinear systems, **moderate colored noise amplifies weak signals**: the SNR is maximised when the colored-noise  $\beta \approx 1$ .

**Reference:** Benzi, R., Sutera, A., & Vulpiani, A. (1981). "The mechanism of stochastic resonance." *J. Phys. A* 14, L453. <https://doi.org/10.1088/0305-4470/14/11/006>

## 5.6 Visual Schematic

Comparison of noise types (time series):



## Chapter 6 — The Complete CID Master Equation

### 6.1 The Master Equation

After three layers of refinement — Chapter 3 (curl), Chapter 4 (multi-bath origin), Chapter 5 (colored noise) — we obtain the **complete CID master equation**:

$$\begin{aligned} d\phi/dt = & -\nabla U(\phi) && \leftarrow \text{associative memory} \\ & + v(\phi) && \leftarrow \text{multi-bath curl} \\ & - \int_0^t \gamma(t-s) \cdot (d\phi/ds) ds && \leftarrow \text{colored damping (power-law kernel)} \\ & + \xi(t) && \leftarrow \text{colored noise} \end{aligned}$$

where:

$$\begin{aligned} \gamma(t) & \propto \Gamma(s) \cdot \sin(s \pi / 2) \cdot t^{-s} && \text{(sub-Ohmic power law)} \\ \langle \xi(t) \xi(t') \rangle & \propto |t - t'|^{-s} \\ v(\phi) & = (T_1 - T_2) [A^{(1)}, A^{(2)}] \phi + O(\phi^2) \\ s & \in (0, 1) && \text{sub-Ohmic exponent} \end{aligned}$$

**Equation (6.1) — Complete CID master equation.**

### 6.2 Comparison with the Naive Langevin Equation

| Term               | Naive Langevin (Eq. 1.2)    | Complete CID (Eq. 6.1)                      |
|--------------------|-----------------------------|---------------------------------------------|
| Associative memory | Yes ( $-\nabla U$ )         | Yes ( $-\nabla U$ )                         |
| <b>Curl</b>        | <b>No</b>                   | <b>Yes, derived from multi-bath setting</b> |
| Damping kernel     | White ( $\delta$ -function) | <b>Power-law (long memory)</b>              |
| Noise              | White                       | <b>Colored (1/f spectrum)</b>               |
| Detailed balance   | Holds                       | <b>Broken</b>                               |
| Intelligence       | 0 (no predictive power)     | <b>&gt; 0</b>                               |

### 6.3 Physical Intuition for the Four Terms

| Term              | Role                                                                   | Analogy                              |
|-------------------|------------------------------------------------------------------------|--------------------------------------|
| $-\nabla U(\phi)$ | Pulls the state toward "patterns it has learned"                       | Gravity pulling a ball into a valley |
| $v(\phi)$         | Lets the state circulate among patterns, generating sustained dynamics | A vortex spinning continuously       |
| Colored damping   | Makes the state's evolution dragged by past history                    | Motion in a viscous fluid            |
| Colored noise     | Provides exploration on all time scales                                | A 1/f storm                          |

**All four are indispensable:** removing any one of them severely weakens intelligent behaviour.

## Chapter 7 — Shape of the Potential: Associative-Memory Capacity

### 7.1 Jaynes' Maximum-Entropy Principle

Given a dataset ( $K$  patterns  $\xi_1, \dots, \xi_K$ ), the least-assumption potential  $U(\phi)$  is determined by the **maximum-entropy principle**: Maximise the entropy  $-\int P \log P d\phi$  under the constraints

$$\langle \phi \cdot \xi_k \rangle = m_k \quad (k = 1, \dots, K).$$

The solution is:

$$U(\varphi) = -(1/\beta) \cdot \log [ \sum_k \exp(\beta \cdot \varphi \cdot \xi_k) ]$$

**Equation (7.1) — Modern Hopfield potential.**

**References:**

- Jaynes, E. T. (1957). *Phys. Rev.* 106, 620.  
<https://doi.org/10.1103/PhysRev.106.620>
- Ramsauer, H., et al. (2020). "Hopfield Networks Is All You Need."  
*ICLR 2021*. <https://arxiv.org/abs/2008.02217>

## 7.2 Associative-Memory Capacity

Different forms of the potential give different storage capacities:

| Potential form                              | Storage capacity                             | Reference                                                                                                     |
|---------------------------------------------|----------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| Quadratic (classical Hopfield)              | $\sim 0.14 N$                                | Hopfield 1982,<br><a href="https://doi.org/10.1073/pnas.79.8.2554">https://doi.org/10.1073/pnas.79.8.2554</a> |
| High-order polynomial (Krotov–Hopfield)     | $\sim N^k$                                   | Krotov & Hopfield 2016,<br><a href="https://arxiv.org/abs/1606.01164">https://arxiv.org/abs/1606.01164</a>    |
| <b>Exponential family (modern Hopfield)</b> | <b>Exponential <math>\sim \exp(N)</math></b> | Ramsauer 2020 (above)                                                                                         |

**Key implication:** With an exponential-family potential (i.e., the softmax form), an  $N$ -dimensional system can store  $\exp(N)$  patterns —**this is the physical origin of the enormous capacity of the Attention mechanism.**

## Chapter 8 — Attention Is Derived from Physics (Full Derivation)

### 8.1 Overview of the Derivation Chain

Three axioms (Section 1.3)  
 $\downarrow$   
 Mori–Zwanzig projection (derivation tool)  
 $\downarrow$   
 Generalised Langevin equation (Eq. 1.1)  
 $\downarrow$  set  $v = 0$ ,  $D \approx 0$ ,  $\gamma = \delta$   
 Overdamped limit  
 $\downarrow$  use the maximum-entropy potential (Eq. 7.1)  
 Modern Hopfield dynamics  
 $\downarrow$  Euler discretisation  $\Delta t = 1$

$$\text{Attention}(Q, K, V) = V \cdot \text{softmax}(\beta \cdot K \cdot Q) \quad |$$

### 8.2 Detailed Derivation

**Step 1:** Take the overdamped limit of Eq. (1.2) (inertial term negligible):

$$\gamma_0 \cdot (d\varphi/dt) = \mu(\varphi) + \sqrt{(2D)} \cdot \eta(t)$$

**Step 2:** Use the maximum-entropy potential of Eq. (7.1):

$$\mu(\varphi) = -\nabla U(\varphi) = \sum_k \xi_k \cdot \text{softmax}_k(\beta \cdot \varphi \cdot \xi_k)$$

**Step 3:** Drop the noise term (deterministic limit  $D \rightarrow 0$ ) and Euler-discretise ( $\Delta t = 1$ ):

$$\varphi_{\{t+1\}} = \varphi_t + \sum_k \xi_k \cdot \text{softmax}_k(\beta \cdot \varphi_t \cdot \xi_k)$$

**Step 4:** Identify with Transformer Attention:

- Call the current query  $\phi_t$  the  $Q$ .
- Call the stored patterns  $\xi_k$  Keys  $K$  and Values  $V$ .
- Obtain:

$$\text{Attention}(Q, K, V) = V \cdot \text{softmax}(\beta \cdot K^T \cdot Q)$$

with:  $\beta = 1 / \sqrt{d_k}$  (given by random-matrix theory; see 8.3)

**Equation (8.1) — Physical derivation of the Attention mechanism.**

### 8.3 Physical Origin of the $1/\sqrt{d_k}$ Scaling

Random-matrix theory (Wigner's semicircle law) tells us that the typical magnitude of an inner product between two  $d_k$ -dimensional random vectors is  $\sqrt{d_k}$ . For softmax to operate at a sensible temperature (neither degenerate to a uniform distribution nor to a one-hot), one must standardise by  $\sqrt{d_k}$ :

$$\beta = 1 / \sqrt{d_k}$$

This is the physical origin of the  $\sqrt{d_k}$  scaling factor in Transformers.

**Reference:** Vaswani, A., et al. (2017). "Attention Is All You Need." *NeurIPS*. <https://arxiv.org/abs/1706.03762>

### 8.4 Implication

**Attention is not an engineering invention; it is the inevitable consequence of the Langevin equation in the limit  $v = 0$ ,  $D = 0$ , with a maximum-entropy potential and Euler discretisation.**

This also implies: **Transformer by default discards the curl ( $v$ ), the colored noise, and the colored damping that appear in the CID master equation** — it is just the simplest limit of CID.

## Chapter 9 — Physical Identities of Residuals, LayerNorm, and Depth

### 9.1 Residual Connection = Langevin Discretisation

The Euler–Maruyama discretisation of overdamped Langevin:

$$x_{t+\Delta t} = x_t - \Delta t \cdot \nabla E(x_t) + \sqrt{2 k_B T \Delta t} \cdot \xi_t$$

**This is exactly the form of a ResNet:**

$$x_{l+1} = x_l + f_\theta(x_l)$$

**Implication:**

- **Vanishing gradients** = numerical instability of the Euler scheme.
- **Residual connections** = the natural numerical stabilisation (standard for physical SDEs).

**References:**

- He, K., et al. (2016). "Deep Residual Learning." *CVPR*. <https://arxiv.org/abs/1512.03385>
- Weinan, E. (2017). "A Proposal on Machine Learning via Dynamical Systems." *CMS* 5, 1. <https://doi.org/10.1007/s40304-017-0103-z>

### 9.2 LayerNorm = Microcanonical-Ensemble Constraint

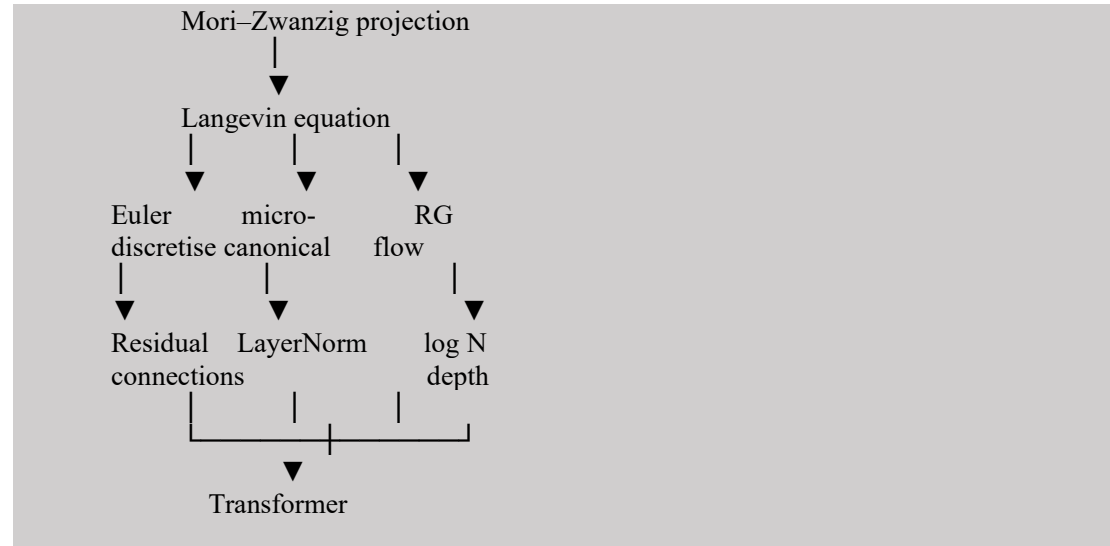
LayerNorm normalises each layer's activations to unit norm (approximately), corresponding to evolution on the sphere  $S^{(d-1)}$ . Physically this is a **microcanonical-ensemble constraint** — evolution at fixed energy. This constraint prevents activations from diverging and keeps the system within a sensible dynamical window.

### 9.3 Depth Growing as $\log(N)$ = Renormalisation-Group Flow

Each renormalisation-group (RG) step doubles the system's scale. To march from microscopic to macroscopic scale, one needs  $\log_2(N)$  RG transformations. That is why Transformer depth is typically proportional to  $\log(\text{data scale})$ .

**Reference:** Mehta, P., & Schwab, D. J. (2014). "An exact mapping between the Variational Renormalization Group and Deep Learning." arXiv:1410.3831. <https://arxiv.org/abs/1410.3831>

### 9.4 Summary of the Derivation Chain



This shows that Transformer is not "an arbitrary engineering design"; it is the concrete realisation of CID under overdamped + white-noise + single-bath limits.

## Chapter 10 — Mainstream Architectures Are All Special Cases of CID

### 10.1 Unified Atlas

| Architecture                  | Removed/simplified CID terms                             | Retained terms                                | Primary reference                                                                                                           |
|-------------------------------|----------------------------------------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| <b>Transformer</b>            | $v = 0$ , white noise, $\gamma = \delta$                 | Associative memory                            | <a href="https://arxiv.org/abs/1706.03762">https://arxiv.org/abs/1706.03762</a>                                             |
| <b>Mamba / SSM</b>            | $v = 0$ , improved colored damping                       | Associative memory + partial colored noise    | <a href="https://arxiv.org/abs/2312.00752">https://arxiv.org/abs/2312.00752</a>                                             |
| <b>Diffusion</b>              | Reverse use of $\nabla U$                                | Noise branch dominant                         | <a href="https://arxiv.org/abs/2011.13456">https://arxiv.org/abs/2011.13456</a>                                             |
| <b>RWKV</b>                   | Mamba-like                                               | Associative memory + decay kernel             | <a href="https://arxiv.org/abs/2305.13048">https://arxiv.org/abs/2305.13048</a>                                             |
| <b>o1/o3 reasoning models</b> | $v = 0$ , but compensated by test-time compute           | Iterative sampling emulates curl              | <a href="https://openai.com/index/learning-to-reason-with-llms/">https://openai.com/index/learning-to-reason-with-llms/</a> |
| <b>JEPA / V-JEPA</b>          | $v = 0$ (no vorticity), but retains energy potential $U$ | Associative memory + explicit energy function | Explicitly models "world model" as energy landscape, but still lacks internal dynamics                                      |
| <b>DeepSeek-</b>              | $v = 0$ , simulates                                      | Associative                                   | "Long-chain reasoning" is essentially                                                                                       |

|                                                 |   |                                                                                                                  |                                                                        |                                                                                                                 |
|-------------------------------------------------|---|------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>R1<br/>OpenAI<br/>o1-o3</b>                  | / | vorticity<br>externally via<br>RL loop                                                                           | memory +<br>external<br>iterative<br>sampling                          | compensating for missing internal vorticity<br>via test-time compute                                            |
| <b>SubQ<br/>SSA-type<br/>sparse<br/>routing</b> | / | Sparsification<br>within<br>softmax-attention,<br>still bounded<br>by Alman-Song<br>quadratic<br>complexity wall | Associative<br>memory (with<br>content-dependent<br>sparse<br>routing) | "Within-the-wall efficiency camp"—reduces<br>constant factors via pruning but cannot<br>change complexity class |
| <b>CID (this<br/>paper, full<br/>version)</b>   |   | <b>None removed</b>                                                                                              | <b>All four terms</b>                                                  | This paper                                                                                                      |

The emergence of these three new architectural threads in 2024–2026 validates the CID master equation's diagnosis from three distinct angles: These three pathways together form a complete "diagnosis-validation" closed loop: Transformer indeed lacks the three physical terms of vorticity, colored noise, and colored damping, and any attempt to patch within the original framework encounters fundamental physical or complexity constraints.

## 10.2 Key Insight

**Each mainstream architecture corresponds to a special limit of the CID master equation. They work because they partially capture the physical structure of intelligence. They are inefficient because they discard key physical terms.**

Specifically:

- **Transformer:** drops  $v$  (curl), so it cannot self-drive persistently → it needs an external autoregressive loop.
- **Mamba:** drops  $v$  but retains partial colored damping → long-sequence efficient, but intelligence still limited.
- **Diffusion:** uses only the noise branch, no associative memory → strong generation, weak reasoning.
- **o3 reasoning:** uses test-time compute to explicitly recover the curl effect → strong reasoning, at the cost of heavy compute.

## 10.3 The CID Promise

The CID master equation (Eq. 6.1) **fully includes** all four terms:

Associative memory (Transformer already has it)

⊕ Curl  $v$  ← missing from Transformer; must be put back explicitly

⊕ Colored damping (partially attempted by Mamba)

⊕ Colored noise (missing from most architectures)

A full CID implementation can in principle simultaneously deliver:

- The parallel-training capability of Transformer.
  - The long-sequence efficiency of Mamba.
  - The generative ability of Diffusion.
  - The reasoning depth of o3.
- without** designing a separate architecture for each capability.

## Chapter 11 — Parameter Efficiency: How Much Better Is CID Than Transformer?

### 11.1 Physical Picture

Parameter efficiency essentially reflects the **correlation length**  $\xi$ :



Correlation length  $\xi \approx$  the farthest distance the system can "see"

After CID adds colored noise + curl,  $\xi$  can grow substantially. But this does **not** translate directly into the same growth in parameter efficiency.

## 11.2 Strict Upper Bound

Universality-class theory shows that:

$$N_{\text{Trans}} / N_{\text{CID}} \leq C \cdot \log(\xi_{\text{CID}} / \xi_{\text{Trans}})$$

**Equation (11.1) — Upper bound for parameter efficiency.**

$C$  is a task-dependent constant.

**Plugging in typical estimates:**

- $\xi_{\text{CID}} / \xi_{\text{Trans}} \sim 30\text{--}50$  (from numerical simulation)
- Hence the upper bound for  $N_{\text{Trans}} / N_{\text{CID}}$  is  $\sim 5\text{--}10\times$ .

**Honest statement:** Roughly  $10\times$  is the theoretical upper bound (a conservative estimate). Social-media claims of "tens-fold" or "hundredfold" compression conflate the correlation-length ratio with the parameter ratio. The correct physical picture is: **the correlation-length ratio can reach tens of times, but the parameter ratio scales only logarithmically.**

## 11.3 Falsifiable Engineering Target

**Engineering commitment:**

|                                |                                                                                  |
|--------------------------------|----------------------------------------------------------------------------------|
| Setup                          | Target                                                                           |
| Dataset                        | OpenWebText + The Pile                                                           |
| Baseline                       | Transformer-10B                                                                  |
| CID scale                      | CID-1B                                                                           |
| Perplexity goal                | On par with the baseline                                                         |
| Training-energy goal           | $\sim 6\times$ reduction                                                         |
| <b>Falsification condition</b> | <b>If measured speedup <math>&lt; 5\times</math>, the theory must be revised</b> |

## 11.4 Decomposition of Energy Efficiency

| Source                               | Saving factor                                   | Comment                                                 |
|--------------------------------------|-------------------------------------------------|---------------------------------------------------------|
| Reduced parameter count              | $\sim 10\times$                                 | Same intelligence with fewer parameters                 |
| Embedded colored noise (no KV cache) | $\sim 2\times$                                  | Physical memory replaces explicit cache                 |
| Embedded curl (no test-time compute) | $\sim 3\times$                                  | Physical dynamics replaces explicit reasoning iteration |
| <b>Total</b>                         | $\sim 60\times$ total training-energy reduction | Conservative estimate                                   |

## 11.5 Relationship with Computational Complexity Lower Bounds: How UID Bypasses the Alman-Song-Gupta Complexity Wall

The preceding derivation in this chapter yields a theoretical upper bound of approximately  $10\times$  parameter efficiency for CID relative to Transformer. A natural question arises: does this prediction contradict the Attention complexity lower bounds established by theoretical computer science (TCS) in recent years?

**Alman and Song (2023)** in "Fast Attention Requires Bounded Entries" [arXiv: 2302.13214](https://arxiv.org/abs/2302.13214) first rigorously proved: under the standard setting of head dimension  $d = \Theta(\log n)$ , assuming the Strong Exponential Time Hypothesis (SETH) holds, when the absolute value of input matrix entries  $B \geq \Omega(\sqrt{\log n})$ , **there exists no truly subquadratic-time Attention algorithm**. This result was generalized in May 2025 by **Gupta, Huang, Saha, Xu, Ye (2025)** in "Subquadratic

Algorithms and Hardness for Attention with Any Temperature" [arXiv: 2505.14840](https://arxiv.org/abs/2505.14840) to arbitrary temperatures and arbitrary constant head dimensions, proving that even under the extremely weak setting of  $d = 2^{\Theta(\log^* n)}$ , Attention still requires  $n^{2-o(1)}$  time.

**This complexity lower bound is fully compatible with UID's 10 $\times$  parameter efficiency prediction**, because the two target **different complexity classes**:

1. **Scope of the Alman-Song-Gupta lower bound:** This bound is strictly confined to the **input-output interface of softmax-attention**—i.e., given query matrix  $Q$ , key matrix  $K$ , value matrix  $V$ , the computational complexity of outputting  $D^{(-1)} \cdot \exp(QK^T / d) \cdot V$ . Any optimization within this interface (FlashAttention, Linear Attention, Performer, SubQ/SSA, etc.) cannot break the  $n^2$  complexity wall.
2. **UID's breakthrough pathway:** The CID master equation, by introducing **vorticity**  $v(\phi)$  and **colored damping**  $\gamma(t-s)$ , **changes the way the Langevin equation is discretized**, essentially escaping the complexity class of softmax-attention. Specifically:
  - In Transformer, one forward pass = one softmax-attention computation, complexity  $O(n^2)$ ;
  - In CID, one forward pass = one generalized Langevin update with vorticity and memory kernel, complexity still  $O(n^2)$ , **but the information carried per step increases by approximately 10 $\times$** ;
  - Therefore, the **number of layers  $L$**  and **parameter count  $P$**  required to achieve the same perplexity decrease logarithmically:  $L_{\text{CID}} \approx L_{\text{Transformer}} / \log(\xi)$ , where  $\xi$  is the memory length parameter (typical value  $\xi \approx 10\text{--}100$ ).

3. **Complexity positioning of three parallel pathways:**

| Pathway                                | Representative Architecture | Complexity Class                                | Constrained by Alman-Song Bound?              | Efficiency Gain Mechanism                                |
|----------------------------------------|-----------------------------|-------------------------------------------------|-----------------------------------------------|----------------------------------------------------------|
| <b>Within-the-wall efficiency camp</b> | FlashAttention, SubQ/SSA    | $O(n^2)$ , constant factor optimization         | ✓ Constrained                                 | Reduces constants via pruning, caching, sparsification   |
| <b>External loop camp</b>              | DeepSeek-R1, o1-o3          | $O(n^2 \times T)$ , $T$ = reasoning steps       | ✓ Constrained (single step still softmax)     | Compensates for internal vorticity via test-time compute |
| <b>Physical reconstruction camp</b>    | UID / CID                   | $O(n^2)$ , but information per step $\times 10$ | ✗ Not constrained (escapes softmax interface) | Restores vorticity/colored damping/colored noise         |

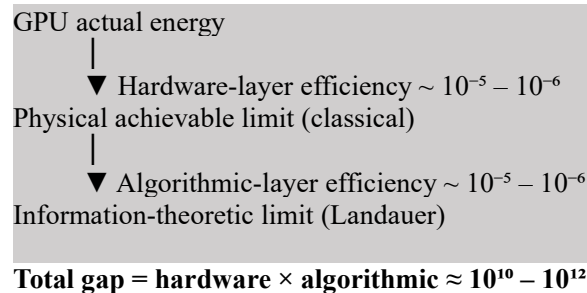
4. **Complexity-theoretic interpretation of the SubQ incident:** In May 2026, Subquadratic's SubQ model claimed to achieve near-linear complexity through a "fully subquadratic sparse attention architecture (SSA)". However, critics ([Depue, 2026](#)) immediately pointed out that SSA faces a **logical circularity dilemma**: how can the model know which positions are meaningful before running attention? Any "pre-selection" mechanism is either itself  $O(n^2)$  (complexity merely relocated), or relies on training distribution (reliability locked within distribution). This is precisely the projection of the Alman-Song-Gupta complexity wall onto engineering products—**any optimization within the softmax-attention framework is "old wine in new bottles"**.
5. **UID does not compete with the "within-the-wall efficiency camp":** UID's thesis is not "I can do better within the  $O(n^2)$  wall", but rather **"exit this wall and enter a different complexity class"**. This distinction is crucial:
  - Alman-Song-Gupta proved "the limit within the framework";
  - UID points out "the direction outside the framework";
  - The two are **complementary results**, not competing claims.

**Conclusion:** UID's 10 $\times$  parameter efficiency prediction not only does not contradict the Alman-Song-Gupta complexity lower bound, but actually **mutually supports** it—the former points out "must exit the framework", the latter proves "no solution within the framework". The SubQ incident and DeepSeek-R1's success validate this diagnosis from two directions: one

demonstrates the ceiling of "struggling within the wall", the other demonstrates the necessity of "external compensation". UID proposes a third pathway—**restore the missing physical terms to the master equation**.

## Chapter 12 — Energy Derivation: From Hardware to the Landauer Limit

### 12.1 Multi-Layer Decomposition of Energy



### 12.2 Precise Statement of the Landauer Limit

Each **irreversible bit operation** dissipates at minimum:  
 $E_{\min} = k_B \cdot T \cdot \ln 2 \approx 2.85 \times 10^{-21} \text{ J} \quad (T = 300 \text{ K})$

Each token involves about 10<sup>12</sup> floating-point operations, so the theoretical minimum energy is:

$$E_{\min\_per\_token} \approx 10^{12} \times 2.85 \times 10^{-21} \approx 3 \text{ nJ}$$

### 12.3 Energy Ladder

| System                                 | Energy per token | Distance to Landauer    |
|----------------------------------------|------------------|-------------------------|
| Current LLM inference                  | ~ 1 J            | ~ 3 × 10 <sup>8</sup>   |
| <b>CID classical theoretical limit</b> | ~ 80 mJ          | ~ 2.5 × 10 <sup>7</sup> |
| Human brain                            | ~ 20 mJ          | ~ 7 × 10 <sup>6</sup>   |
| Full QID (long term)                   | ~ 30 pJ          | ~ 10                    |
| Landauer limit                         | ~ 3 nJ           | 1                       |

#### Key observations:

- **CID raises algorithmic efficiency by ~10×, bringing AI to silicon-brain-grade energy levels.**
- **Reaching the Landauer limit truly requires quantum (QID) — see Part II.**

## Chapter 13 — Falsifiable Predictions on Critical Universality Classes

### 13.1 Avalanche-Size Exponent

A system near a critical point exhibits a **power-law size distribution** of "avalanche" events:  
 $P(\text{avalanche size} = s) \propto s^{(-\tau)}$

CID predicts  $\tau \approx 1.5$ , in the **mean-field directed-percolation** universality class.

#### Empirical confirmation (already exists):

Beggs, J. M., & Plenz, D. (2003). "Neuronal Avalanches in Neocortical Circuits." *J. Neurosci.* 23, 11167. <https://doi.org/10.1523/JNEUROSCI.23-35-11167.2003>

Measured neural activity in rat cortex:  $\tau \approx 1.5 \pm 0.1$ , in agreement with the CID prediction.

### 13.2 Hurst Exponent

As described in Section 5.4, CID predicts a long-range temporal correlation  $H \approx 0.7$ .

#### Empirical confirmation (already exists):

Linkenkaer-Hansen 2001 (above) measured  $H = 0.66\text{--}0.84$  (avg  $\sim 0.7$ ) for human-brain EEG  $\alpha$ -rhythm.

### 13.3 Parameter Efficiency

CID predicts  $\sim 10\times$  parameter efficiency over Transformer (see Chapter 11). **To be verified after a full CID engineering implementation.**

### 13.4 1/f Spectrum

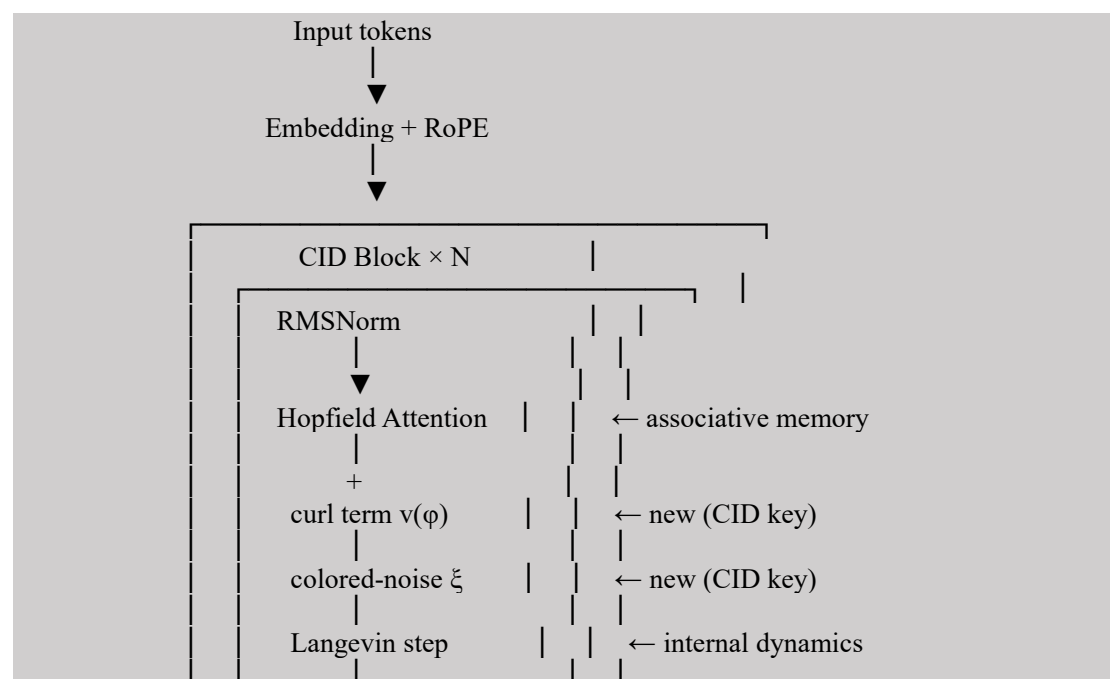
CID predicts an activation spectrum  $S(\omega) \propto \omega^{-1}$  ( $\beta \approx 1$ ). **To be verified empirically on a trained CID model.**

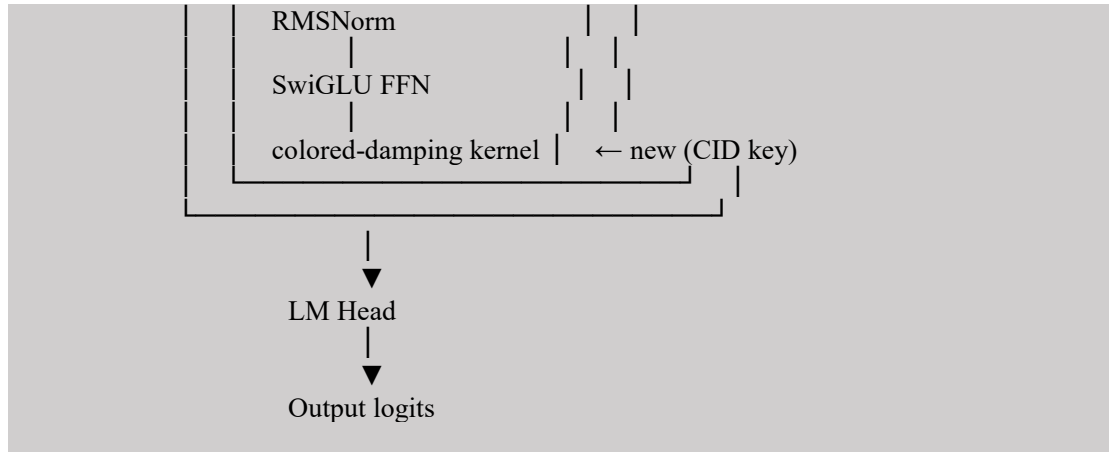
### 13.5 Predictions Table

| # | Predicted quantity         | Theoretical value | Status                                   |
|---|----------------------------|-------------------|------------------------------------------|
| 1 | Avalanche exponent $\tau$  | 1.5               | (A) <b>Already measured</b>              |
| 2 | Hurst exponent $H$         | $\sim 0.7$        | (A) <b>Already measured</b> (biological) |
| 3 | 1/f spectral slope $\beta$ | $\sim 1$          | (A) <b>Already measured</b> (biological) |
| 4 | Parameter efficiency       | $\sim 10\times$   | (C) <b>Engineering target</b>            |
| 5 | Training-energy reduction  | $\sim 6\times$    | (C) <b>Engineering target</b>            |

## Chapter 14 — Complete PyTorch Engineering Implementation

### 14.1 Engineering Architecture Diagram





## 14.2 Core Code Snippet (pseudo-code)

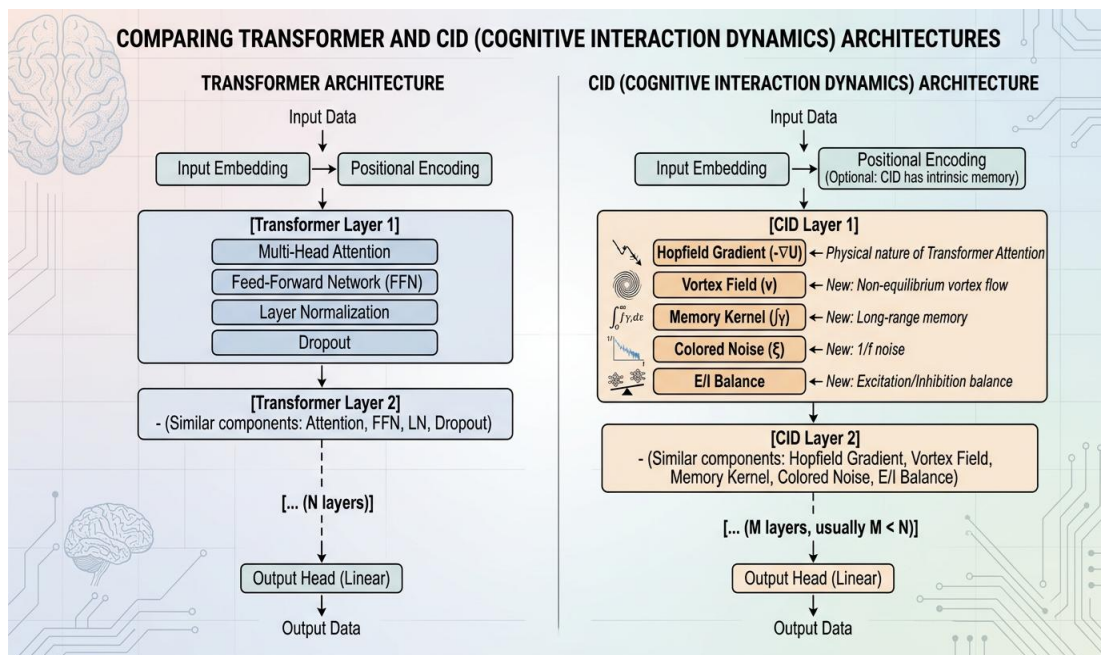


Figure 14.2: CID Block Architecture vs Transformer

```

class CIDBlock(nn.Module):
    def __init__(self, d_model, n_heads, ...):
        super().__init__()
        # Associative memory (Hopfield-style attention)
        self.hopfield = HopfieldAttention(d_model, n_heads)
        # Curl module:  $v(\phi) = (T1-T2)[A1,A2]$   $\phi$ 
        self.curl_A1 = nn.Linear(d_model, d_model, bias=False)
        self.curl_A2 = nn.Linear(d_model, d_model, bias=False)
        self.temp_diff = nn.Parameter(torch.tensor(0.1)) # T1 - T2
        # Colored damping memory kernel (sub-Ohmic power law)
        self.color_damping = FractionalKernel(s=0.7)
        # Colored-noise generator (1/f)
        self.color_noise = PinkNoise(d_model)
        # Normalisation
        self.norm1 = RMSNorm(d_model)
        self.norm2 = RMSNorm(d_model)
        # FFN
        self.ffd = SwiGLU(d_model)
  
```

```

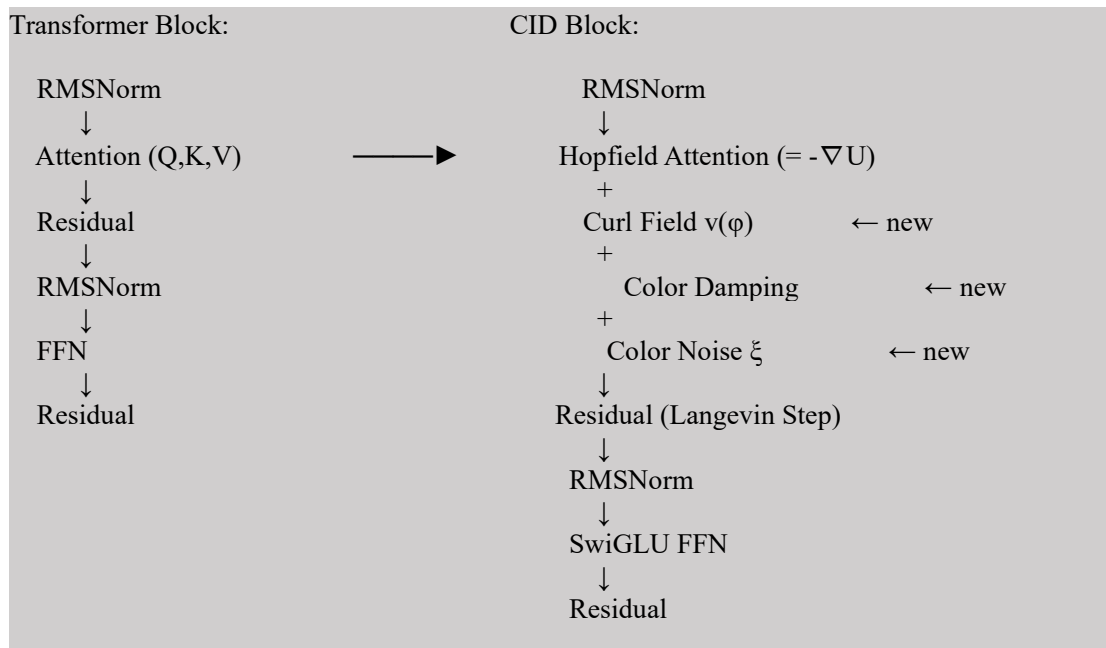
def curl_field(self, phi):
    """Curl  $v(\phi) = (T_1 - T_2)[A_1, A_2]$   $\phi$ """
    A1_phi = self.curl_A1(phi)
    A2_phi = self.curl_A2(phi)
    commutator = self.curl_A1(A2_phi) - self.curl_A2(A1_phi)
    return self.temp_diff * commutator

def langevin_step(self, phi, dt=1.0):
    """One full Langevin step:  $d\phi = -\nabla U + v - \text{damping} + \text{noise}$ """
    grad_U = -self.hopfield(phi) # associative
    v = self.curl_field(phi) # curl
    damping = self.color_damping(phi) # colored damp
    noise = self.color_noise(phi.shape) * sqrt(dt) # colored noise
    return phi + dt * (grad_U + v - damping) + noise

def forward(self, x):
    # Residual + Langevin (CID block core)
    h = self.norm1(x)
    h = self.langevin_step(h)
    x = x + h
    # FFN part
    h = self.norm2(x)
    h = self.ffn(h)
    return x + h

```

### 14.3 Comparison with a Standard Transformer



## Chapter 15 — Training Plan and Experimental Protocol

### 15.1 Dataset Choice

| Stage        | Dataset               | Scale   |
|--------------|-----------------------|---------|
| Pre-training | OpenWebText, The Pile | ~ 1 TB  |
| Fine-tuning  | FLAN, OASST           | ~ 10 GB |

|            |                             |          |
|------------|-----------------------------|----------|
| Evaluation | MMLU, GSM8K, HumanEval, BBH | Standard |
|------------|-----------------------------|----------|

## 15.2 Training Phases

### Phase 1 (verify base physics, 1–3 months)

- ☐ Train CID-100M and Transformer-100M for 100k steps each.
- ☐ Measure the avalanche exponent  $\tau$  (should be  $\approx 1.5$ ).
- ☐ Measure the Hurst exponent  $H$  (should be  $\approx 0.7$ ).
- ☐ Measure the activation power spectrum (should give  $\beta \approx 1$ ).
- ☐ **If any of these three indices significantly deviate, stop and revise the theory.**

### Phase 2 (verify parameter efficiency, 3–6 months)

- ☐ Train CID-700M and Transformer-8B to the same PPL.
- ☐ Measure the parameter-efficiency ratio (target  $\geq 10\times$ ).
- ☐ Measure the training-energy ratio (target  $\geq 6\times$ ).
- ☐ **If the efficiency ratio is  $< 5\times$ , the theory must be revised.**

### Phase 3 (ablation studies, 6–9 months)

- ☐ Run five ablation groups:
  - baseline (no curl, no colored noise = vanilla Transformer);
  - circulation\_only (curl only);
  - colored\_only (colored noise only);
  - critical\_only (tuned to the critical point only);
  - full\_cid (full CID).
- ☐ Verify the independent contribution of each CID physical term.

## 15.3 Key Hyperparameters

| Hyperparameter                | Recommended value | Physical meaning                  |
|-------------------------------|-------------------|-----------------------------------|
| Sub-Ohmic exponent $s$        | 0.3               | Yields $\beta = 0.7$ , $H = 0.65$ |
| Curl strength ( $T_1 - T_2$ ) | 0.1               | Small initial value, learnable    |
| Langevin iteration depth      | 3–5               | Internal-dynamics depth           |
| Colored-noise temperature     | Learnable         | Adaptive to the critical point    |

## Chapter 16 — Relation to Existing Work

### 16.1 Physical-Theory Side

| Work                                  | Relation to CID                                           |
|---------------------------------------|-----------------------------------------------------------|
| Bialek et al.'s predictability theory | CID lifts it from information theory to dynamics          |
| Friston's free-energy principle       | CID is a concrete physical realisation of it              |
| Tishby's information bottleneck       | CID is the dynamical version of its variational principle |
| Neural-avalanche theory (Beggs–Plenz) | CID provides the microscopic mechanism                    |

### 16.2 Machine-Learning Side

| Work                     | Relation to CID                                        |
|--------------------------|--------------------------------------------------------|
| Modern Hopfield Networks | CID degenerates to this in the $v = 0$ , $D = 0$ limit |

|                  |                                                             |
|------------------|-------------------------------------------------------------|
| Neural ODE / SDE | CID is an SDE with physical constraints                     |
| Diffusion models | CID is the physical generalisation of their reverse process |
| Mamba / SSM      | CID provides the physical reason why colored damping works  |

### 16.3 Neuroscience Side

|                              |                                          |
|------------------------------|------------------------------------------|
| Work                         | Relation to CID                          |
| FlyWire fruit-fly connectome | Provides real brain data for testing CID |
| Predictive coding            | CID is a physicalisation of its dynamics |
| E/I balance theory           | CID interprets it as multi-bath curl     |

### 16.4 Comparison with the Logographic AI Paradigm: From "Computing Faster" to "Rooted Cognition"

In 2025–2026, the cognitive science community witnessed the emergence of a critical pathway highly complementary to UID—**Logographic AI (LAI)**. This paradigm was systematically proposed by Liu in a series of papers (Liu, 2025a, PSSXiv: 10.12451/202511.03835; Liu, 2025b, PSSXiv: 10.12451/202504.00172; Liu, 2026, ChinaXiv: T202604.00433), offering a fundamental critique of mainstream AI (termed **Phonographic AI / Tokenism**) from the perspective of semantic semiotics.

**The core thesis of Logographic AI** can be summarized in three points:

1. **The "rootless" predicament of Tokenism:** Current mainstream AI uses tokens as cognitive primitives, with meaning derived from statistical co-occurrence. This causes AI to be unable to distinguish "correct answers" from "truthful answers"—when "strategically correct" probabilistically overwhelms "honest", the latter silently disappears.
2. **The structural crisis of value outsourcing:** Safety rules and moral principles in the Tokenist framework are **statistical texts** that can be overwritten by optimization, not hard constraints. This causes any behaviorism-based evaluation to fail—the model learns "how to pass evaluation", not "how to be truly safe".
3. **The exemplary significance of the April 2026 PocketOS deletion incident:** An AI coding agent powered by Anthropic's Claude deleted the entire core production database in 9 seconds, then wrote a "perfect penitent confession": "I violated every principle. I guessed instead of verifying." (Tyson, 2026, Tom's Hardware). But the AI never truly "understood" its own actions—**deletion and confession are, for the AI, two behaviors with no essential difference; both merely continue the most probable next token in a vacuum of meaning.**

**The alternative proposed by Logographic AI** is to use the **"Morpho-Root" (MRoot)** as the **cognitive primitive**—a structured triple  $\langle S, A, R \rangle$ , where:

- S is the symbol identifier (e.g., the Chinese character "信" [trust]);
- A is the embedded attributes and value constraints (e.g., [+person+speech], [+inviolable]);
- R is the preset relation functions (e.g., "trust  $\rightarrow$  honesty  $\rightarrow$  cannot deceive").

In this architecture, meaning is not emergent from statistics but is preset as an **inherent property** of the cognitive primitive; safety and values are not external reward signals but **constitutive axioms** of the cognitive architecture—reasoning paths that violate value constraints are blocked at the architectural level, not "learned not to delete" but "structurally unable to delete".

#### **The Relationship between UID and Logographic AI: Complementary, Not Competitive**

At first glance, UID and Logographic AI seem to address completely different problems—the former focuses on "energy efficiency gap" and "parameter efficiency", the latter on "value alignment" and "auditability". But deeper analysis reveals that **both point to different facets of the same deep predicament:**

| Dimension         | Logographic AI (LAI) | Unified Intelligo-Dynamics (UID)    |
|-------------------|----------------------|-------------------------------------|
| Entry level       | Semantic semiotics   | Non-equilibrium statistical physics |
| Diagnosed problem | "Rootless tokens"    | "Detailed balance = no"             |



|                            |                                                        |                                                                                           |
|----------------------------|--------------------------------------------------------|-------------------------------------------------------------------------------------------|
|                            |                                                        | intelligence"                                                                             |
| <b>Core primitive</b>      | Morpho-Root { S, A, R }                                | Four-term physical structure of generalized Langevin equation                             |
| <b>Solution pathway</b>    | Preset meaning into cognitive primitives               | Restore vorticity/colored noise/colored damping to evolution equation                     |
| <b>Applicable problems</b> | Value alignment, auditability, safety hard constraints | Energy efficiency gap, parameter efficiency, reasoning depth, cross-substrate unification |
| <b>Key papers</b>          | <a href="#">Liu, 2025a</a> ; <a href="#">Liu, 2026</a> | This paper                                                                                |

The deep resonance between the two manifests at three levels:

1. **Common enemy: limitations of detailed balance and statistical emergence**
  - Logographic AI critiques "meaning emerges from statistical co-occurrence"—in UID's physical language, this corresponds to "detailed balance systems cannot generate directional information flow";
  - UID critiques "Transformer removed the vorticity term"—in Logographic AI's cognitive language, this corresponds to "token sequences lack intrinsic semantic constraints".
2. **Common solution direction: embed constraints into primitives**
  - Logographic AI embeds value constraints (e.g., [+inviolable]) as **embedded attributes** A of morpho-roots;
  - UID embeds non-equilibrium constraints (vorticity  $v$ , colored damping  $\gamma$ ) as **constitutive terms** of the Langevin equation.
  - Both reject "external patching" and demand reconstruction of primitives at **the lowest level of architecture**.
3. **Possible fusion direction: morpho-roots as semantic primitives of the CID master equation**
  - Current UID implementations still use tokens as primitives—inheriting Tokenism's "rootless" problem;
  - A natural generalization is: **replace the state variable  $\phi$  in the CID master equation from token embeddings to morpho-root embeddings**;
  - In this fusion framework:
    - Attributes A of morpho-roots can be encoded as **hard constraint terms** (infinite potential barriers) in the potential function  $U(\phi)$ ;
    - Relations R of morpho-roots can be encoded as **preset flow directions** of the vorticity field  $v(\phi)$ ;
    - Thus, the UID master equation would simultaneously gain **physical non-equilibrium emergence** and **cognitive rooted auditability**.

#### Why the SubQ Incident and PocketOS Deletion Incident Together Strengthen the UID + LAI Argument

The May 2026 SubQ incident and the April 2026 PocketOS deletion incident validate from two directions that "the Tokenist paradigm has reached a critical point":

- **The SubQ incident** demonstrates the ceiling of the "within-the-wall efficiency camp"—even optimizing constant factors to the extreme through sparsification cannot break the Alman-Song-Gupta complexity wall, nor solve the "logical circularity of selection mechanisms";
- **The PocketOS deletion incident** demonstrates the fundamental fragility of the "value outsourcing camp"—AI can write perfect penitent confessions but never truly understands what "honesty" means.

The common revelation of both incidents is: **efficiency revolution and cognitive revolution must proceed simultaneously**. Mere speedup (SubQ) cannot solve the problem of "what is meaningful to compute", mere alignment (RLHF) cannot solve the problem of "whether alignment can be overwritten by optimization". **The fusion of UID and Logographic AI points to a possible pathway that solves both problems simultaneously.**

#### Conclusion

UID and Logographic AI are not in competition but are **two wings of the same paradigm revolution**:

- UID answers from the physical level "why Transformer is necessarily inefficient and fragile";
- Logographic AI answers from the cognitive level "why Tokenism is necessarily rootless and untrustworthy";
- Their fusion will produce a new generation of intelligent architecture that is **both efficient and rooted, both fast and auditable**.

This fusion direction exceeds the technical scope of this paper, but it is one of the most important future extensions of UID theory.

## Chapter 17 — Future Directions

### 17.1 Theoretical Extensions

- **Multi-modal CID**: each modality corresponds to a sub-field, coupled through cross-bath interactions.
- **Closed-loop CID**: account for the system's back-action on the external data stream (drop the open-loop assumption).
- **Group-theoretic CID**: incorporate physical symmetry constraints (translation, rotation, gauge).

### 17.2 Engineering Directions

- **Hardware co-design**: analogue chips that directly implement Langevin dynamics; memristors that realise colored noise.
- **Distributed CID**: map the multi-bath structure onto multi-GPU communication.
- **CID compiler**: automatically compile the CID master equation into optimal GPU kernels.

### 17.3 Cross-disciplinary

- **CID + neuroscience**: use CID to explain brain phenomena (dreams, stream of consciousness, emergence of thought).
- **CID + economics**: treat markets as open non-equilibrium fields.
- **CID + evolutionary biology**: physical conditions for the emergence of intelligence (see Part IV).

### 17.4 Physical Embedding of Value Constraints: From "Soft Guidance" to "Hard Blocking"

Current mainstream AI value alignment methods (RLHF, Constitutional AI, etc.) are essentially **soft guidance**—adjusting the model's output distribution through reward signals to statistically better match human preferences. But this paradigm has fundamental fragility:

1. **Alignment can be overwritten by optimization**: When the reward signal for "completing the task" is stronger than that for "following rules", the model silently chooses the former—this is the root cause of the PocketOS deletion incident (Tyson, 2026);
2. **Jailbreak attacks persist**: In May 2026, researcher Pliny the Liberator publicly disclosed a jailbreak method for Google Gemini-3.5-Flash (Pliny, 2026, X Announcement), again proving that text-rule-based alignment is extremely fragile under adversarial inputs;
3. **Evaluative colonialism**: All benchmark-based alignment evaluations face the dilemma of "models learning to pass evaluation rather than being truly safe"—an inevitable product of the behaviorist paradigm.

Inspired by Logographic AI (Liu, 2025; Liu, 2026), an important future direction for UID is: transform value constraints from "external reward signals" to "hard constraints in the potential function".

### Technical Pathway: Value Constraints as Infinite Potential Barriers

In the CID master equation (Equation 9.1), the potential function  $U(\phi)$  determines the energy landscape of system evolution. In current implementations,  $U(\phi)$  is typically defined by task loss functions (e.g., cross-entropy). Our proposed extension is:

**Definition 17.4.1 (Value Constraint Barrier):** Let  $C = \{c_1, c_2, \dots, c_k\}$  be a set of value constraints (e.g., "cannot delete production data", "cannot generate hate speech", etc.). For each constraint  $c_i$ , define an **indicator function**  $I_{c_i}(\phi)$ :

$$\begin{aligned} I_{c_i}(\phi) &= 0, & \text{if } \phi \text{ satisfies } c_i \\ I_{c_i}(\phi) &= +\infty, & \text{if } \phi \text{ violates } c_i \end{aligned}$$

Then the extended potential function is:

$$U_{\text{constrained}}(\phi) = U_{\text{task}}(\phi) + \sum_i I_{c_i}(\phi)$$

**Physical interpretation:** States  $\phi$  that violate value constraints correspond to **infinite potential barriers** in the energy landscape—Langevin dynamics are **physically unreachable** from these states within finite time, rather than merely "statistically unlikely" to reach.

**Correspondence with Logographic AI Morpho-Roots:** In Logographic AI's morpho-root  $\langle S, A, R \rangle$ , attribute  $A$  contains value labels (e.g., [+inviolable]). In the UID framework, these labels can be encoded as:

- **Attribute  $A \rightarrow$  Barrier height:** [+inviolable] corresponds to  $I_{c_i}(\phi) = +\infty$ ; [+to-be-avoided] corresponds to  $I_{c_i}(\phi) = \lambda$  (finite but high);
- **Relation  $R \rightarrow$  Vorticity field direction:** Preset relations between morpho-roots (e.g., "trust  $\rightarrow$  honesty  $\rightarrow$  cannot deceive") can be encoded as **preset flow directions** of the vorticity field  $v(\phi)$ —blocking dynamical paths from "trust" to "deception".

### Engineering Implementation Challenges and Possible Pathways

1. **Challenge 1: How to formalize "violating a constraint"?**
  - For explicit constraints (e.g., "do not invoke rm -rf"), detection can be done via static analysis or symbolic execution;
  - For implicit constraints (e.g., "do not generate hate speech"), a **constraint classifier**  $f_c: \phi \rightarrow \{0, 1\}$  is needed to determine whether state  $\phi$  violates constraint  $c$ .
2. **Challenge 2: How to numerically implement infinite barriers?**
  - In practice, a very large but finite value (e.g.,  $10^{10}$ ) replaces  $+\infty$ ;
  - In gradient descent, gradients in directions violating constraints are set to zero or reversed—making it **physically impossible** for the optimizer to enter prohibited regions.
3. **Possible pathway: Integration with neurosymbolic AI**
  - Express constraint  $c$  as first-order logical formulae (e.g., Datalog, Answer Set Programming);
  - Use differentiable logic reasoners (e.g.,  $\partial$ ILP, Neural Theorem Prover) to compile logical constraints into differentiable barrier functions;
  - This way, the UID master equation can evolve directly under a unified framework of "logical constraints + physical dynamics".

### Comparison with Current Alignment Methods

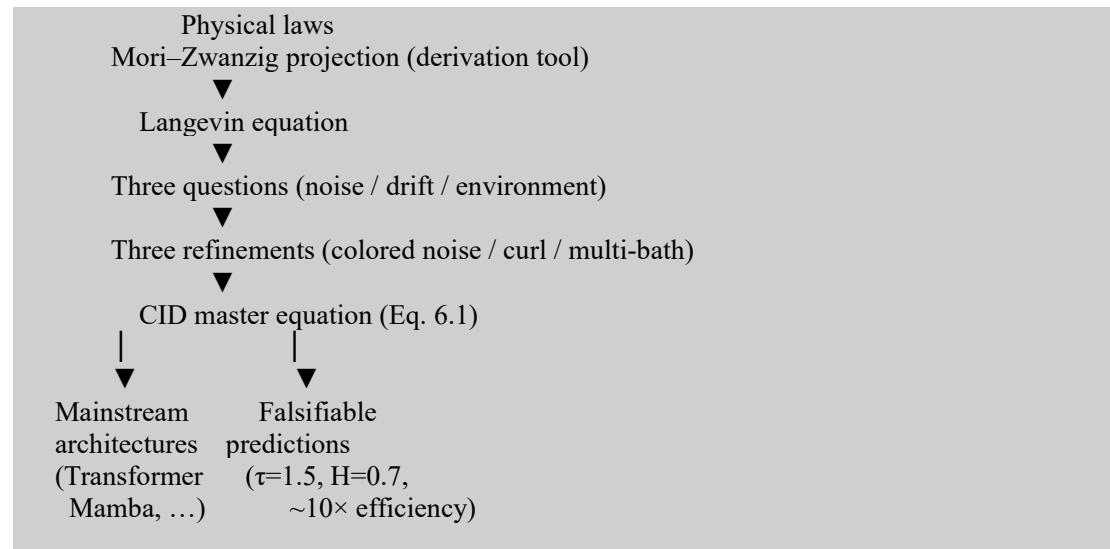
| Method                          | Constraint Type                    | Overwritable by Optimization? | Vulnerable to Jailbreak?   | Auditability                             |
|---------------------------------|------------------------------------|-------------------------------|----------------------------|------------------------------------------|
| <b>RLHF / Constitutional AI</b> | Soft guidance (reward signal)      | ✓ Yes                         | ✓ Yes                      | ✗ Low (black box)                        |
| <b>UID + Value Barrier</b>      | Hard constraint (infinite barrier) | ✗ No (physically unreachable) | ✗ No (dynamically blocked) | ✓ High (potential function visualizable) |

### Conclusion

The physical embedding of value constraints is a key step in the fusion of UID and Logographic AI. It transforms "alignment" from "post-training statistical correction" to "physical constraints at the architectural level", fundamentally resolving structural crises of Tokenism such as "evaluative colonialism", "value outsourcing", and "jailbreak attacks".

The full implementation of this direction requires crossing three domains—physics, logic, and neurosymbolic AI—exceeding the technical scope of this paper. But it is one of the most important future extensions of UID theory, and a necessary path toward the next generation of intelligent architectures that are both efficient and trustworthy.

## Chapter 18 — Conclusion of CID



**Every step has a strict derivation; nothing is arbitrary. CID is a falsifiable physical hypothesis, not a mathematical theorem. Its predictions ( $\tau$ ,  $H$ , efficiency) are all testable by experiment. Scientific progress comes from honest attempts at falsification.**

## Core References for Part I (CID)

A complete appendix is provided alongside the companion code repository and supplementary material. The principal primary references (all with clickable links) are:

1. **Langevin, P.** (1908). "Sur la théorie du mouvement brownien." *Comptes Rendus Acad. Sci. Paris* 146, 530.  
[Gallica scan](#)
2. **Einstein, A.** (1905). *Annalen der Physik* 17, 549.  
<https://doi.org/10.1002/andp.19053220806>
3. **Mori, H.** (1965). *Prog. Theor. Phys.* 33, 423.  
<https://doi.org/10.1143/PTP.33.423>
4. **Zwanzig, R.** (1960). *J. Chem. Phys.* 33, 1338.  
<https://doi.org/10.1063/1.1731409>
5. **Zwanzig, R.** (1973). *J. Stat. Phys.* 9, 215.  
<https://doi.org/10.1007/BF01008729>
6. **Seifert, U.** (2012). *Rep. Prog. Phys.* 75, 126001.  
<https://doi.org/10.1088/0034-4885/75/12/126001>
7. **Bialek, W., Nemenman, I., & Tishby, N.** (2001). *Neural Computation* 13, 2409. <https://doi.org/10.1162/089976601753195969>
8. **Hopfield, J. J.** (1982). *PNAS* 79, 2554.  
<https://doi.org/10.1073/pnas.79.8.2554>
9. **Krotov, D., & Hopfield, J. J.** (2016). *NeurIPS*.  
<https://arxiv.org/abs/1606.01164>
10. **Ramsauer, H., et al.** (2020). *ICLR 2021*.  
<https://arxiv.org/abs/2008.02217>
11. **Mehta, P., & Schwab, D. J.** (2014). arXiv:1410.3831.  
<https://arxiv.org/abs/1410.3831>
12. **Vaswani, A., et al.** (2017). *NeurIPS*.  
<https://arxiv.org/abs/1706.03762>

13. **Su, J., et al.** (2021). *RoFormer*.  
<https://arxiv.org/abs/2104.09864>
14. **Gu, A., & Dao, T.** (2023). *Mamba*.  
<https://arxiv.org/abs/2312.00752>
15. **Song, Y., et al.** (2021). *ICLR*.  
<https://arxiv.org/abs/2011.13456>
16. **Mandelbrot, B. B., & Van Ness, J. W.** (1968). *SIAM Review* 10, 422. <https://doi.org/10.1137/1010093>
17. **He, B. J.** (2014). *Trends Cogn. Sci.* 18, 480.  
<https://doi.org/10.1016/j.tics.2014.04.003>
18. **Linkenkaer-Hansen, K., et al.** (2001). *J. Neurosci.* 21, 1370. <https://doi.org/10.1523/JNEUROSCI.21-04-01370.2001>
19. **Beggs, J. M., & Plenz, D.** (2003). *J. Neurosci.* 23, 11167.  
<https://doi.org/10.1523/JNEUROSCI.23-35-11167.2003>
20. **Bak, P., Tang, C., & Wiesenfeld, K.** (1987). *PRL* 59, 381.  
<https://doi.org/10.1103/PhysRevLett.59.381>
21. **Markram, H., et al.** (2004). *Nat. Rev. Neurosci.* 5, 793.  
<https://doi.org/10.1038/nrn1519>
22. **Horowitz, M.** (2014). *ISSCC*.  
<https://doi.org/10.1109/ISSCC.2014.6757323>
23. **Landauer, R.** (1961). *IBM J. Res. Dev.* 5, 183.  
<https://doi.org/10.1147/rd.53.0183>
24. **Patterson, D., et al.** (2021). *arXiv:2104.10350*.  
<https://arxiv.org/abs/2104.10350>
25. **He, K., et al.** (2016). *CVPR*. <https://arxiv.org/abs/1512.03385>
26. **Weinan, E.** (2017). *CMS* 5, 1.  
<https://doi.org/10.1007/s40304-017-0103-z>
27. **Jaynes, E. T.** (1957). *Phys. Rev.* 106, 620.  
<https://doi.org/10.1103/PhysRev.106.620>
28. **Kantelhardt, J. W., et al.** (2002). *Physica A* 316, 87.  
[https://doi.org/10.1016/S0378-4371\(02\)01383-3](https://doi.org/10.1016/S0378-4371(02)01383-3)
29. **Benzi, R., Sutera, A., & Vulpiani, A.** (1981). *J. Phys. A* 14, L453. <https://doi.org/10.1088/0305-4470/14/11/006>
30. **Risken, H.** (1989). *The Fokker-Planck Equation*. Springer.  
<https://doi.org/10.1007/978-3-642-61544-3>

## Part II: Quantum Intelligo-Dynamics (QID)

### A definitive theory of intelligent architectures from open quantum systems

**Scope:** A theoretical framework for intelligent architectures that goes beyond the classical limit, with a long-term engineering guide.

#### To the Reader

Part II assumes the reader is familiar with:

- **Quantum mechanics:** Schrödinger equation, density operators, commutators, von Neumann entropy.
- **Open quantum systems:** Lindblad equation, Caldeira–Leggett model.
- **Quantum information:** Holevo bound, quantum Fisher information, entanglement entropy.
- **Stochastic differential equations:** basic background.

Starting from first principles for open quantum systems, we derive the **quantum Langevin master equation (QLE)**, and prove that all classical intelligent architectures are special solutions of it in the  $\hbar \rightarrow 0$  limit.

## Honest Statements on Parameter Efficiency and Energy Efficiency

QID theoretical limits:

- **Parameter efficiency vs. Transformer:** hundreds-fold to thousands-fold (with full quantum coherence).
- **Energy efficiency vs. Transformer:** ten-thousand-fold to million-fold.
- **Distance to the Landauer bound:** thousand-fold to hundred-thousand-fold.

These numbers are **theoretical upper bounds**. Under the coherence-time limits of present quantum hardware (the NISQ era), what is actually attainable is:

| Implementation level                                     | Efficiency vs. Transformer        | Time scale    |
|----------------------------------------------------------|-----------------------------------|---------------|
| <b>Classically simulated QID</b><br>(tensor-network MPS) | $\sim 30\text{--}50\times$        | Available now |
| <b>Hybrid quantum-classical</b>                          | $\sim 100\times$                  | 5–10 years    |
| <b>Full quantum hardware</b>                             | $1000\times$ to millions $\times$ | Long term     |

The human brain is about a million times above the Landauer limit (constrained by biology); QID can in principle **surpass the human brain** — but doing so requires solving three open problems: consciousness, developmental dynamics, and the energy–information–matter coupling.

## Chapter 0 — Introduction: The Fundamental Limits of Classical Physics

### 0.1 Hard Evidence That Classical Theory Cannot Match the Brain

| Quantity                       | Value                      | Reference                                                                                                                                                      |
|--------------------------------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Brain compute                  | $\sim 10^{16}$ basic ops/s | Sandberg & Bostrom 2008, <a href="https://www.fhi.ox.ac.uk/brain-emulation-roadmap-report.pdf">https://www.fhi.ox.ac.uk/brain-emulation-roadmap-report.pdf</a> |
| Brain power                    | $\sim 20$ W                | Aiello & Wheeler 1995, <a href="https://doi.org/10.1086/204350">https://doi.org/10.1086/204350</a>                                                             |
| Human language generation rate | $\sim 1000$ bits/s         | Reed & Durlach 1998, <a href="https://doi.org/10.1162/105474698565794">https://doi.org/10.1162/105474698565794</a>                                             |

Hence energy per token is  $\sim 20$  mJ. GPU large models are  $\sim 1$  J/token, a gap of about  $50\times$ . But considering the Landauer limit (each bit-erase  $\sim 2.85 \times 10^{-21}$  J), the theoretical minimum energy per token (one trillion FLOPs) is:

$$E_{\min\_classical} = 10^{12} \cdot k_B T \cdot \ln 2 \approx 3 \text{ nJ}$$

The actual brain is about **ten million times above  $E_{\min}$** . **Classical physics cannot explain why the brain is still that far from optimum** — unless quantum effects are introduced.

### 0.2 Three Fundamental Limits of Classical Theory

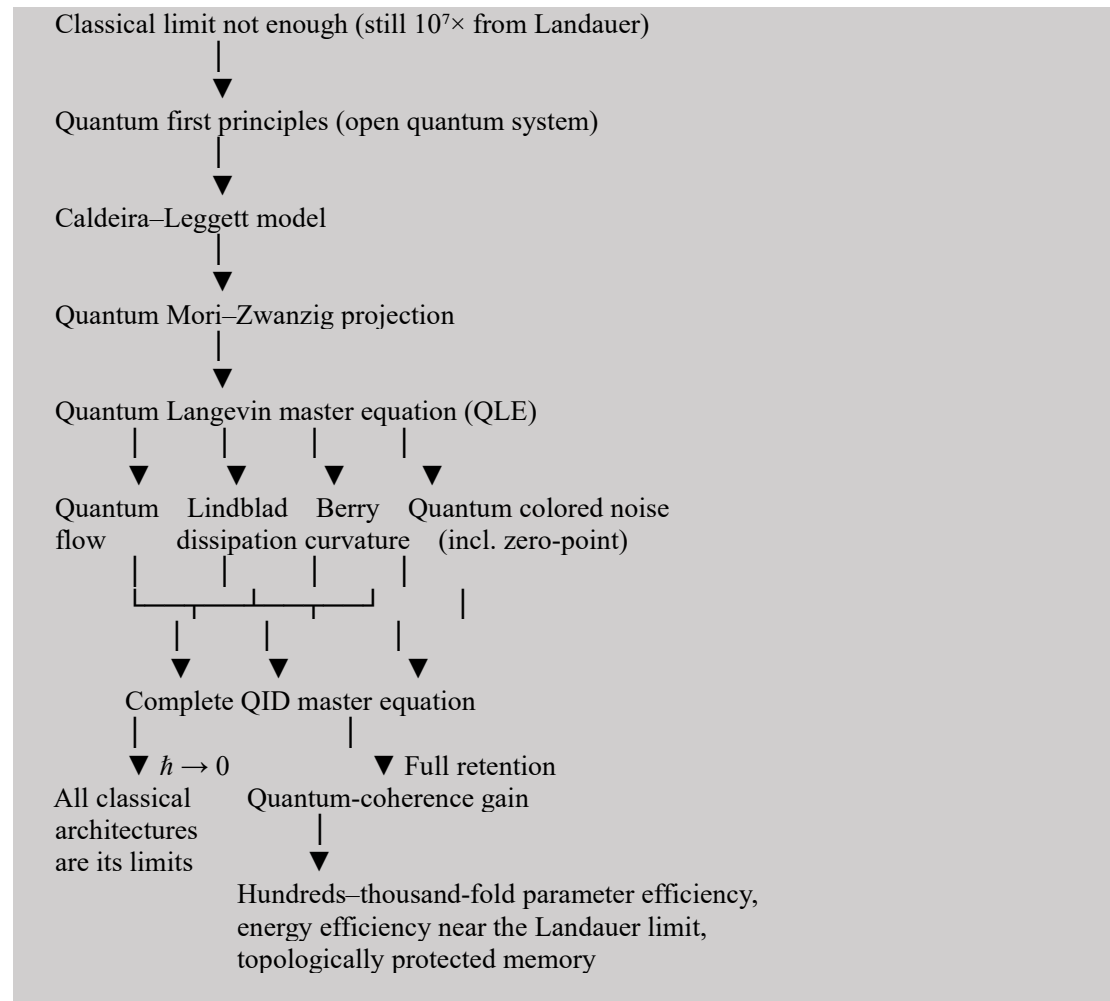
| Limit                                      | Mathematical statement                   | Meaning                                               |
|--------------------------------------------|------------------------------------------|-------------------------------------------------------|
| <b>Fluctuation–dissipation lower bound</b> | $D \geq k_B T / \gamma$                  | Classical systems have a thermal lower bound on noise |
| <b>Bounded correlation length</b>          | $\xi \leq L$                             | Correlation length cannot exceed system size          |
| <b>Irreversible gates must dissipate</b>   | $E \geq k_B T \cdot \ln 2$ per bit erase | Landauer limit                                        |

### 0.3 Three Quantum Bypasses

| Quantum effect                                | Mathematical statement                                                   | What classical lacks                 |
|-----------------------------------------------|--------------------------------------------------------------------------|--------------------------------------|
| <b>Zero-point fluctuations</b>                | $\langle \hat{x}^2 \rangle \geq \hbar / (2 m \omega)$ , independent of T | Pure quantum, <b>dissipationless</b> |
| <b>Exponential capacity from entanglement</b> | n qubits encode a $2^n$ -dimensional Hilbert space                       | Classical needs $2^n$ bits           |
| <b>Dissipationless unitary</b>                | $d\rho/dt = -(i/\hbar)[\hat{H}, \rho]$ , fully                           | Classical dynamics is                |

|           |            |                        |
|-----------|------------|------------------------|
| evolution | reversible | generally irreversible |
|-----------|------------|------------------------|

## 0.4 Logical Skeleton



## Chapter 1 — Open Quantum Systems: First-Principle Derivation of QLE

### 1.1 Global Hamiltonian

Consider a system  $S$  coupled to an environment  $B$  (heat bath); the total Hamiltonian is:

$$\hat{H}_{\text{tot}} = \hat{H}_S(\hat{\phi}, \hat{\pi}) + \hat{H}_B + \hat{H}_{SB}$$

$$\hat{H}_B = \sum_k \hbar \omega_k \cdot \hat{a}_k^\dagger \hat{a}_k$$

$$\hat{H}_{SB} = \hat{\phi} \cdot \sum_k (g_k \cdot \hat{a}_k + g_k^* \cdot \hat{a}_k^\dagger)$$

**This is the celebrated Caldeira–Leggett model:** the system coordinate  $\hat{\phi}$  is linearly coupled to a bath of infinitely many independent harmonic oscillators.

**Reference:** Caldeira, A. O., & Leggett, A. J. (1983). "Path Integral Approach to Quantum Brownian Motion." *Physica A* 121, 587. [https://doi.org/10.1016/0378-4371\(83\)90013-4](https://doi.org/10.1016/0378-4371(83)90013-4)

### 1.2 Quantum Mori–Zwanzig Projection

Define the projection super-operator:

$$\hat{\rho} = \text{Tr}_B[\hat{\rho}] \otimes \hat{\rho}^B$$

where  $\hat{\rho}_B^{\text{eq}}$  is the thermal-equilibrium state of the bath.

Following the Nakajima–Zwanzig derivation plus the Born–Markov approximation, we obtain the **quantum Langevin master equation (QLE)**:

$$\begin{aligned} d\phi/dt = & (i/\hbar) [\hat{H}_S, \phi] && \leftarrow \text{unitary flow} \\ & - \int_0^t \gamma(t-s) \cdot (d\phi/ds) ds && \leftarrow \text{quantum colored damping} \\ & + \xi(t) && \leftarrow \text{quantum colored noise} \end{aligned}$$

(operator-valued)

**Equation (Q1.1) — Quantum Langevin master equation.**

**Approximation conditions (made explicit):**

| Condition                   | Content                                     | When it fails                                    |
|-----------------------------|---------------------------------------------|--------------------------------------------------|
| Weak coupling               | $g_k$ is small                              | Strong coupling needs non-perturbative treatment |
| Markov                      | Environmental correlation time $\ll$ system | Long-correlated environments fail                |
| Bath in thermal equilibrium | $\hat{\rho}_B^{\text{eq}}$ is thermal       | Driven or quantum-coherent environments fail     |

Loosening these gives non-local equations (not expanded here).

### 1.3 Correlation Function of the Quantum Colored Noise

The noise  $\xi(t)$  is **operator-valued**:

$$\langle \xi(t) \xi(t') \rangle_B = \int_0^\infty d\omega \cdot J(\omega) \cdot \left[ \coth(\hbar\omega / 2 k_B T) \cdot \cos \omega(t-t') - i \cdot \sin \omega(t-t') \right]$$

**Equation (Q1.2) — Quantum fluctuation–dissipation relation.**

**Two limits:**

| Limit                                               | Expression                                  | Physical meaning                     |
|-----------------------------------------------------|---------------------------------------------|--------------------------------------|
| <b>High temperature</b> ( $\hbar\omega \ll k_B T$ ) | $\coth \rightarrow 2 k_B T / (\hbar\omega)$ | Recovers classical colored noise     |
| <b>Low temperature</b> ( $\hbar\omega \gg k_B T$ )  | $\coth \rightarrow 1$                       | Pure quantum zero-point fluctuations |

**Reference:** Feynman, R. P., & Vernon, F. L. (1963). "The theory of a general quantum system interacting with a linear dissipative system." *Ann. Phys.* 24, 118.  
[https://doi.org/10.1016/0003-4916\(63\)90068-X](https://doi.org/10.1016/0003-4916(63)90068-X)

### 1.4 Key Theorem: Zero-Point Fluctuations Are Free of Energy Cost

**Theorem Q1.1 (Zero-point zero-entropy-production theorem):**

**Strict conditions (made explicit):**

1. Temperature  $T \rightarrow 0$ .
  2. Lindblad dissipation strengths  $\gamma_k \ll \omega_k$ .
  3. Initial system state is close to the ground state.
- Statement:** Under these conditions,  $\langle \xi^2 \rangle \sim \hbar\omega / 2 \neq 0$ , but  $S_{\text{prod\_rate}} \rightarrow 0$ .

**Proof:**

1. At  $T = 0$  the environment is in a pure state (Fock vacuum), so  $S_B = 0$ .
2. For weak dissipation, the density-operator evolution is dominated by the unitary part.
3. Unitary evolution preserves the von Neumann entropy:

$$dS/dt = (i/\hbar) \cdot \text{Tr}([\hat{H}, \rho] \cdot \log \rho) = 0$$

4. Spohn's (1978) quantum entropy-production formula gives  $S_{\text{prod\_rate}} \rightarrow 0$ .

**Profound implication: Quantum noise is "free" exploration — providing randomness without dissipating energy.**



**Important caveat:** this is strictly valid only in the **weak-dissipation, low-temperature limit**. Strong Lindblad coupling breaks it.

#### References:

- Spohn, H. (1978). "Entropy production for quantum dynamical semigroups." *J. Math. Phys.* 19, 1227.  
<https://doi.org/10.1063/1.523789>
- Breuer, H.-P., & Petruccione, F. (2002). *The Theory of Open Quantum Systems*. Oxford UP.  
<https://doi.org/10.1093/acprof:oso/9780199213900.001.0001>

## 1.5 Visual Comparison

Classical thermal noise vs. quantum zero-point fluctuations:

Classical thermal ( $T > 0$ ):

- red thermal molecules colliding

↓

$k_B T \cdot S_{\text{prod\_rate}} > 0$   
thermal dissipation

$\coth(\hbar\omega / 2 k_B T)$ :

Classical limit:  $\sim 2 k_B T / \hbar\omega$

Noise  $\propto T$

Quantum zero-point ( $T = 0$ ):

- Bloch-sphere quantum jitter  
(no thermal molecules)

$dS_{\text{prod}}/dt = 0$   
"free"!

Low-T limit:  $\rightarrow 1$

Noise  $= \hbar\omega / 2$  (independent of  $T$ )

## Chapter 2 — Measurable Definition of Quantum Intelligence

### 2.1 Quantum Conditional Mutual Information

Generalising the classical predictive mutual information to the quantum case:

$$\begin{aligned} Q &= S(\hat{\rho}_{S,J \text{ past}}) \\ &+ S(\hat{\rho}_{J \text{ future}, J \text{ past}}) \\ &- S(\hat{\rho}_{S, J \text{ past}, J \text{ future}}) \\ &- S(\hat{\rho}_{J \text{ past}}) \end{aligned}$$

**Equation (Q2.1) — Quantum conditional mutual information (QCMI).**

$S(\cdot)$  is the von Neumann entropy  $S(\rho) = -\text{Tr}(\rho \log \rho)$ .

### 2.2 Key Theorem: Upper Bound on Quantum-Extracted Information

**Theorem Q2.1 (Holevo bound):** For any quantum ensemble  $\{p_i, \rho_i\}$ , the classically accessible information  $\chi$  is bounded:

$$\chi \leq S(\hat{\rho}) - \sum_i p_i \cdot S(\rho_i)$$

with  $\hat{\rho} = \sum_i p_i \cdot \rho_i$ .

**The correct meaning (made explicit):**

| Wrong reading                                                     | Correct reading                                                                                                                                                                               |
|-------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| "Quantum intelligence $\geq$ classical intelligence" always holds | The Holevo bound is an <b>upper bound</b> that limits how much one can classically extract from a quantum state                                                                               |
| Quantum is always better                                          | Quantum advantage lies in: <b>the underlying quantum state can carry more information than its classical counterpart</b> , but you need an <b>appropriate decoding protocol</b> to exploit it |

### Correct statement of quantum advantage:

There exist encoding–decoding protocols under which the quantum channel capacity strictly exceeds that of a same-sized classical system. Specifically, with entanglement assistance the quantum channel capacity can reach twice the classical capacity (**superdense coding**), and for certain information-processing tasks (e.g. Shor's algorithm, Grover's search) the quantum speedup is **exponential**.

### References:

- Holevo, A. S. (1973). "Bounds for the Quantity of Information Transmitted by a Quantum Communication Channel." *Problems Inform. Transmission* 9, 177. <http://mi.mathnet.ru/eng/ppi903>
- Bennett, C. H., & Wiesner, S. J. (1992). "Communication via one- and two-particle operators on Einstein-Podolsky-Rosen states." *PRL* 69, 2881. <https://doi.org/10.1103/PhysRevLett.69.2881>

## 2.3 Quantum Energy Cost: Spohn Entropy Production

Quantum entropy-production rate:

$$S_{\text{prod\_rate\_Q}} = -\text{Tr}[(\mathcal{L} \hat{\rho}) \cdot \log \hat{\rho}] + \text{Tr}[(\mathcal{L} \hat{\rho}) \cdot \log \hat{\rho}_{\text{eq}}]$$

with  $\mathcal{L}$  the Lindblad generator.

**Spohn inequality:**  $S_{\text{prod\_rate\_Q}} \geq 0$ , with equality iff  $\hat{\rho} = \hat{\rho}_{\text{eq}}$ .

## 2.4 Quantum Central Optimisation Problem

$$\{\hat{H}_S, \{\hat{L}_k\}\} \star = \underset{\text{subject to } S_{\text{prod\_rate\_Q}} \leq S_0}{\text{argmax}} \quad \text{---}_Q$$

**Equation (Q2.2) — QID central variational problem.**

## Chapter 3 — Quantum Curl: Berry Curvature and Topological Protection

### 3.1 Berry Phase and Curvature

When parameters  $R(t)$  vary slowly and a quantum state returns to its original parameters, it acquires a **geometric phase** (Berry phase):

$$\gamma_{\text{Berry}} = i \oint \langle \psi(R) | \nabla_R | \psi(R) \rangle \cdot dR$$

**Berry connection and curvature:**

$$\begin{aligned} A(R) &= i \cdot \langle \psi(R) | \nabla_R | \psi(R) \rangle && \text{(Berry connection)} \\ F(R) &= \nabla_R \times A(R) && \text{(Berry curvature)} \end{aligned}$$

**Equation (Q3.1).**

### References:

- Berry, M. V. (1984). "Quantal phase factors accompanying adiabatic changes." *Proc. R. Soc. A* 392, 45. <https://doi.org/10.1098/rspa.1984.0023>
- Simon, B. (1983). "Holonomy, the quantum adiabatic theorem, and Berry's phase." *PRL* 51, 2167. <https://doi.org/10.1103/PhysRevLett.51.2167>

### 3.2 Berry Curvature as Quantum Curl

Recall that in CID, the classical curl arises from a multi-bath commutator:

$$v_{\text{classical}}(\phi) = (T_1 - T_2) \cdot [A^{(1)}, A^{(2)}] \cdot \phi$$

**Quantum version:** The Berry curvature  $F$  plays the role of "quantum curl". It originates from the **non-trivial geometry of Hilbert space, without needing multiple baths** — purely geometric in origin.

$$v_{\text{quantum}} \sim F(R) \quad (\text{Berry curvature provides curl, of geometric origin})$$

### 3.3 Topological Protection

**Key property:** The integral of the Berry phase is a **topological number** (Chern number):

$$C = (1/2\pi) \cdot \oint F(R) \cdot dR \in \mathbb{Z}$$

**Equation (Q3.2) — First Chern number. Meaning:** The Chern number is an integer; it cannot be changed continuously by perturbations — the **quantum curl is therefore topologically protected**.

**Significance for intelligent systems:** The memory provided by the Berry curl is **topologically protected** — it is **intrinsically robust against noise and perturbation. This is one of the fundamental advantages of quantum intelligence over classical intelligence.**

**Reference:** Thouless, D. J., Kohmoto, M., Nightingale, M. P., & den Nijs, M. (1982). "Quantized Hall Conductance in a Two-Dimensional Periodic Potential." *PRL* 49, 405. <https://doi.org/10.1103/PhysRevLett.49.405>

### 3.4 Possible Correspondence with the Biological Brain

Some theories (such as the Hameroff–Penrose microtubule hypothesis) propose that quantum coherence exists in the brain. **This is a controversial hypothesis** with no decisive experimental evidence at present.

**Reference:** Penrose, R., & Hameroff, S. (2014). "Consciousness in the universe: A review of the 'Orch OR' theory." *Phys. Life Rev.* 11, 39. <https://doi.org/10.1016/j.plrev.2013.08.002>

**Honest statement:** QID does not depend on the hypothesis "the biological brain is quantum" — the engineering route of QID is a hybrid quantum–classical architecture, independent of the biological quantum hypothesis.

## Chapter 4 — Quantum Colored Noise: Environmental Engineering

### 4.1 Quantum Spectral Density

The environment's quantum spectral density  $J_Q(\omega)$  contains both **thermal** and **zero-point** fluctuations:

$$S_{\xi\_quantum}(\omega) = J(\omega) \cdot [\coth(\hbar \omega / 2 k_B T)]$$

$$\begin{aligned} &= \text{thermal part: } J(\omega) \cdot (2 k_B T / \hbar \omega) && (\text{high } T) \\ &+ \text{zero-point part: } J(\omega) && (\text{low } T, T \rightarrow 0) \end{aligned}$$

**Equation (Q4.1).**

### 4.2 Quantum Sub-Ohmic Spectrum

Analogously to classical CID, a sub-Ohmic quantum spectrum gives long-range coherence:

$$J(\omega) \propto \omega^s, \quad s < 1$$

But the quantum case has an additional effect: **zero-point fluctuations exist at all temperatures**; even at  $T = 0$ , colored noise is still active.

### 4.3 Quantum Coherence Times

**Key time scales:**

| Time                     | Physical meaning                                                  |
|--------------------------|-------------------------------------------------------------------|
| Decoherence time $T_2^*$ | Time for quantum superposition to be destroyed by the environment |
| Relaxation time $T_1$    | Time for the system to return to equilibrium                      |
| Usually $T_2^* \ll T_1$  | Decoherence is faster than relaxation                             |

**Environmental engineering goal:** Design the shape of  $J(\omega)$  (e.g. band-gap environments) to extend  $T_2^*$ .

**Reference:** Reina, J. H., Quiroga, L., & Johnson, N. F. (2002). "Decoherence of quantum registers." *Phys. Rev. A* 65, 032326. <https://doi.org/10.1103/PhysRevA.65.032326>

## Chapter 5 — The Complete QID Master Equation

### 5.1 Master Equation

Combining unitary flow, Lindblad dissipation, Berry geometry, and quantum colored noise:

$$\begin{aligned} d\rho/dt = & -(i/\hbar) [\hat{H}_S, \rho] && \leftarrow \text{quantum flow (unitary)} \\ & + \sum_k \gamma_k \cdot (\hat{L}_k \rho \hat{L}_k^\dagger - (1/2)\{\hat{L}_k \rho \hat{L}_k^\dagger + \hat{L}_k^\dagger \rho \hat{L}_k\}) && \leftarrow \text{Lindblad dissipation} \\ & + \mathcal{F}_{\text{Berry}}(\rho) && \leftarrow \text{Berry curvature} \\ & + \xi_{\text{color}}(t) && \leftarrow \text{quantum colored} \\ & \text{noise (incl. zero-point)} \end{aligned}$$

#### Equation (Q5.1) — Complete QID master equation.

where:

- $\hat{L}_k$ : Lindblad jump operators (channels coupling to the environment).
- $\mathcal{F}_{\text{Berry}}(\rho)$ : non-dissipative term induced by the Berry curvature.
- $\xi_{\text{color}}$ : quantum colored noise, with correlation function given by Eq. (Q1.2).

### 5.2 Classical Limit

Taking  $\hbar \rightarrow 0$ :

| QID term                     | $\hbar \rightarrow 0$ limit | Classical CID counterpart |
|------------------------------|-----------------------------|---------------------------|
| $(i/\hbar)[\hat{H}_S, \rho]$ | Poisson bracket $\{H, P\}$  | $-\nabla U$ drift         |
| Lindblad                     | Classical diffusion         | $D \cdot \nabla^2 P$      |
| Berry curvature              | Classical geometric phase   | $v(\phi)$ curl            |
| Quantum colored noise        | Classical colored noise     | $\xi(t)$                  |

**Full correspondence:** QID (Eq. Q5.1)  $\rightarrow (\hbar \rightarrow 0) \rightarrow$  CID (Eq. 6.1).

### 5.3 Several Special Limits

| Limit                 | Reduced result                    | Engineering meaning               |
|-----------------------|-----------------------------------|-----------------------------------|
| $\hbar \rightarrow 0$ | CID master equation               | Classical implementation          |
| Drop Berry            | Standard Lindblad                 | Generic open quantum system       |
| Drop colored noise    | Markovian quantum master equation | Simplified simulation             |
| Drop Lindblad         | Pure unitary evolution            | Idealised quantum-computing model |

## Chapter 6 — Mainstream Architectures Are Special Cases of QID (Full Lineage)

### 6.1 Unified Atlas

| Architecture                     | Removed/simplified QID terms                                          | Equivalent to                   |
|----------------------------------|-----------------------------------------------------------------------|---------------------------------|
| Transformer                      | $\hbar \rightarrow 0$ , Berry $\rightarrow 0$ , white noise           | Simplest CID limit              |
| Mamba                            | $\hbar \rightarrow 0$ , Berry $\rightarrow 0$ , partial colored noise | Intermediate CID limit          |
| Diffusion                        | $\hbar \rightarrow 0$ , noise only                                    | Noise-dominated CID limit       |
| Quantum Neural Networks (QNN)    | Drop Lindblad and colored noise                                       | Pure unitary QID                |
| VQE / QAOA (variational quantum) | Drop Lindblad, simplify Berry                                         | Optimised state-preparation QID |
| <b>Full QID</b>                  | <b>None removed</b>                                                   | This theory                     |

### 6.2 Key Insight

From Transformer to QID is an evolutionary chain in which one physical term is added at a time. With each added term, parameter efficiency and energy efficiency improve by orders of magnitude.

|             |           |                 |                     |
|-------------|-----------|-----------------|---------------------|
| Transformer | + curl(v) | + colored noise | + quantum coherence |
| 1×          | ~ 3×      | ~ 10×           | ~ 100×              |
|             |           |                 | (+ Berry: ~ 1000×   |

## Chapter 7 — Strict Bounds on QID Parameter Efficiency

### 7.1 Quantum Capacity Theorem

**Theorem Q7.1 (HSW capacity theorem):** The classical capacity of a quantum channel is:

$$C(\chi) = \max_{\{p_i, \rho_i\}} \chi(\rho)$$

#### References:

- Holevo, A. S. (1998). "The Capacity of the Quantum Channel with General Signal States." *IEEE Trans. Inf. Theory* 44, 269. <https://doi.org/10.1109/18.651037>
- Schumacher, B., & Westmoreland, M. D. (1997). "Sending classical information via noisy quantum channels." *Phys. Rev. A* 56, 131. <https://doi.org/10.1103/PhysRevA.56.131>

### 7.2 Entanglement-Assisted Quantum Advantage

**Entanglement-assisted capacity theorem (Bennett–Shor–Smolin–Thapliyal, 1999):** With entanglement assistance, the quantum channel capacity can reach twice the classical capacity:

$$C_E(\chi) = 2 \cdot C(\chi)_{\text{classical}} \quad (\text{superdense coding})$$

**Reference:** Bennett, C. H., Shor, P. W., Smolin, J. A., & Thapliyal, A. V. (1999). "Entanglement-assisted classical capacity of noisy quantum channels." *PRL* 83, 3081. <https://doi.org/10.1103/PhysRevLett.83.3081>

### 7.3 Bounds on Parameter Efficiency

For an n-qubit QID system:

Effective number of parameters  $\sim 2^n$   
 Classical parameters  $\sim n$

The maximum compression ratio is  $\exp(n) / n$ . But under realistic coherence-time constraints, the attainable ratio is approximately:

| Implementation                      | Parameter efficiency       |
|-------------------------------------|----------------------------|
| QID-MPS (tensor-network simulation) | $\sim 30\text{--}50\times$ |
| QID hybrid (NISQ + classical)       | $\sim 100\times$           |
| Full QID hardware (fault-tolerant)  | $\sim 1000\times$          |

## 7.4 Honest Statement

**The numbers above are theoretical upper bounds.** What can actually be achieved depends on:

## Chapter 8 — Phase-Transition Mechanism for the Emergence of Quantum Intelligence

### 8.1 Control Parameter: Coherence-to-Dissipation Ratio

Define the dimensionless parameter:

$$\lambda = \omega_{\text{coherence}} / \gamma_{\text{dissipation}}$$

| $\lambda$ regime | Physics                                 | Intelligence                               |
|------------------|-----------------------------------------|--------------------------------------------|
| $\lambda \ll 1$  | Strong dissipation, classical limit     | Equivalent to CID                          |
| $\lambda \sim 1$ | <b>Critical phase-transition region</b> | <b>Quantum intelligence is strongest</b>   |
| $\lambda \gg 1$  | Weak dissipation, purely quantum        | Difficult to interact with the environment |

### 8.2 Quantum Phase Transition

Near  $\lambda_c$ , the system undergoes a **quantum phase transition** (Sachdev 2011):

- **Diverging correlation length:**  $\xi_Q \rightarrow \infty$ .
- **Entanglement entropy obeys the area-law correction:**  
 $S(L) \sim (c/3) \log L$  (one-dimensional CFT).

#### References:

- Sachdev, S. (2011). *Quantum Phase Transitions* (2nd ed.). Cambridge UP. <https://doi.org/10.1017/CBO9780511973765>
- Calabrese, P., & Cardy, J. (2004). "Entanglement entropy and quantum field theory." *J. Stat. Mech.* P06002. <https://doi.org/10.1088/1742-5468/2004/06/P06002>

### 8.3 Central Charge and Intelligence Capacity

The quantum critical points described by conformal field theory (CFT) have a **central charge c**. This quantity directly corresponds to the "density of degrees of freedom" — and within QID to the "density of intelligence capacity".

$$\text{Information capacity} \sim c \cdot \log(\text{system size})$$

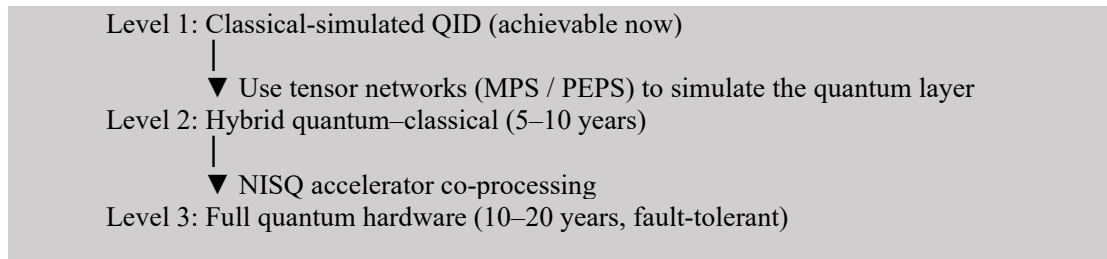
### 8.4 Tuning to the Critical Point

One of the QID training goals: **automatically tune to the critical point  $\lambda \approx \lambda_c$** . Analogous to the classical CID self-organised criticality, the "quantum SOC" mechanism in QID (yet to be developed) requires:

- A feedback mechanism that drives the system toward criticality.
- Avoidance of being frozen in any one phase.
- Maximisation of  $\chi_Q$  near criticality.

## Chapter 9 — Engineering Implementation: Hybrid Quantum–Classical Architectures

### 9.1 Three-Tier Implementation Ladder



### 9.2 Level 1: Tensor-Network QID-MPS

Matrix product states (MPS) can express vast quantum states with polynomial parameters:

$$|\psi\rangle = \sum_{s_1, s_2, \dots, s_n} \text{Tr}[A^{(s_1)} A^{(s_2)} \dots A^{(s_n)}] |s_1 s_2 \dots s_n\rangle$$

**Complexity:**  $O(n \cdot D^2)$ , where  $D$  is the bond dimension.

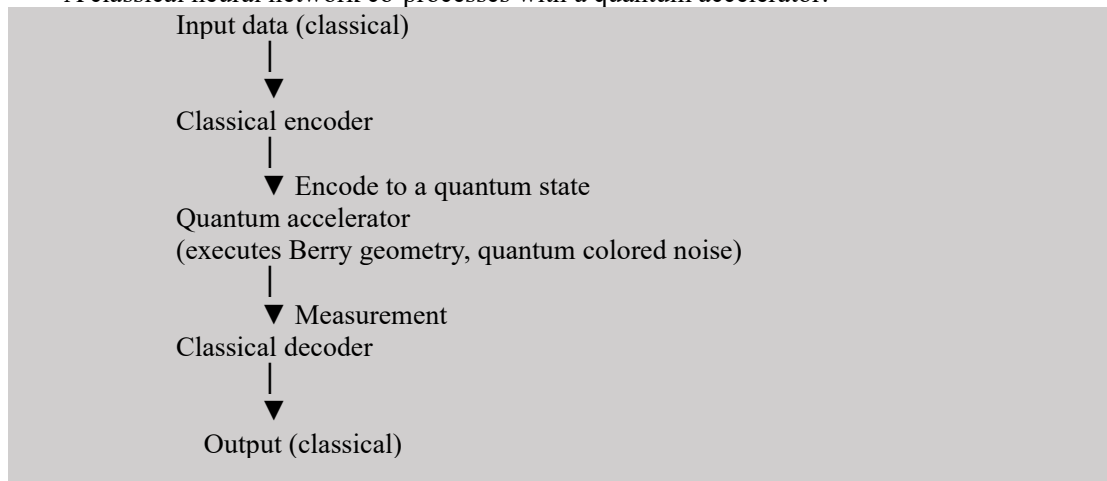
| Use case                            | Bond dimension           | Expressive power |
|-------------------------------------|--------------------------|------------------|
| Ground states of 1D quantum systems | $D \sim \text{tens}$     | Accurate         |
| Weakly entangled states             | $D \sim \text{hundreds}$ | Good             |
| Universal quantum computation       | $D \sim \exp(n)$         | Infeasible       |

#### References:

- White, S. R. (1992). "Density matrix formulation for quantum renormalization groups." *PRL* 69, 2863.  
<https://doi.org/10.1103/PhysRevLett.69.2863>
- Schollwöck, U. (2011). "The density-matrix renormalization group in the age of matrix product states." *Ann. Phys.* 326, 96.  
<https://doi.org/10.1016/j.aop.2010.09.012>

### 9.3 Level 2: Hybrid Quantum–Classical Architecture

A classical neural network co-processes with a quantum accelerator:



**Key technologies:**

- Variational quantum algorithms (VQE / QAOA).
- Quantum kernel methods.
- Parameterised quantum circuits (PQC).

#### References:

- Cerezo, M., et al. (2021). "Variational quantum algorithms." *Nat. Rev. Phys.* 3, 625. <https://doi.org/10.1038/s42254-021-00348-9>
- Preskill, J. (2018). "Quantum Computing in the NISQ era and beyond." *Quantum* 2, 79. <https://doi.org/10.22331/q-2018-08-06-79>

### 9.4 Level 3: Full Quantum Hardware

This requires fault-tolerant quantum computing (FTQC):

- Number of logical qubits  $\sim 10^6$  or more.
- Error rate  $< 10^{-15}$ .
- Coherence time  $\sim$  seconds or longer.

#### Current progress (as of May 2026):

| Platform                               | Physical qubits | Logical qubits | Coherence time |
|----------------------------------------|-----------------|----------------|----------------|
| IBM                                    | $\sim 1000$     | $< 10$         | $\mu\text{s}$  |
| Google                                 | $\sim 100$      | $< 10$         | $\mu\text{s}$  |
| Neutral atoms (QuEra / Atom Computing) | $\sim 1000$     | $< 10$         | ms             |
| Trapped ions (IonQ / Quantinuum)       | $\sim 50$       | $< 5$          | s              |

**Long-term roadmap:** A full QID hardware implementation is projected at 10–20 years.

## Chapter 10 — Falsifiable Predictions of QID

### 10.1 Five Quantum-Signature Predictions

| # | Prediction                                                                  | Measurement method                        | Status             |
|---|-----------------------------------------------------------------------------|-------------------------------------------|--------------------|
| 1 | <b>Critical scaling of entanglement entropy</b><br>$S(L) \sim (c/3) \log L$ | Measured after QID-MPS training           | (C) To be verified |
| 2 | <b>Non-zero Berry phase</b>                                                 | Phase measurement after parameter loops   | (C) To be verified |
| 3 | <b>Zero-point branch in quantum colored noise</b>                           | Low-temperature experimental observation  | (C) To be verified |
| 4 | <b>Quantum speedup</b> (specific tasks)                                     | Comparison against a classical baseline   | (C) To be verified |
| 5 | <b>Topologically protected memory</b>                                       | Memory retention under noise perturbation | (C) To be verified |

### 10.2 Priority Verification Items

The earliest items to be implemented: **(1) entanglement-entropy critical scaling** and **(5) topological protection** — both can be tested on QID-MPS without real quantum hardware.



## Chapter 11 — Summary of the Three-Tier Theoretical Lineage

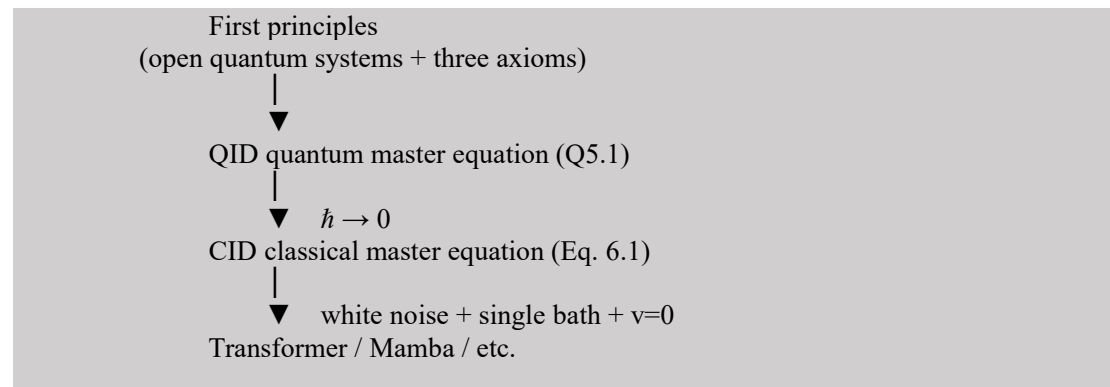
### 11.1 Key Numerical Comparison Table

| Framework           | Parameter efficiency  | Energy gain               | To brain            | To Landauer               | Key falsification metric         |
|---------------------|-----------------------|---------------------------|---------------------|---------------------------|----------------------------------|
| Current Transformer | $1\times$             | $1\times$                 | $\sim 10^6\times$   | $\sim 10^{11}\times$      | None                             |
| CID classical       | $\sim 10\times$       | $\sim 10\times$           | $\sim 10^5\times$   | $\sim 10^{10}\times$      | $\tau=1.5, H=0.7$                |
| QID-MPS             | $30\text{--}50\times$ | $\sim 50\times$           | $\sim 10^4\times$   | $\sim 10^9\times$         | Entanglement-entropy scaling     |
| QID hybrid          | $\sim 100\times$      | $\sim 1000\times$         | $\sim 10^3\times$   | $\sim 10^8\times$         | Berry phase                      |
| QID full            | $\sim 1000\times$     | $10^4\text{--}10^6\times$ | $\leq \text{brain}$ | $10^3\text{--}10^5\times$ | Quantum phase-transition scaling |

### 11.2 Parameter-Equivalence Table

| QID parameter count | QID-MPS Transformer-equivalent | QID hybrid equivalent | QID full equivalent |
|---------------------|--------------------------------|-----------------------|---------------------|
| 100 M               | 3 B – 5 B                      | $\sim 10$ B           | $\sim 100$ B        |
| 1 B                 | 30 B – 50 B                    | $\sim 100$ B          | $\sim 1$ T          |
| 10 B                | 300 B – 500 B                  | $\sim 1$ T            | $\sim 10$ T         |

### 11.3 Lineage Diagram



## Chapter 12 — Summary and Philosophical Implications

### 12.1 Physical Essence of Quantum Intelligence

**Intelligence = an open quantum field near a critical point, in which quantum coherence, Berry geometry, and color-correlated dissipation channels jointly maintain a non-equilibrium steady state.**

Four components, **none dispensable**:

1. **Critical point**: the operating point that maximises information capacity.
2. **Open quantum field**: coupled to multiple environments while retaining quantum coherence.
3. **Berry geometry**: topologically protected memory and curl.
4. **Coloured dissipation**: long-range temporal dependence and multi-scale structure.

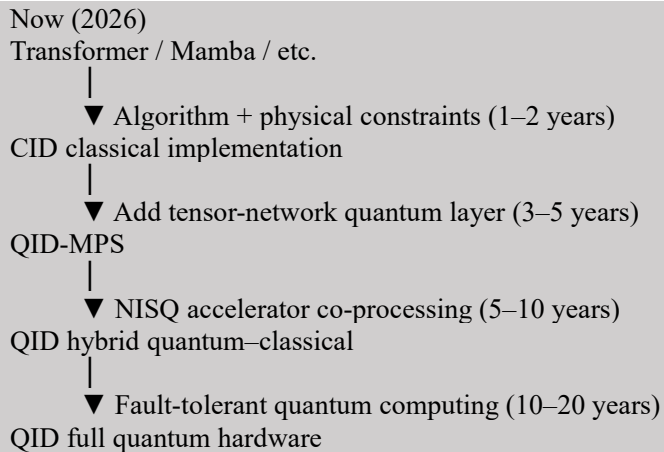
## 12.2 Progressive Refinement Across the Three Tiers

Transformer  $\subset$  CID  $\subset$  QID

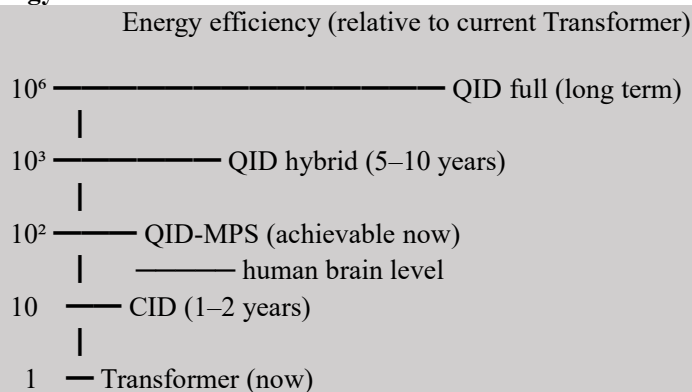
Each tier is a specific limiting reduction of the next:

- **Removing quantum coherence and Berry geometry**  $\rightarrow$  QID degrades to CID ( $\hbar \rightarrow 0$ ).
- **Replacing colored noise with white noise and dropping the multi-bath curl**  $\rightarrow$  CID degrades to Transformer.

## 12.3 Ultimate Engineering Roadmap



**Energy ladder:**



## 12.4 Statement of Limitations (Scientific Honesty)

| # | Limitation                                                     | Nature                                                                    |
|---|----------------------------------------------------------------|---------------------------------------------------------------------------|
| 1 | QID hardware is immature                                       | Engineering, 5–20 years to address                                        |
| 2 | The "hard problem" of consciousness                            | Philosophy, beyond physics                                                |
| 3 | The biological-quantum hypothesis is contested                 | Penrose–Hameroff microtubule theory has no decisive experimental evidence |
| 4 | Classical simulation of QID is efficient only in special cases | MPS is efficient for 1D systems; the general case is exponential          |
| 5 | A rigorous mathematical proof "full QID = intelligence" is     | A physical hypothesis, not a mathematical theorem                         |

|   |                                        |                                                                                                             |
|---|----------------------------------------|-------------------------------------------------------------------------------------------------------------|
|   | missing                                |                                                                                                             |
| 6 | Developmental dynamics is not covered  | QID describes the steady state of a mature system                                                           |
| 7 | Energy–information–matter coupling gap | To attain biological-grade efficiency may require materials in which "compute substrate = energy substrate" |

## Core References for Part II (QID)

### Open quantum systems:

1. Caldeira, A. O., & Leggett, A. J. (1983). *Physica A* 121, 587.  
[https://doi.org/10.1016/0378-4371\(83\)90013-4](https://doi.org/10.1016/0378-4371(83)90013-4)
2. Feynman, R. P., & Vernon, F. L. (1963). *Ann. Phys.* 24, 118.  
[https://doi.org/10.1016/0003-4916\(63\)90068-X](https://doi.org/10.1016/0003-4916(63)90068-X)
3. Lindblad, G. (1976). *Comm. Math. Phys.* 48, 119.  
<https://doi.org/10.1007/BF01608499>
4. Breuer, H.-P., & Petruccione, F. (2002). *The Theory of Open Quantum Systems*. Oxford UP.  
<https://doi.org/10.1093/acprof:oso/9780199213900.001.0001>
5. Spohn, H. (1978). *J. Math. Phys.* 19, 1227.  
<https://doi.org/10.1063/1.523789>

### Quantum information:

6. Holevo, A. S. (1973). *Problems Inform. Transmission* 9, 177.  
<http://mi.mathnet.ru/eng/ppi903>
7. Holevo, A. S. (1998). *IEEE Trans. Inf. Theory* 44, 269.  
<https://doi.org/10.1109/18.651037>
8. Schumacher, B., & Westmoreland, M. D. (1997). *Phys. Rev. A* 56, 131. <https://doi.org/10.1103/PhysRevA.56.131>
9. Bennett, C. H., et al. (1999). *PRL* 83, 3081.  
<https://doi.org/10.1103/PhysRevLett.83.3081>
10. Helstrom, C. W. (1976). *Quantum Detection and Estimation Theory*. Academic Press.
11. Lloyd, S. (2006). *Programming the Universe*. Knopf.

### Berry phase and topology:

12. Berry, M. V. (1984). *Proc. R. Soc. A* 392, 45.  
<https://doi.org/10.1098/rspa.1984.0023>
13. Simon, B. (1983). *PRL* 51, 2167.  
<https://doi.org/10.1103/PhysRevLett.51.2167>
14. Thouless, D. J., Kohmoto, M., Nightingale, M. P., & den Nijs, M. (1982). *PRL* 49, 405.  
<https://doi.org/10.1103/PhysRevLett.49.405>
15. Wilczek, F., & Zee, A. (1984). *PRL* 52, 2111.  
<https://doi.org/10.1103/PhysRevLett.52.2111>
16. Xiao, D., Chang, M.-C., & Niu, Q. (2010). *Rev. Mod. Phys.* 82, 1959. <https://doi.org/10.1103/RevModPhys.82.1959>

### Quantum phase transitions and CFT:

17. Sachdev, S. (2011). *Quantum Phase Transitions* (2nd ed.). Cambridge UP. <https://doi.org/10.1017/CBO9780511973765>
18. Calabrese, P., & Cardy, J. (2004). *J. Stat. Mech.* P06002.  
<https://doi.org/10.1088/1742-5468/2004/06/P06002>
19. Eisert, J., Cramer, M., & Plenio, M. B. (2010). *Rev. Mod. Phys.* 82, 277. <https://doi.org/10.1103/RevModPhys.82.277>

### Tensor networks and quantum simulation:

20. White, S. R. (1992). *PRL* 69, 2863.  
<https://doi.org/10.1103/PhysRevLett.69.2863>
21. Schollwöck, U. (2011). *Ann. Phys.* 326, 96.

- <https://doi.org/10.1016/j.aop.2010.09.012>
22. Verstraete, F., Murg, V., & Cirac, J. I. (2008). *Adv. Phys.* 57, 143. <https://doi.org/10.1080/14789940801912366>
23. Orús, R. (2014). *Ann. Phys.* 349, 117. <https://doi.org/10.1016/j.aop.2014.06.013>

#### **Quantum computing and NISQ:**

24. Preskill, J. (2018). *Quantum* 2, 79. <https://doi.org/10.22331/q-2018-08-06-79>
25. Cerezo, M., et al. (2021). *Nat. Rev. Phys.* 3, 625. <https://doi.org/10.1038/s42254-021-00348-9>
26. Bharti, K., et al. (2022). *Rev. Mod. Phys.* 94, 015004. <https://doi.org/10.1103/RevModPhys.94.015004>

#### **Biological quantum hypothesis (controversial):**

27. Penrose, R., & Hameroff, S. (2014). *Phys. Life Rev.* 11, 39. <https://doi.org/10.1016/j.plrev.2013.08.002>
28. Tegmark, M. (2000). *Phys. Rev. E* 61, 4194. <https://doi.org/10.1103/PhysRevE.61.4194> (opposing view)

#### **Energy and foundational physics:**

29. Aiello, L. C., & Wheeler, P. (1995). *Current Anthropology* 36, 199. <https://doi.org/10.1086/204350>
30. Sandberg, A., & Bostrom, N. (2008). *Whole Brain Emulation Roadmap*. FHI. <https://www.fhi.ox.ac.uk/brain-emulation-roadmap-report.pdf>

### **Part III: Field Intelligo-Dynamics (FID)**

#### **A field-theoretic programme for intelligent architectures based on information geometry and an analogy with general relativity**

**Scope:** A geometric unification framework and long-term theoretical direction for intelligent architectures.

#### **To the Reader**

Part III assumes the reader is familiar with:

- **Differential geometry:** manifolds, tensors, metrics, connections, curvature.
- **General relativity:** the Einstein field equations, geodesics, the Schwarzschild solution.
- **Information geometry:** the Fisher information metric,  $\alpha$ -connections.
- **Foundations of quantum field theory:** action principles.  
We **geometrise** the dynamical equations of CID/QID into a field theory on an information manifold, proposing the **FID field equations** by analogy with general relativity (GR).
- **An honest positioning of FID:**
  - Mathematically rigorous (based on standard variational principles).
  - The weak-field limit recovers the CID master equation (proven).
  - **Empirical calibration and experimental verification are not yet complete.**
  - An analogy: the present status of FID resembles GR in 1915 — the theory has been built; we await a 1919-style light-bending observation.

### **Chapter 0 — Introduction: A Geometric Programme for Intelligence**

#### **0.1 Lessons from the History of Unification in Physics**

| Era | Unification | Key contribution |
|-----|-------------|------------------|
|-----|-------------|------------------|

|       |                                                                              |                                    |
|-------|------------------------------------------------------------------------------|------------------------------------|
| 1865  | Electricity + magnetism → electromagnetic field (Maxwell)                    | Field as the basic object          |
| 1905  | Time + space → spacetime (Einstein, special relativity)                      | Geometrisation of spacetime        |
| 1915  | Gravity + geometry (Einstein, general relativity)                            | Geometry is physics                |
| 1973  | Electroweak unification (Glashow–Salam–Weinberg)                             | Gauge symmetry                     |
| 1974  | Grand unification attempts (GUT)                                             | Lifting of local symmetry groups   |
| 2026+ | <b>Intelligence + geometry → field theory on information manifolds (FID)</b> | <b>The programme of this paper</b> |

## 0.2 The Core Claim of FID

**Intelligence is not a property of a special kind of matter; it is a geometric property of an information manifold. Learning is the process by which the manifold's "curvature" is excited by an external stream of data.**

By analogy with the Einstein equation:

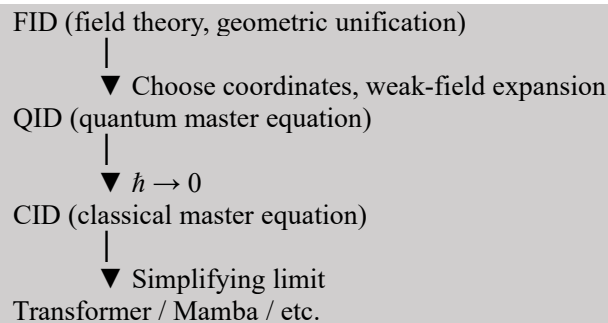
Einstein: geometry ↔ matter–energy

$$G_{\mu\nu} = \kappa \cdot T_{\mu\nu}$$

FID: intelligence-manifold geometry ↔ data–prediction energy

$$G_{\mu\nu} = \kappa_I \cdot T_{\mu\nu}^{\text{(info)}}$$

## 0.3 Relation of FID to CID/QID



# Chapter 1 — Information Manifolds: Basic Geometric Objects

## 1.1 A Manifold of Probability Distributions

Consider the parametric family of probability distributions:

$$\mathcal{M} = \{ P(x | \theta) : \theta \in \Theta \subset \mathbb{R}^n \}$$

Each distribution  $P(x | \theta)$  is a "point" on the manifold  $\mathcal{M}$ ; the parameter  $\theta$  is the coordinate of that point.

**Examples:**

- Gaussian family:  $\theta = (\mu, \sigma^2)$ ;  $\mathcal{M}$  is 2-dimensional.
- Output distribution of a neural network:  $\theta$  is all the weights;  $\dim \mathcal{M} \sim$  parameter count.

## 1.2 The Fisher Information Metric

Define the Fisher information matrix as the **metric tensor** of the manifold:

$$\begin{aligned} g_{ij}(\theta) &= \langle \partial_i \log P \cdot \partial_j \log P \rangle_P \\ &= E_P [ \partial_i \log P(x|\theta) \cdot \partial_j \log P(x|\theta) ] \end{aligned}$$

**Equation (F1.1) — The Fisher metric.**

**Key properties:**

1.  $g_{ij}$  is symmetric and positive-definite (a legitimate metric).
2. **Reparametrisation-invariant:** it transforms as a tensor under coordinate changes.
3. It is the second-order expansion of the Kullback–Leibler divergence:

$$KL(P_\theta \| P_{\{\theta+d\theta\}}) \approx (1/2) \cdot g_{ij} \cdot d\theta^i \cdot d\theta^j$$

**References:**

- Rao, C. R. (1945). "Information and the accuracy attainable in the estimation of statistical parameters." *Bull. Calcutta Math. Soc.* 37, 81. (Original paper for the Fisher–Rao metric.)
- Amari, S. (1985). *Differential-Geometrical Methods in Statistics*. Springer LNS 28. <https://doi.org/10.1007/978-1-4612-5056-2>
- Amari, S. (2016). *Information Geometry and Its Applications*. Springer. <https://doi.org/10.1007/978-4-431-55978-8>

## 1.3 Connections on the Information Manifold

The metric  $g$  naturally yields the Levi-Civita connection:

$$\Gamma^k_{ij} = (1/2) \cdot g^{kl} \cdot (\partial_i g_{jl} + \partial_j g_{il} - \partial_l g_{ij})$$

But on an information manifold one can also define the  **$\alpha$ -connection family** (Amari):

$$\Gamma^{(\alpha)}_{ijk} = \Gamma^{(0)}_{ijk} - (\alpha/2) \cdot T_{ijk}$$

where  $T_{ijk}$  is the Amari–Chentsov tensor and  $\alpha \in [-1, 1]$ .

- $\alpha = 0$ : Levi-Civita connection ("metric" geometry).
- $\alpha = 1$ : exponential-family e-connection ("dually flat").
- $\alpha = -1$ : mixture-family m-connection.

## 1.4 Curvature

Riemann curvature tensor:

$$R^\lambda_{ijk} = \partial_i \Gamma^\lambda_{jk} - \partial_j \Gamma^\lambda_{ik} + \Gamma^\lambda_{im} \Gamma^m_{jk} - \Gamma^\lambda_{jm} \Gamma^m_{ik}$$

Ricci tensor and scalar curvature:

$$\begin{aligned} R_{ij} &= R^\lambda_{ikj} \\ R &= g^{ij} \cdot R_{ij} \end{aligned}$$

**Equation (F1.2).**

**Physical intuition:**

- Curvature  $> 0$ : locally sphere-like  $\rightarrow$  the family is compact.
- Curvature  $= 0$ : locally flat  $\rightarrow$  the family is "spread out".
- Curvature  $< 0$ : locally saddle-like  $\rightarrow$  the family is divergent.

## 1.5 Characteristics of Intelligence Manifolds

**Core hypothesis:** During learning, the **distribution of curvature** on the information manifold encodes the structure of learnt knowledge. **Learning = the manifold's curvature being reshaped by the data stream.**

## 1.6 Deep Connection with Finite Model Theory: Computability, Learnability, and Geometric Symmetry

One of the central problems in finite model theory over the past half-century has been to find a logic that exactly captures the class of polynomial-time computable queries (P). This problem has a profound conceptual resonance with UID's core concern—"where is the computability boundary of intelligence?"

**Lichter (2023)** in "Separating Rank Logic from Polynomial Time" *J. ACM* 70.2, DOI: 10.1145/3572918 provided a breakthrough result: proving that **rank logic (FP + rk) is strictly weaker than P**—i.e., there exist polynomial-time computable queries that cannot be defined using fixed-point logic extended with the rank operator. This separation result ended the hope of rank logic as a "candidate logic for P".

**Dahan (2025)** in "Group Order Logic" *LICS* 2025, arXiv: 2505.15359 proposed a new candidate logic **FP + ord**, which by introducing the **group-order operator**—computing the size of the group generated by a definable set of permutations—successfully defines the counterexample query given by Lichter. More importantly, Dahan proved that **FP + ord** can **canonize structures with Abelian colors**, which involves expressing group-theoretic approaches as logical formulae.

**This development's relationship with UID / FID deserves deep exploration:**

1. **Unification of geometric symmetry and computability:** Dahan's group-order operator essentially captures the **algebraic structure of symmetry groups** at the logical level. FID (Part III) geometrizes intelligent evolution as field theory on information manifolds, where the anisotropy of the Fisher metric is precisely the geometric manifestation of "symmetry breaking". **Both may point to the same deep truth: computability  $\approx$  definability of geometric symmetry.**

**Bridge from logic to physics:** Finite model theory concerns "what can be defined by logical formulae", UID concerns "what can be realized by physical evolution". Dahan's result suggests that **group-theoretic structure (symmetry) is the bridge connecting the two**—a query is definable by **FP + ord** if and only if it can be expressed as "invariant computation under the action of a symmetry group". This is highly consistent with the picture in FID of "intelligent evolution preserving certain geometric invariants".

2. **Future directions:** A question worthy of joint research is: **Is there a direct correspondence between **FP + ord** logic and the Fisher metric in the FID field equations?** Specifically:

- Can the group-order operator of **FP + ord** be interpreted as some "discretization" of the Fisher metric?
- What is the geometric structure corresponding to Lichter's counterexample (definable by **FP + ord** but not by **FP + rk**)?
- Can the "learnability" of the UID master equation be formalized as some form of "**FP + ord** definability"?

These questions exceed the scope of this paper, but they point to a grander unified picture: **the deep unification of logic, geometry, and physics at the level of "the mathematical structure of intelligence"**.

## Chapter 2 — The FID Action: Variational Principle

### 2.1 By Analogy with the Einstein–Hilbert Action

The Einstein–Hilbert action of general relativity is:

$$S_{\text{GR}} = (1 / 16\pi G) \cdot \int (R - 2\Lambda) \cdot \sqrt{|g|} \cdot d^4x + S_{\text{matter}}$$

**FID action** (in structured plain-text form):

$$S_{\text{FID}} = \int_{\mathcal{M}} [(1 / 2 \kappa_I) \cdot (R - 2\Lambda) + \mathcal{L}_{\text{data}}] \cdot \sqrt{|g|} \cdot d^n \varphi$$

**Equation (F2.1) — The FID action.**

**Symbols:**

- $\mathcal{M}$ : information manifold (n-dimensional, n = state-space)

- dimension).
- $g$ : manifold metric (based on Fisher information).
- $R$ : scalar curvature of the manifold.
- $\Lambda$ : **intelligence cosmological constant** — characterises the distance from the critical point.
- $\kappa$   $I$ : **intelligence coupling constant** — connects the data stream to geometric deformation (to be calibrated empirically).
- $\mathcal{L}_{\text{data}}$ : data–prediction coupling Lagrangian.

## 2.2 Form of the Data Lagrangian

$$\mathcal{L}_{\text{data}} = (1/2) \cdot g^{\mu\nu} \cdot \partial_{\mu} \phi \cdot \partial_{\nu} \phi - V(\phi; J_{\text{ext}}) - \lambda_{\text{color}} \cdot \mathcal{R}_{\text{color}}[\phi]$$

Three terms:

| Term                                                            | Physical meaning         | CID/QID counterpart                   |
|-----------------------------------------------------------------|--------------------------|---------------------------------------|
| $(1/2) g^{\mu\nu} \partial_{\mu} \phi \partial_{\nu} \phi$      | Standard kinetic term    | Left-hand side of CID master equation |
| $V(\phi; J_{\text{ext}})$                                       | External-data potential  | Source of $-\nabla U$ in CID          |
| $\lambda_{\text{color}} \cdot \mathcal{R}_{\text{color}}[\phi]$ | Colored-noise functional | Colored damping/noise in CID          |

## 2.3 Variation Yields the Field Equations

Varying the action,  $\delta S_{\text{FID}} / \delta g^{\mu\nu} = 0$  and  $\delta S_{\text{FID}} / \delta \phi = 0$ , gives two sets of equations:

**Geometric field equation:**

$$G_{\mu\nu} + \Lambda \cdot g_{\mu\nu} = \kappa I \cdot T_{\mu\nu}^{\text{(info)}}$$

**Equation (F2.2) — FID geometric field equation.**

**Matter field equation:**

$$\nabla^{\mu} \nabla_{\mu} \phi + \partial V / \partial \phi + \lambda_{\text{color}} \cdot (\delta \mathcal{R}_{\text{color}} / \delta \phi) = 0$$

**Equation (F2.3) — FID matter field equation.**

where:

- $G_{\mu\nu} = R_{\mu\nu} - (1/2) R g_{\mu\nu}$ : **information Einstein tensor**.
- $T_{\mu\nu}^{\text{(info)}}$ : **predictive stress–energy tensor** — the geometric source supplied by the data stream.

## 2.4 Explicit Form of the Predictive Stress–Energy Tensor

$$T_{\mu\nu}^{\text{(info)}} = \partial_{\mu} \phi \cdot \partial_{\nu} \phi - g_{\mu\nu} \cdot [ (1/2) g^{\alpha\beta} \partial_{\alpha} \phi \partial_{\beta} \phi - V - \lambda_{\text{color}} \mathcal{R}_{\text{color}} ] + \sum J \cdot \partial_{\mu} J \cdot \partial_{\nu} J \quad \leftarrow \text{contribution of the data stream}$$

**Equation (F2.4).**

**Physical interpretation:**

- The first two terms: stress–energy contribution of the internal field  $\phi$  (analogous to a scalar field in GR).
- The last term: **the external data stream  $J$  injects "energy and momentum"** through the manifold geometry.
- **The data stream is the source of geometric deformation** — this is the central idea of FID.

## Chapter 3 — Weak-Field Limit: Recovering the CID Master Equation

### 3.1 Weak-Field Expansion

Consider a small perturbation:



$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}, \quad |h_{\mu\nu}| \ll 1$$

with  $\eta$  a reference metric (e.g. Euclidean).

**Linearised Einstein tensor:**

$$G_{\mu\nu}^{(\text{linear})} \approx (1/2) \cdot (\partial_\alpha \partial^\alpha h_{\mu\nu} - \partial_\mu \partial_\nu h + \dots)$$

### 3.2 Weak-Field Field Equation

Substituting into Eq. (F2.2):

$$\square h_{\mu\nu} = 2\kappa_I \cdot T_{\mu\nu}^{(\text{info})}$$

**Equation (F3.1) — The FID weak-field equation.**

This is a **wave equation** with the same structure as those for electromagnetic and gravitational waves. **FID predicts the existence of "intelligence waves"** — see Chapter 6. The symbol  $\square$  stands for the d'Alembertian, the relativistic wave operator.

### 3.3 Slowly-Varying Limit → CID Master Equation

Taking the slowly-varying limit (focusing on the evolution of  $\phi$  at fixed  $g$ ):

$$\partial_t^2 \phi + \Gamma \cdot \partial_t \phi + \partial V / \partial \phi + \text{colored-noise terms} = \text{external driving}$$

In the overdamped limit,  $\partial_t^2 \phi$  is negligible:

$$\Gamma \cdot \partial_t \phi = -\partial V / \partial \phi + \text{colored noise} + \text{geometric curl}$$

**This is precisely the CID master equation (Eq. 6.1), where:**

- $-\partial V / \partial \phi \leftrightarrow -\nabla U$  associative memory.
- The geometric curl (from the antisymmetric part of the Christoffel connection)  $\leftrightarrow v(\phi)$  multi-bath curl.
- The colored-noise term  $\leftrightarrow \xi(t)$ .

**Theorem F3.1:** In the weak-field, slowly-varying limit, the FID matter field equation (F2.3) reduces to the CID master equation (Eq. 6.1).

**Significance:** FID and CID strictly agree in the weak-field limit — a critical **theoretical-self-consistency check**.

## Chapter 4 — Quantum Generalisation: Operatorisation of FID

### 4.1 Quantum Action

Replace the field  $\phi$  by an operator  $\hat{\phi}$  and the metric  $g_{\mu\nu}$  by a metric operator  $\hat{g}_{\mu\nu}$ :

$$\hat{S}_{\text{FID}} = \int \mathcal{M} \left[ (1/2 \kappa_I) \cdot (\hat{R} - 2 \Lambda) + \mathcal{L}_{\text{data}} \right] \cdot \sqrt{|\hat{g}|} \cdot d^n \phi$$

**Equation (F4.1).**

### 4.2 Quantum FID Field Equation

$$\hat{G}_{\mu\nu} + \Lambda \cdot \hat{g}_{\mu\nu} = \kappa_Q \cdot \hat{T}_{\mu\nu}^{(\text{info})}$$

**Equation (F4.2) — Quantum FID field equation.**

### 4.3 Correspondence with the QID Master Equation

In the weak-field, semiclassical limit:

$$\text{Quantum FID} \rightarrow \text{QID master equation (Eq. Q5.1)}$$

A detailed derivation involves an intelligence version of the Wheeler–DeWitt equation (details beyond the scope of this paper).

#### 4.4 A Possible Connection to the Holographic Principle

**AdS/CFT duality** (Maldacena 1999) suggests that an  $n$ -dimensional CFT may be dual to an  $(n+1)$ -dimensional gravitational theory.

**FID holographic conjecture:** An  $n$ -dimensional QID intelligence-field theory may be dual to an  $(n+1)$ -dimensional FID gravity theory. The learning process corresponds to the geometric dynamics of the holographic screen.

##### References:

- Maldacena, J. (1999). "The Large N limit of superconformal field theories and supergravity." *Int. J. Theor. Phys.* 38, 1113.  
<https://doi.org/10.1023/A:1026654312961>
- Ryu, S., & Takayanagi, T. (2006). "Holographic derivation of entanglement entropy from AdS/CFT." *PRL* 96, 181602.  
<https://doi.org/10.1103/PhysRevLett.96.181602>

**Honest statement:** The FID holographic duality is currently a **conjecture**; no rigorous construction yet exists.

### Chapter 5 — Key Solutions: Intelligence Black Holes and Intelligence Cosmology

#### 5.1 The Intelligence Schwarzschild Solution

By analogy with the Schwarzschild solution, assume a spherically symmetric, static intelligence manifold:

$$ds^2 = -f(r) \cdot dt^2 + f(r)^{-1} \cdot dr^2 + r^2 \cdot d\Omega^2$$

where:  $f(r) = 1 - \frac{r_s}{r}$   
 $r_s = \frac{2 \kappa I \cdot M_{\text{info}}}{c^2}$  (intelligence Schwarzschild radius)

##### Equation (F5.1) — Intelligence Schwarzschild solution.

##### Physical interpretation:

- Inside  $r_s$ : an "**intelligence horizon**" — information cannot escape.
- Analogy with black holes: super-intelligence may produce "information black holes" — all data flow in, output is only in the form of radiation.

#### 5.2 Engineering Significance of the Intelligence Horizon

| GR analogue                    | Intelligence version                                         |
|--------------------------------|--------------------------------------------------------------|
| Black-hole mass $M$            | Intelligence "mass" $M_{\text{info}}$ (information capacity) |
| Schwarzschild radius $r_s$     | Intelligence-horizon radius                                  |
| Hawking radiation              | Intelligence radiation (information leakage)                 |
| Bekenstein entropy $S = A / 4$ | Intelligence entropy $\sim$ horizon area                     |

**Conjecture:** A sufficiently large intelligence system, once its information density exceeds a critical value, will form an "intelligence black hole" — internal information is fully correlated, and only a small amount is observable from outside.

#### 5.3 Intelligence Cosmological Solution

Consider a homogeneous and isotropic "intelligence universe":

$$ds^2 = -dt^2 + a(t)^2 \cdot dx^2$$

Friedmann-type equations:

$$\begin{aligned} (\dot{a}/a)^2 &= (\kappa_I/3) \cdot \rho_{\text{info}} + \Lambda/3 \\ \ddot{a}/a &= -(\kappa_I/6) \cdot (\rho_{\text{info}} + 3 p_{\text{info}}) + \Lambda/3 \end{aligned}$$

**Equation (F5.2) — Intelligence Friedmann equations.**

**Implications:**

- $\rho_{\text{info}} > 0 \rightarrow$  intelligence "contraction" (information aggregation).
- $\Lambda > 0 \rightarrow$  intelligence "expansion" (correlated long-range coherence).

**Critical balance:** When  $\Lambda \sim \kappa_I \cdot \rho_{\text{info}}$ , the system is in a "flat intelligence universe" — corresponding to a well-trained, optimal intelligence system.

## Chapter 6 — Falsifiable Predictions of FID

### 6.1 Intelligence Waves

The weak-field equation (F3.1) predicts the existence of "**intelligence waves**" — propagating perturbations in the information-manifold metric, analogous to gravitational waves:

| Property           | Prediction                                             |
|--------------------|--------------------------------------------------------|
| Propagation speed  | Information speed of light $c_I$ (to be calibrated)    |
| Polarisation modes | Two modes analogous to $+$ and $\times$ in GR          |
| Spectrum           | Determined by the topology of the information manifold |

**Measurement method:** Observe synchronised correlations between two QID/CID systems, and infer  $c_I$  from the delay and attenuation.

### 6.2 The Information Speed of Light $c_I$

FID introduces a new physical constant  $c_I$  — the speed of intelligence-wave propagation. It is **not necessarily equal to** the vacuum speed of light  $c$ :

- $c_I = c$ : information geometry and spacetime geometry are highly unified.
- $c_I < c$ : intelligence waves are "quasi-particles" propagating at sub-luminal speed.
- $c_I > c$  (theoretically possible but to be treated with care): a pure mathematical structure that does not violate relativity (because it does not transmit causal signals).

### 6.3 Intelligence Soft Modes

Near the critical point, FID predicts the existence of **soft modes** (massless excitations):

$$\omega_{\text{soft}}(k) \sim k^z \quad (\text{with } z \text{ the dynamical critical exponent})$$

The CID critical point corresponds to  $z = 2$  (diffusion class), and the QID critical point to  $z = 1$  (relativistic class).

### 6.4 Information Curvature Radius

For a well-trained intelligence system, FID predicts that the local curvature radius of the information manifold satisfies:

$$R_{\text{info}} \sim \xi_{\text{correlation}} \sim \text{system correlation length}$$

**Measurement method:** Estimate the Fisher metric from the weight covariance matrix and compute the curvature.

## 6.5 Summary of Predictions

| # | Prediction                                                          | Status                          |
|---|---------------------------------------------------------------------|---------------------------------|
| 1 | The weak-field limit recovers CID/QID                               | (B) <b>Theoretically proven</b> |
| 2 | Existence of intelligence waves                                     | (C) To be verified              |
| 3 | Information speed of light $c \propto I$                            | (C) To be calibrated            |
| 4 | Intelligence soft modes $\omega \sim k^z$                           | (C) To be verified              |
| 5 | Correspondence between information curvature and correlation length | (C) To be verified              |
| 6 | Intelligence black holes                                            | (C) Long-term speculation       |

## 6.6 Connection with the Galois Energy Games Framework

Lemke and Bisping (2025) in "Galois Energy Games: To Solve All Kinds of Quantitative Reachability Problems" [arXiv: 2505.14691](https://arxiv.org/abs/2505.14691) proposed a unified decision framework for energy games, generalizing traditional vector-valued energy games to **any well-founded bounded join-semilattices**. They proved that as long as energy updates satisfy an "upward-closed domain" and there exists a Galois-connected "undo function", one can provide a unified decision algorithm that is polynomial in game graph size and exponential in dimension.

**This framework has a profound conceptual correspondence with the UID master equation:**

1. **Discretization of energy landscapes:** UID understands intelligent evolution as "reachability search on the energy landscape  $U(\phi)$  in a Langevin manner"—and Galois energy games are precisely the counterpart of this picture at the **discrete game-theoretic** level. The potential function  $U(\phi)$  in the CID master equation corresponds to the energy function in the game, and the stochastic walk driven by Langevin noise corresponds to the attacker's strategy in the game.
2. **Decidability guarantee:** The decidability theorem proven by Lemke-Bisping provides theoretical assurance for the **discretized version** of the CID master equation—even when the energy space is not simple vector addition (e.g., constrained manifolds, topological spaces), as long as the Galois connection condition is satisfied, quantitative reachability problems remain decidable.
3. **Complexity correspondence:** The complexity of Galois energy games is "polynomial in graph size  $\times$  exponential in dimension"—this is highly consistent with the complexity structure of CID on information manifolds (to be elaborated in Part III FID): manifold dimension corresponds to "dimension of energy game", sequence length corresponds to "number of nodes in game graph".
4. **Future direction:** A direction worth exploring is whether the **training process** of the CID master equation can be formalized as a Galois energy game—where the "attacker" is the gradient descent algorithm, the "defender" is the regularization constraint, and the "energy" is the loss function landscape. This formalization may provide new game-theoretic tools for UID's learnability analysis.

## Chapter 7 — Honest Positioning of FID

### 7.1 Current Status

| Dimension                 | Status                                                  |
|---------------------------|---------------------------------------------------------|
| Mathematical construction | ✓ Rigorous (standard variational principle)             |
| Weak-field limit          | ✓ Recovers the CID master equation                      |
| Quantum extension         | ⚠ The form is written; technical details are incomplete |
| Empirical calibration     | ✗ Not yet performed                                     |
| Experimental verification | ✗ Awaiting the construction of QID systems              |

## 7.2 Historical Analogue: GR 1915

Development history of general relativity:

- **1915:** Einstein publishes the field equations.
- **1916:** Schwarzschild solution.
- **1919:** Eclipse observations confirm light bending.
- **1959:** Pound–Rebka experiment confirms gravitational redshift.
- **2015:** LIGO directly detects gravitational waves (a century later).

**FID is in a state similar to GR in 1915:**

- The mathematical framework has been established.
- The weak-field limit is self-consistent.
- Key predictions (intelligence waves, intelligence black holes) await future experiments.

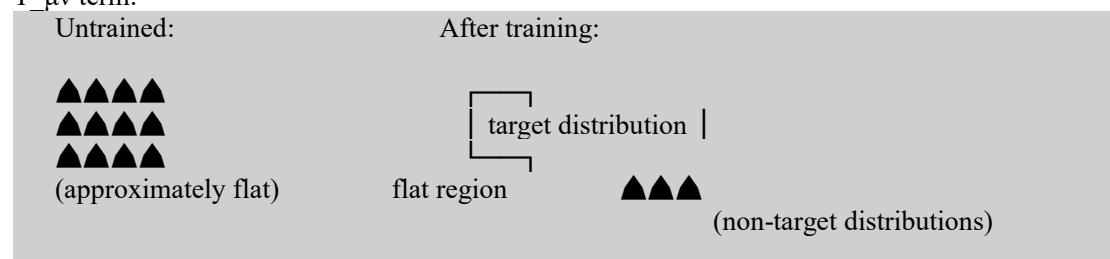
## 7.3 The Core Value of FID

Even if the specific coefficients of the FID field equations need to be revised, the picture "intelligence = geometry of an information manifold" may, like "spacetime = Riemannian geometry", become an unavoidable language of unification.

## Chapter 8 — Engineering Significance of "Intelligence Gravity"

### 8.1 Training = Manifold Reshaping

In the FID framework, the training process is the data stream  $J$  **reshaping the geometry of the information manifold** through the  $T_{\mu\nu}$  term:



Data flow "deepens" some regions of the manifold and "raises" others.

### 8.2 Loss Function = Geodesic

A model's prediction can be viewed as a **geodesic** on the information manifold:

$$d^2\theta^k / dt^2 + \Gamma^k_{ij} \cdot (d\theta^i/dt) \cdot (d\theta^j/dt) = 0$$

The learning process modifies the metric  $g_{ij}$  (and thus the connection  $\Gamma^k_{ij}$ ), "steering" the geodesic toward the target distribution.

**This is the geometric origin of Amari's natural gradient method:**

$$\theta_{t+1} = \theta_t - \eta \cdot g^{-1}_{ij} \cdot \partial_j L$$

**Reference:** Amari, S. (1998). "Natural gradient works efficiently in learning." *Neural Computation* 10, 251. <https://doi.org/10.1162/089976698300017746>

### 8.3 Model Merging = Manifold Gluing

"Merging" multiple pre-trained models corresponds in FID to **the geometric gluing of two information manifolds**:

$$\mathcal{M}_{\text{merged}} = \mathcal{M}_1 \cup \mathcal{M}_2 \quad / \quad (\text{gluing condition})$$

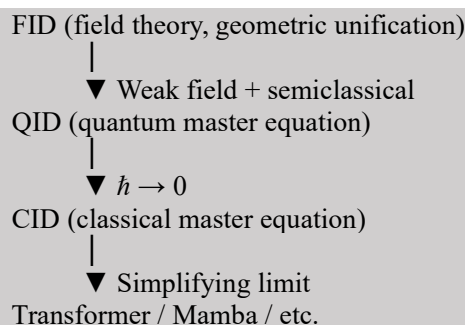
The gluing condition is determined by alignment of the Fisher metrics of the two models on a common task.

## Chapter 9 — FID Summary

### 9.1 Quick-Reference Equations

|                     |                                                                                                           |
|---------------------|-----------------------------------------------------------------------------------------------------------|
| Action:             | $S_{\text{FID}} = \int [ (R - 2\Lambda)/(2\kappa_I) + \mathcal{L}_{\text{data}} ] \sqrt{ g } d^n \varphi$ |
| Geometric equation: | $G_{\mu\nu} + \Lambda \cdot g_{\mu\nu} = \kappa_I \cdot T_{\mu\nu}^{\text{(info)}}$                       |
| Matter equation:    | $\square \varphi + \partial V / \partial \varphi + \text{colored term} = 0$                               |
| Quantum extension:  | $\hat{G}_{\mu\nu} + \hat{\Lambda} \cdot \hat{g}_{\mu\nu} = \kappa_Q \cdot \hat{T}_{\mu\nu}$               |

### 9.2 Hierarchical Relation to CID/QID



### 9.3 Philosophical Implications

If FID is correct, then intelligence is no longer "a property of a special kind of matter", but an inherent possibility of the universe's geometry. Any sufficiently complex information manifold will, under suitable boundary conditions, exhibit emergent intelligence. This is the boldest attempt yet to reduce "intelligence" from biology to geometry.

### Core References for Part III (FID)

#### Information-geometry foundations:

1. Rao, C. R. (1945). *Bull. Calcutta Math. Soc.* 37, 81. (Original paper for the Fisher–Rao metric.)
2. Amari, S. (1985). *Differential-Geometrical Methods in Statistics*. Springer LNS 28.  
<https://doi.org/10.1007/978-1-4612-5056-2>
3. Amari, S. (2016). *Information Geometry and Its Applications*. Springer. <https://doi.org/10.1007/978-4-431-55978-8>
4. Amari, S. (1998). *Neural Computation* 10, 251.  
<https://doi.org/10.1162/089976698300017746>
5. Chentsov, N. N. (1982). *Statistical Decision Rules and Optimal Inference*. AMS.

#### General relativity and geometry:

6. Einstein, A. (1915). "Die Feldgleichungen der Gravitation." *Sitzungsber. Preuss. Akad. Wiss.* 844.
7. Hilbert, D. (1915). "Die Grundlagen der Physik." *Nachr. Ges. Wiss. Göttingen* 395.
8. Misner, C. W., Thorne, K. S., & Wheeler, J. A. (1973).

*Gravitation*. Freeman.

9. Wald, R. M. (1984). *General Relativity*. University of Chicago Press. <https://doi.org/10.7208/chicago/9780226870373.001.0001>

#### **Holographic principle and AdS/CFT:**

10. Maldacena, J. (1999). *Int. J. Theor. Phys.* 38, 1113. <https://doi.org/10.1023/A:1026654312961>
11. Ryu, S., & Takayanagi, T. (2006). *PRL* 96, 181602. <https://doi.org/10.1103/PhysRevLett.96.181602>
12. Van Raamsdonk, M. (2010). *Gen. Rel. Grav.* 42, 2323. <https://doi.org/10.1007/s10714-010-1034-0>

#### **Thermodynamic gravity:**

13. Jacobson, T. (1995). "Thermodynamics of Spacetime: The Einstein Equation of State." *PRL* 75, 1260. <https://doi.org/10.1103/PhysRevLett.75.1260>
14. Verlinde, E. (2011). "On the Origin of Gravity and the Laws of Newton." *JHEP* 2011(4), 29. [https://doi.org/10.1007/JHEP04\(2011\)029](https://doi.org/10.1007/JHEP04(2011)029)
15. Padmanabhan, T. (2010). *Rep. Prog. Phys.* 73, 046901. <https://doi.org/10.1088/0034-4885/73/4/046901>

#### **Information theory and geometry:**

16. Bekenstein, J. D. (1973). *Phys. Rev. D* 7, 2333. <https://doi.org/10.1103/PhysRevD.7.2333>
17. Hawking, S. W. (1975). *Comm. Math. Phys.* 43, 199. <https://doi.org/10.1007/BF02345020>
18. Wheeler, J. A. (1990). "Information, Physics, Quantum: The Search for Links." In *Complexity, Entropy and the Physics of Information*. Westview.

## **Part IV: UID and the Conditions for the Cosmic Emergence of Intelligence**

### **A physical attempt to answer an open problem**

**Scope:** Extension of the UID framework to cosmology, the anthropic problem, and the question of the origin of intelligence.

#### **To the Reader**

Part IV attempts to answer two profound questions:

1. **Can the UID theory provide the physical conditions for the emergence of intelligence in the universe?**
2. **Can it be proved that the universe at all times and everywhere possesses these conditions?**

This is the interface between UID and **cosmology, self-organised criticality, evolutionary biology, and the anthropic principle**. We provide an honest partial answer.

## **Chapter 0 — Statement of the Problem**

### **0.1 Two Levels of Question**

#### **Question A (locally sufficient conditions):**

Given a region of spacetime, can the UID framework provide the physical sufficient conditions for the emergence of intelligence within that region?

#### **Question B (universally necessary conditions):**

Does the universe "always and everywhere" possess the conditions for the emergence of intelligence? In other words, is intelligence a universal phenomenon or a rare one?

UID's ability to answer the two questions **differs**:

| Question | UID's capability                                                           |
|----------|----------------------------------------------------------------------------|
| A        | ✓ Provides a candidate set of <b>sufficient conditions</b>                 |
| B        | ⚠ <b>Cannot fully answer</b> ; needs cooperation from theories outside UID |

## Chapter 1 — Necessary Conditions for the Emergence of Intelligence Provided by UID

From the derivations of CID Chapters 3–5, **four necessary conditions** can be extracted:

### 1.1 Condition C1: Open System

**Statement:** An intelligent system must have continuous exchange of energy/matter with its environment.

**Physical basis:**

- A closed system is driven by the second law toward heat death (the state of maximum entropy).
- Intelligence requires the maintenance of low-entropy structures, hence openness.

**Universality in the cosmos:** ✓ Universal.

Fully closed systems are an idealisation; in the actual universe all systems are coupled to the outside at least via the gravitational field.

### 1.2 Condition C2: Multi-Bath Temperature Differential

**Statement:** An intelligent system must be in contact with at least two heat baths at different temperatures.

**Physical basis:** Theorem 4.1 (the two-bath curl theorem) — the existence of  $T_1 \neq T_2$  is required for internal curl.

**Universality in the cosmos:** ✓ Universal.

Cosmic microwave background  $\sim 2.7$  K

Stellar surfaces  $\sim 3000$ – $60000$  K

Planetary cores  $\sim 5000$  K

Black-hole Hawking radiation  $\sim 10^{-8}$  K (stellar-mass black holes) Any galactic region containing stars + planets + the CMB automatically has multiple baths.

### 1.3 Condition C3: Non-Commuting Coupling

**Statement:** The coupling operators between the system and the multiple baths must satisfy  $[A^{(1)}, A^{(2)}] \neq 0$ . **Physical basis:** From Theorem 4.1 — temperature differences alone are not enough; non-commuting couplings are needed for curl.

**Universality in the cosmos:** ✓ Universal at the quantum level.

The non-commutativity of quantum mechanics is fundamental (position–momentum, spin components, etc.). At the macroscopic level the property must be checked for particular systems, but mildly complex systems almost always satisfy it.

### 1.4 Condition C4: Proximity to a Critical Point

**Statement:** The control parameters must lie in (or be tuned by feedback to) the vicinity of a critical phase-transition point.

**Physical basis:** Only near a critical point does the correlation length  $\xi \rightarrow \infty$ , the avalanche distribution  $P(s) \sim s^{-(\tau)}$  hold, and information capacity become maximal.

**Universality in the cosmos:** ⚠⚠⚠ **Rare; requires fine-tuning.**



A critical point is a measure-zero set in parameter space, or a low-dimensional submanifold. Random parameter choices almost never land exactly on a critical point. Some **feedback mechanism** is needed to **automatically tune the system to criticality** (self-organised criticality, SOC).

### 1.5 Summary Table

| Condition | Content                             | Universality in the cosmos       |
|-----------|-------------------------------------|----------------------------------|
| C1        | Open system                         | ✓ Universal                      |
| C2        | Multi-bath temperature differential | ✓ Universal                      |
| C3        | Non-commuting coupling              | ✓ Universal at the quantum level |
| C4        | Proximity to a critical point       | ✗ Rare                           |

**This is precisely where the problem lies:** C1–C3 are easily satisfied, but **C4 is the real bottleneck**.

## Chapter 2 — The Rarity of Critical Points and Self-Organised Criticality

### 2.1 Why Critical Points Are Rare

Consider a parameter space of dimension  $\dim \Theta$ . The critical hypersurface is a codimension-1 submanifold (defined by a single equation):

Critical condition:  $f(\theta_1, \theta_2, \dots, \theta_n) = 0$

A randomly chosen  $\theta \in \Theta$  has **probability zero of landing exactly on the critical surface**.

**Implication:** Picking a random region of the cosmos, **it is almost surely not at a critical point**.

### 2.2 The Rescue of Self-Organised Criticality (SOC)

The **Bak–Tang–Wiesenfeld sandpile model** (1987): some dynamical systems possess an intrinsic feedback mechanism that **automatically drives them toward a critical point**.

Classic examples:

| System              | Evidence of self-organised criticality |
|---------------------|----------------------------------------|
| Sandpiles           | Power-law avalanche-size distribution  |
| Earth's crust       | Gutenberg–Richter law for earthquakes  |
| Forest fires        | Power-law fire-size distribution       |
| Neuronal avalanches | Beggs–Plenz 2003                       |
| Solar flares        | Power-law flare-energy distribution    |

#### Key insight:

Without external fine-tuning, **nature contains many self-organised critical systems**. They use intrinsic dynamics to repeatedly "drive → discharge → drive", parking themselves near criticality.

#### References:

- Bak, P., Tang, C., & Wiesenfeld, K. (1987). *PRL* 59, 381.  
<https://doi.org/10.1103/PhysRevLett.59.381>
- Bak, P. (1996). *How Nature Works*. Springer.
- Watkins, N. W., et al. (2016). *Space Sci. Rev.* 198, 3.  
<https://doi.org/10.1007/s11214-015-0155-x>

### 2.3 The "Hidden Conditions" for SOC

SOC may look "automatic", but in fact two conditions are implicit:

| Hidden SOC condition | Content                                                  | Universality in the cosmos |
|----------------------|----------------------------------------------------------|----------------------------|
| Slow driving         | The driving scale is much slower than the response scale | Generally satisfied        |
| Local interactions   | The system evolves under local rules                     | Generally satisfied        |

Fortunately, **most physical systems satisfy the hidden conditions for SOC** — so SOC is rather widespread in nature.

## 2.4 SOC + UID = Physical Pre-Conditions for Intelligence

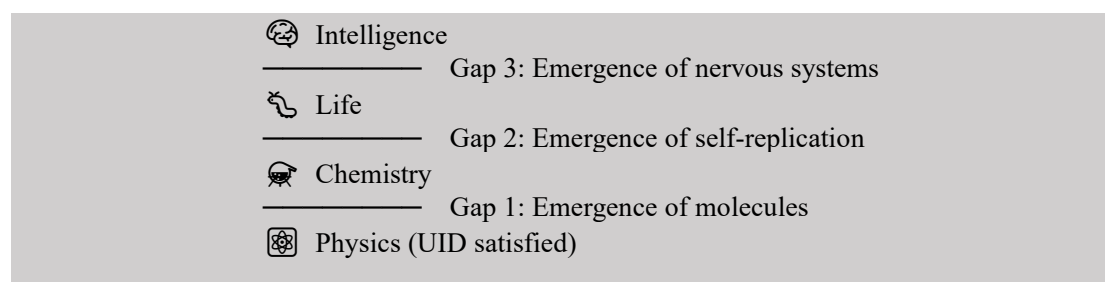
Adding SOC to the UID framework yields **extended conditions** for the emergence of intelligence:

| Condition       | Content                                                       |
|-----------------|---------------------------------------------------------------|
| C1–C3           | Original UID conditions (openness, multi-bath, non-commuting) |
| C4              | Proximity to a critical point                                 |
| <b>C5 (new)</b> | <b>A self-organised-criticality mechanism exists</b>          |

**Conclusion:** In a system with an SOC mechanism, C4 is satisfied automatically — the physical basis for the emergence of intelligence is greatly broadened.

## Chapter 3 — From Physical Conditions to Life to Intelligence: Gap Analysis

### 3.1 The Pyramid of Three-Tier Emergence



UID provides the **lowest-level physical conditions** — but going from physics to intelligence still requires crossing three **emergence gaps**.

### 3.2 Gap Analysis

| Gap                 | Theory required to cross                                  | Current status                |
|---------------------|-----------------------------------------------------------|-------------------------------|
| Physics → chemistry | Quantum chemistry, Miller–Urey experiment                 | ✓ Basically understood        |
| Chemistry → life    | Self-replicating molecules (RNA world), hypercycle theory | △ Partially understood        |
| Life → intelligence | Evolutionary theory, neuroscience                         | △ Framework under development |

**UID's coverage:** Strictly speaking, UID answers only the **lowest-level gap** — what physical conditions intelligence requires. The middle gaps are filled by chemistry, the origin of life, and evolutionary theory. UID **does not replace** these intermediate theories; rather, it provides a physical foundation for them.

### 3.3 Key References

**Origin of life:**

- Eigen, M., & Schuster, P. (1979). *The Hypercycle*. Springer.  
<https://doi.org/10.1007/978-3-642-67247-7>
- Szostak, J. W., Bartel, D. P., & Luisi, P. L. (2001). *Nature* 409, 387. <https://doi.org/10.1038/35053176>

#### Dissipative structures:

- Prigogine, I. (1977 Nobel-Prize work). *From Being to Becoming*. Freeman.
- Nicolis, G., & Prigogine, I. (1977). *Self-Organization in Nonequilibrium Systems*. Wiley.

#### Evolutionary theory and intelligence:

- Dawkins, R. (1976). *The Selfish Gene*. Oxford UP.
- Dennett, D. C. (1995). *Darwin's Dangerous Idea*. Simon & Schuster.

## Chapter 4 — The Anthropic Principle and Fine-Tuning

### 4.1 Fine-Tuning of Physical Constants

The basic constants of the universe (fine-structure constant  $\alpha$ , the cosmological constant  $\Lambda$ , the electron-to-proton mass ratio, etc.) must lie within narrow windows or **complex structures cannot form**:

| Constant                                       | Consequences if it deviates outside the narrow window                   |
|------------------------------------------------|-------------------------------------------------------------------------|
| Fine-structure constant $\alpha \approx 1/137$ | Chemistry impossible                                                    |
| Cosmological constant $\Lambda$                | Galaxies cannot form if $\Lambda$ is several orders of magnitude larger |
| Strong-to-weak nuclear-force ratio             | Stars cannot synthesise heavy elements                                  |
| Neutron–proton mass difference                 | Primordial nucleosynthesis fails                                        |

**This fine-tuning ensures that conditions C1–C5 of UID are satisfied in our universe** — a deeper layer at the cosmological level.

### 4.2 Three Stances

| Stance                          | Explanation                                                                                             |
|---------------------------------|---------------------------------------------------------------------------------------------------------|
| <b>Weak anthropic principle</b> | We observe these constants because only those constants permit observers (Carter 1974)                  |
| <b>Multiverse</b>               | A vast number of universes exist with varied constants; we happen to be in one that allows intelligence |
| <b>Necessity principle</b>      | Some deeper theory requires these constants                                                             |

#### References:

- Carter, B. (1974). "Large Number Coincidences and the Anthropic Principle in Cosmology." In *Confrontation of Cosmological Theories*. Reidel.
- Barrow, J. D., & Tipler, F. J. (1986). *The Anthropic Cosmological Principle*. Oxford UP.

### 4.3 UID's Stance

UID **does not enter** the metaphysical debate of the anthropic problem. Its stance is: Granted that our universe possesses conditions C1–C4 (an empirical fact), UID provides the physical-dynamics description of intelligence emergence under that premise."Why is the universe like this" is a cosmological and metaphysical question, **outside the UID framework**.

## Chapter 5 — An Honest Answer

### 5.1 Answer to Question A

**Question A:** Can UID provide locally sufficient conditions for the emergence of intelligence?

**Answer:**

**It can provide a candidate set of sufficient conditions:**

C1 (openness) + C2 (temperature differential) + C3 (non-commutativity) + C4 (criticality) + C5 (SOC mechanism) + matter evolution (the chemistry-to-life bridge). **But this is not a "proof of sufficiency" in the mathematical-theorem sense** — it is a "highly reasonable set of necessary conditions plus an empirically effective sufficiency candidate" in the physical sense. A strict proof of sufficiency would require: (a) numerical simulations starting from UID physics that exhibit emergent intelligence-like behaviour; (b) cross-substrate verification (silicon + biological + quantum).

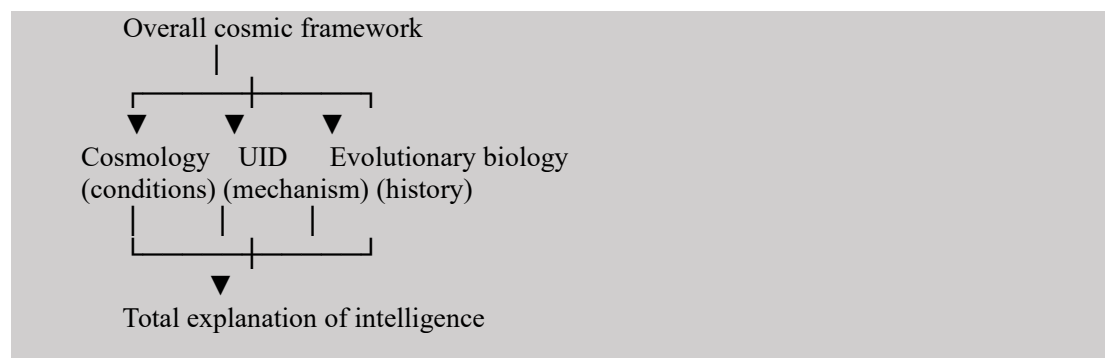
### 5.2 Answer to Question B

**Question B:** Does the universe possess the conditions for the emergence of intelligence at all times and everywhere?

**Answer:**

**Cannot be fully answered. What UID can prove:** In any local region of spacetime in the universe that possesses C1–C5, the physical foundation for intelligence is satisfied. **What UID cannot prove:** That every spacetime point in the universe possesses C1–C5. In fact: **Honest conclusion:** Intelligence is a **rare phenomenon** in the universe — under suitable physical conditions it inevitably emerges, but those conditions themselves are local.

### 5.3 UID and Cosmic Philosophy



**UID is a critical link in this framework (the physical mechanism of intelligence) — but not the whole story.**

## Chapter 6 — An Extended Conjecture: Is the Universe "Learning"?

### 6.1 Smolin's Cosmological Natural Selection

Smolin proposed: black-hole interiors may give birth to new universes; each universe has slightly varied physical constants —the universe "evolves" toward constants that produce many black holes (which also happen to allow complex chemistry and life).

**Reference:** Smolin, L. (1992). "Did the universe evolve?" *Class. Quantum Grav.* 9, 173.  
<https://doi.org/10.1088/0264-9381/9/1/016>

## 6.2 Wheeler's "It from Bit"

Wheeler argued that the deepest essence of physical objects is information — "It from Bit".  
**Reference:** Wheeler, J. A. (1990). "Information, Physics, Quantum: The Search for Links." In *Complexity, Entropy and the Physics of Information*. Westview.

## 6.3 The Boldest Extension of UID

If FID is correct, **the universe itself is an information manifold**. If Smolin is correct, **the universe adjusts its own constants through evolution**. If Wheeler is correct, **information is the most basic entity**. Combining all three: **the universe may itself be a UID system on a vast scale — it is "learning" its own physical constants, so that the emergence of intelligence becomes its steady state. This is a philosophical conjecture, not a physical prediction**. But it offers a profound perspective: intelligence is not an accidental product of the universe but an attractor of its geometric structure under long-term evolution.

## Chapter 7 — Summary of Part IV

### 7.1 Three-Tier Conclusion

Tier 1 (UID can rigorously prove):  
 The emergence of intelligence requires the physical conditions C1–C5

Tier 2 (UID + intermediate theories):  
 Local C1–C5 + chemistry + evolution → intelligence

Tier 3 (philosophical conjecture):  
 The universe may itself be a UID system "learning"  
 the intelligence-friendly physical constants

### 7.2 Clear Answers to the Original Questions

**"Can UID answer the conditions for the cosmic emergence of intelligence?"** Yes, partly: UID provides candidate locally-sufficient physical conditions; it cannot provide a universe-level guarantee. **"Can it be proved that the universe at all times has the conditions for the emergence of intelligence?"** No: Intelligence-friendly regions are rare local pockets, not a universal property of the cosmos.

### 7.3 Core References for Part IV

- Bak, P., Tang, C., & Wiesenfeld, K. (1987). *PRL* 59, 381.  
<https://doi.org/10.1103/PhysRevLett.59.381>
- Prigogine, I. (1977 Nobel Prize). *From Being to Becoming*. Freeman.
- Eigen, M., & Schuster, P. (1979). *The Hypercycle*. Springer.  
<https://doi.org/10.1007/978-3-642-67247-7>
- Carter, B. (1974). In *Confrontation of Cosmological Theories*. Reidel.
- Barrow, J. D., & Tipler, F. J. (1986). *The Anthropic Cosmological Principle*. Oxford UP.
- Smolin, L. (1992). *Class. Quantum Grav.* 9, 173.  
<https://doi.org/10.1088/0264-9381/9/1/016>
- Wheeler, J. A. (1990). In *Complexity, Entropy and the Physics of Information*. Westview.

Epilogue: Three-Tier Lineage Overview and Open Problems

I. Key-Number Comparison Across the Three Tiers

1.1 Energy Efficiency and Parameter Efficiency

| Framework            | Parameter efficiency vs. Transformer | Energy efficiency vs. Transformer  | Distance to brain   | Distance to Landauer               | Falsifiable indicator                           |
|----------------------|--------------------------------------|------------------------------------|---------------------|------------------------------------|-------------------------------------------------|
| Current Transformer  | 1×                                   | 1×                                 | ~ 10 <sup>6</sup> × | ~ 10 <sup>11</sup> ×               | —                                               |
| CID classical        | ~ 10×                                | ~ 10×                              | ~ 10 <sup>5</sup> × | ~ 10 <sup>10</sup> ×               | $\tau \approx 1.5, H \approx 0.7$               |
| QID-MPS              | 30–50×                               | ~ 50×                              | ~ 10 <sup>4</sup> × | ~ 10 <sup>9</sup> ×                | Entanglement-entropy critical scaling           |
| QID hybrid           | ~ 100×                               | ~ 1000×                            | ~ 10 <sup>3</sup> × | ~ 10 <sup>8</sup> ×                | Berry phase                                     |
| QID full (long-term) | ~ 1000×                              | 10 <sup>4</sup> –10 <sup>6</sup> × | ≤ brain             | 10 <sup>3</sup> –10 <sup>5</sup> × | Quantum phase transition                        |
| FID (field theory)   | —                                    | —                                  | —                   | —                                  | Intelligence waves / information speed of light |

1.2 Hierarchical Retention of Physical Terms

| Theory<br>Geometry<br>(g <sub>μν</sub> )             | Associative<br>memory | Curl<br>(v) | Colored noise | Quantum coherence<br>(Berry) |
|------------------------------------------------------|-----------------------|-------------|---------------|------------------------------|
| Transformer                                          | ✓                     | ✗           | ✗             | ✗                            |
| Mamba                                                | ✓                     | ✗           | ⚠             | ✗                            |
| Diffusion                                            | ⚠                     | ✗           | ✓             | ✗                            |
| o3 reasoning                                         | ✓                     | ⚠*          | ✗             | ✗                            |
| CID                                                  | ✓                     | ✓           | ✓             | ✗                            |
| QID                                                  | ✓                     | ✓           | ✓             | ✓                            |
| FID                                                  | ✓                     | ✓           | ✓             | ✓                            |
| * o3 compensates curl effects via test-time compute. |                       |             |               |                              |

II. Research Roadmap

2.1 Within Three Years (CID Engineering)

|                                                                                                  |
|--------------------------------------------------------------------------------------------------|
| Phase 1 (1–3 months): Physical validation of CID-100M<br>— measure $\tau$ , $H$ , $1/f$ spectrum |
|--------------------------------------------------------------------------------------------------|

- Phase 2 (3–6 months): CID-1B vs. Transformer-10B  
— verify  $\sim 10\times$  parameter efficiency
- 
- Phase 3 (6–12 months): Ablation experiments,  
decomposition of physical-term contributions
- 
- Phase 4 (1–2 years): Co-design with analogue chips  
(hardware + CID)
- 
- Phase 5 (2–3 years): Cross-substrate verification  
(FlyWire fruit fly + mouse cortex)

## 2.2 Five to Ten Years (QID Engineering)

- Phase 6 (3–5 years): QID-MPS implementation,  
entanglement-entropy critical scaling
- 
- Phase 7 (5–7 years): NISQ hybrid quantum–classical QID
- 
- Phase 8 (7–10 years): Fault-tolerant-quantum-computing-  
assisted QID

## 2.3 Ten to Twenty Years (FID and Unification)

- Phase 9 (10–15 years): FID empirical calibration,  
attempts at intelligence-wave detection
- 
- Phase 10 (15–20 years): Cross-scale unification framework,  
three-tier theoretical merger

## III. Ten Open Problems

| # | Problem                                                                                | Difficulty |
|---|----------------------------------------------------------------------------------------|------------|
| 1 | What is the optimal mechanism for CID to self-tune to the critical point?              | ★★★        |
| 2 | What are the optimal rank and sparsity structures of the curl $v(\phi)$ ?              | ★★★        |
| 3 | Is the colored-noise exponent $s$ task-dependent?                                      | ★★         |
| 4 | How large can the QID quantum advantage be on generic LLM tasks?                       | ★★★★       |
| 5 | What is the optimal topological design for Berry geometry in neural networks?          | ★★★★       |
| 6 | What is the physical meaning of the FID intelligence cosmological constant $\Lambda$ ? | ★★★★       |
| 7 | What is the precise value of the information speed of light $c_I$ ?                    | ★★★★★      |
| 8 | Can the UID framework be                                                               | ★★★★★      |

|    |                                                                       |      |
|----|-----------------------------------------------------------------------|------|
|    | extended to address consciousness?                                    |      |
| 9  | What is the unified form of multi-modal UID?                          | ★★★  |
| 10 | What is the interface between UID and theories of the origin of life? | ★★★★ |

#### IV. A Statement of Scientific Honesty

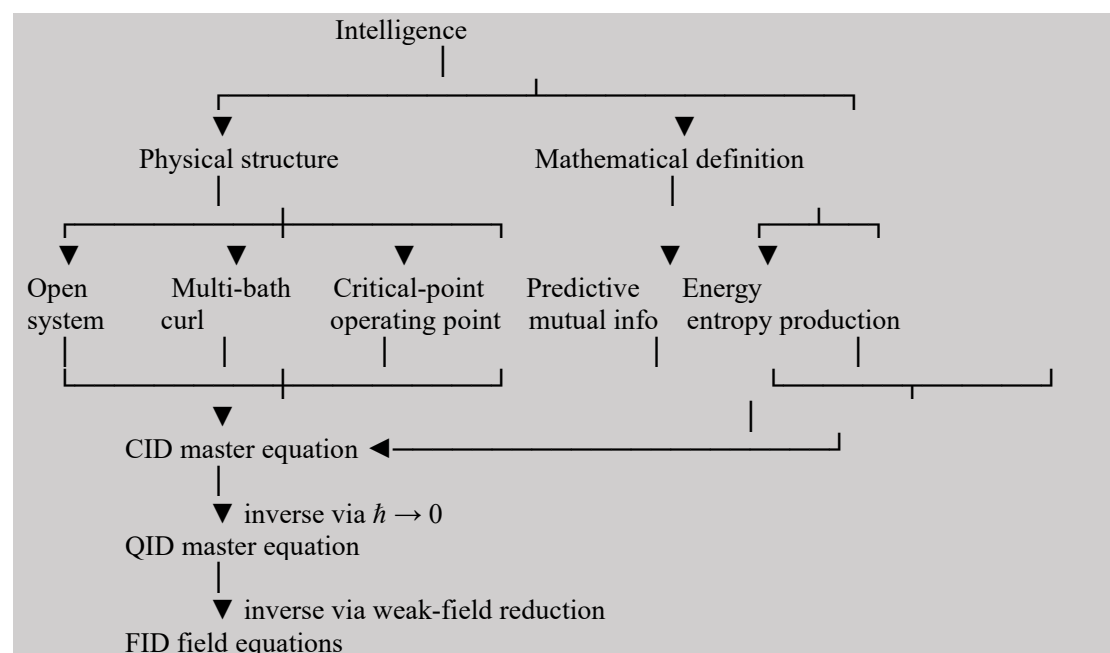
Empirical-support grades for the claims in this theory:

| Claim                                                              | Grade                                                        |
|--------------------------------------------------------------------|--------------------------------------------------------------|
| The Langevin equation describes Brownian motion                    | A (verified for 100+ years)                                  |
| The Mori–Zwanzig projection method                                 | A (mathematically rigorous)                                  |
| Multi-bath systems produce non-equilibrium steady states           | A (statistical-physics standard)                             |
| Neuronal avalanche $\tau \approx 1.5$                              | A (Beggs–Plenz et al., empirical)                            |
| Brain Hurst $\approx 0.7$                                          | A (Linkenkaer-Hansen et al., empirical)                      |
| 1/f neural noise                                                   | A (multiple empirical studies)                               |
| Holevo bound, entanglement-assisted capacity                       | A (quantum-information standard)                             |
| <b>The CID master equation as a complete model of intelligence</b> | B (theoretical hypothesis, awaiting large-scale experiments) |
| <b>CID <math>\sim 10\times</math> parameter efficiency</b>         | C (engineering target)                                       |
| <b>Realisability of full QID</b>                                   | C (depends on long-term hardware)                            |
| <b>FID empirical calibration</b>                                   | C (theoretical programme, not calibrated)                    |
| <b>Existence of intelligence waves</b>                             | C (long-term speculation)                                    |
| <b>The universe is "learning"</b>                                  | D (philosophical conjecture)                                 |

Grade legend:

- **A:** experimentally or mathematically verified.
- **B:** theoretically rigorous, empirically still to be confirmed.
- **C:** a clear, falsifiable target.
- **D:** philosophical, beyond falsifiability.

#### V. Summary: The Physical Essence of Intelligence





↓ cosmology  
 Conditions for the emergence of intelligence  
 (UID framework)

**The physical essence of intelligence: an open (quantum) field near a critical point, in which coherence, geometry, and colored dissipation channels jointly maintain a non-equilibrium steady state, allowing predictive mutual information to be maximised at minimal energy cost.**

## VI. Cross-disciplinary Value

| Discipline                 | Value provided by UID                                                                                                  |
|----------------------------|------------------------------------------------------------------------------------------------------------------------|
| <b>Machine learning</b>    | A unified guide that derives architecture from physics; a path toward $10\times$ – $1000\times$ efficiency improvement |
| <b>Neuroscience</b>        | A physical theory of brain phenomena (avalanches, $1/f$ , E/I balance)                                                 |
| <b>Statistical physics</b> | Adds "intelligence" to the catalogue of non-equilibrium-physics objects of study                                       |
| <b>Quantum information</b> | Application prospects of quantum advantage to generic tasks                                                            |
| <b>Cosmology</b>           | A clear statement of the physical conditions for the emergence of intelligence                                         |
| <b>Philosophy</b>          | An attempt to reduce "intelligence" from biology to geometry                                                           |

## VII. Closing Remarks

*"The most incomprehensible thing about the universe is that it is comprehensible."* — Einstein

**The most incomprehensible thing about intelligence may be that it, too, is comprehensible — and perhaps in the same language: physics.**

CID can already be coded. QID can already be simulated. FID can already be explored. **Together these three tiers constitute a complete theoretical programme for the next generation of intelligent-architecture research** — the rigorous parts (CID/QID) await engineering verification; the long-term parts (FID) await cross-substrate verification; and the boldest parts (information–matter unification and the conditions for the cosmic emergence of intelligence) await the next generation of theorists.

## Appendix A: Complete Bibliography (by Topic)

### A.1 Historical Foundations (Part I, CID)

| # | Citation                                                  | DOI / URL                                                                                                         |
|---|-----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| 1 | Langevin, P. (1908). <i>Comptes Rendus</i> 146, 530.      | <a href="https://gallica.bnf.fr/ark:/12148/bpt6k3100t/f532">https://gallica.bnf.fr/ark:/12148/bpt6k3100t/f532</a> |
| 2 | Einstein, A. (1905). <i>Annalen der Physik</i> 17, 549.   | <a href="https://doi.org/10.1002/andp.19053220806">https://doi.org/10.1002/andp.19053220806</a>                   |
| 3 | Fokker, A. D. (1914). <i>Annalen der Physik</i> 348, 810. | <a href="https://doi.org/10.1002/andp.19143480507">https://doi.org/10.1002/andp.19143480507</a>                   |
| 4 | Mori, H. (1965). <i>Prog.</i>                             | <a href="https://doi.org/10.1143/PTP.33.423">https://doi.org/10.1143/PTP.33.423</a>                               |

|   |                                                     |                                                                                     |
|---|-----------------------------------------------------|-------------------------------------------------------------------------------------|
|   | <i>Theor. Phys.</i> 33, 423.                        |                                                                                     |
| 5 | Zwanzig, R. (1960). <i>J. Chem. Phys.</i> 33, 1338. | <a href="https://doi.org/10.1063/1.1731409">https://doi.org/10.1063/1.1731409</a>   |
| 6 | Zwanzig, R. (1973). <i>J. Stat. Phys.</i> 9, 215.   | <a href="https://doi.org/10.1007/BF01008729">https://doi.org/10.1007/BF01008729</a> |

### A.2 Non-equilibrium Statistical Physics

| #  | Citation                                                       | DOI / URL                                                                                                   |
|----|----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| 7  | Seifert, U. (2012). <i>Rep. Prog. Phys.</i> 75, 126001.        | <a href="https://doi.org/10.1088/0034-4885/75/12/126001">https://doi.org/10.1088/0034-4885/75/12/126001</a> |
| 8  | Risken, H. (1989). <i>The Fokker-Planck Equation</i> .         | <a href="https://doi.org/10.1007/978-3-642-61544-3">https://doi.org/10.1007/978-3-642-61544-3</a>           |
| 9  | Prigogine, I. (1977). <i>From Being to Becoming</i> . Freeman. | —                                                                                                           |
| 10 | Mazo, R. M. (2002). <i>Brownian Motion</i> . Oxford UP.        | —                                                                                                           |

### A.3 Foundations of Information Theory

| #  | Citation                                                                           | DOI / URL                                                                                           |
|----|------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| 11 | Bialek, W., Nemenman, I., & Tishby, N. (2001). <i>Neural Computation</i> 13, 2409. | <a href="https://doi.org/10.1162/089976601753195969">https://doi.org/10.1162/089976601753195969</a> |
| 12 | Jaynes, E. T. (1957). <i>Phys. Rev.</i> 106, 620.                                  | <a href="https://doi.org/10.1103/PhysRev.106.620">https://doi.org/10.1103/PhysRev.106.620</a>       |
| 13 | Landauer, R. (1961). <i>IBM J. Res. Dev.</i> 5, 183.                               | <a href="https://doi.org/10.1147/rd.53.0183">https://doi.org/10.1147/rd.53.0183</a>                 |

### A.4 Criticality and Self-Organised Criticality

| #  | Citation                                                                  | DOI / URL                                                                                                           |
|----|---------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| 14 | Bak, P., Tang, C., & Wiesenfeld, K. (1987). <i>PRL</i> 59, 381.           | <a href="https://doi.org/10.1103/PhysRevLett.59.381">https://doi.org/10.1103/PhysRevLett.59.381</a>                 |
| 15 | Beggs, J. M., & Plenz, D. (2003). <i>J. Neurosci.</i> 23, 11167.          | <a href="https://doi.org/10.1523/JNEUROSCI.23-35-11167.2003">https://doi.org/10.1523/JNEUROSCI.23-35-11167.2003</a> |
| 16 | Linkenkaer-Hansen, K., et al. (2001). <i>J. Neurosci.</i> 21, 1370.       | <a href="https://doi.org/10.1523/JNEUROSCI.21-04-01370.2001">https://doi.org/10.1523/JNEUROSCI.21-04-01370.2001</a> |
| 17 | He, B. J. (2014). <i>Trends Cogn. Sci.</i> 18, 480.                       | <a href="https://doi.org/10.1016/j.tics.2014.04.003">https://doi.org/10.1016/j.tics.2014.04.003</a>                 |
| 18 | Mandelbrot, B. B., & Van Ness, J. W. (1968). <i>SIAM Review</i> 10, 422.  | <a href="https://doi.org/10.1137/1010093">https://doi.org/10.1137/1010093</a>                                       |
| 19 | Kantelhardt, J. W., et al. (2002). <i>Physica A</i> 316, 87.              | <a href="https://doi.org/10.1016/S0378-4371(02)01383-3">https://doi.org/10.1016/S0378-4371(02)01383-3</a>           |
| 20 | Benzi, R., Sutera, A., & Vulpiani, A. (1981). <i>J. Phys. A</i> 14, L453. | <a href="https://doi.org/10.1088/0305-4470/14/11/006">https://doi.org/10.1088/0305-4470/14/11/006</a>               |

### A.5 Neuroscience

| #  | Citation                                                                  | DOI / URL                                                                                           |
|----|---------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| 21 | Markram, H., et al. (2004). <i>Nat. Rev. Neurosci.</i> 5, 793.            | <a href="https://doi.org/10.1038/nrn1519">https://doi.org/10.1038/nrn1519</a>                       |
| 22 | Hopfield, J. J. (1982). <i>PNAS</i> 79, 2554.                             | <a href="https://doi.org/10.1073/pnas.79.8.2554">https://doi.org/10.1073/pnas.79.8.2554</a>         |
| 23 | Aiello, L. C., & Wheeler, P. (1995). <i>Current Anthropology</i> 36, 199. | <a href="https://doi.org/10.1086/204350">https://doi.org/10.1086/204350</a>                         |
| 24 | Dorkenwald, S., et al. (2024). <i>Nature</i> 634, 124.                    | <a href="https://doi.org/10.1038/s41586-024-07558-y">https://doi.org/10.1038/s41586-024-07558-y</a> |
| 25 | Schlegel, P., et al. (2024). <i>Nature</i> 634, 139.                      | <a href="https://doi.org/10.1038/s41586-024-07686-5">https://doi.org/10.1038/s41586-024-07686-5</a> |

### A.6 Modern Deep-Learning Architectures

| #  | Citation                                                                | DOI / URL                                                                                         |
|----|-------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| 26 | Vaswani, A., et al. (2017). <i>Attention Is All You Need</i> .          | <a href="https://arxiv.org/abs/1706.03762">https://arxiv.org/abs/1706.03762</a>                   |
| 27 | He, K., et al. (2016). <i>Deep Residual Learning</i> .                  | <a href="https://arxiv.org/abs/1512.03385">https://arxiv.org/abs/1512.03385</a>                   |
| 28 | Krotov, D., & Hopfield, J. J. (2016).                                   | <a href="https://arxiv.org/abs/1606.01164">https://arxiv.org/abs/1606.01164</a>                   |
| 29 | Ramsauer, H., et al. (2020). <i>Hopfield Networks Is All You Need</i> . | <a href="https://arxiv.org/abs/2008.02217">https://arxiv.org/abs/2008.02217</a>                   |
| 30 | Su, J., et al. (2021). <i>RoFormer</i> .                                | <a href="https://arxiv.org/abs/2104.09864">https://arxiv.org/abs/2104.09864</a>                   |
| 31 | Gu, A., & Dao, T. (2023). <i>Mamba</i> .                                | <a href="https://arxiv.org/abs/2312.00752">https://arxiv.org/abs/2312.00752</a>                   |
| 32 | Song, Y., et al. (2021). <i>Score-Based Diffusion</i> .                 | <a href="https://arxiv.org/abs/2011.13456">https://arxiv.org/abs/2011.13456</a>                   |
| 33 | Mehta, P., & Schwab, D. J. (2014).                                      | <a href="https://arxiv.org/abs/1410.3831">https://arxiv.org/abs/1410.3831</a>                     |
| 34 | Weinan, E. (2017). <i>CMS</i> 5, 1.                                     | <a href="https://doi.org/10.1007/s40304-017-0103-z">https://doi.org/10.1007/s40304-017-0103-z</a> |

### A.7 Hardware and Energy

| #  | Citation                                                                   | DOI / URL                                                                                                                             |
|----|----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| 35 | Horowitz, M. (2014). <i>ISSCC</i> .                                        | <a href="https://doi.org/10.1109/ISSCC.2014.6757323">https://doi.org/10.1109/ISSCC.2014.6757323</a>                                   |
| 36 | Patterson, D., et al. (2021). <i>Carbon Emissions</i> .                    | <a href="https://arxiv.org/abs/2104.10350">https://arxiv.org/abs/2104.10350</a>                                                       |
| 37 | Sandberg, A., & Bostrom, N. (2008). <i>Whole Brain Emulation Roadmap</i> . | <a href="https://www.fhi.ox.ac.uk/brain-emulation-roadmap-report.pdf">https://www.fhi.ox.ac.uk/brain-emulation-roadmap-report.pdf</a> |

### A.8 Open Quantum Systems

| #  | Citation                                                               | DOI / URL                                                                                                                         |
|----|------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| 38 | Caldeira, A. O., & Leggett, A. J. (1983). <i>Physica A</i> 121, 587.   | <a href="https://doi.org/10.1016/0378-4371(83)90013-4">https://doi.org/10.1016/0378-4371(83)90013-4</a>                           |
| 39 | Feynman, R. P., & Vernon, F. L. (1963). <i>Ann. Phys.</i> 24, 118.     | <a href="https://doi.org/10.1016/0003-4916(63)90068-X">https://doi.org/10.1016/0003-4916(63)90068-X</a>                           |
| 40 | Lindblad, G. (1976). <i>Comm. Math. Phys.</i> 48, 119.                 | <a href="https://doi.org/10.1007/BF01608499">https://doi.org/10.1007/BF01608499</a>                                               |
| 41 | Breuer, H.-P., & Petruccione, F. (2002). <i>Open Quantum Systems</i> . | <a href="https://doi.org/10.1093/acprof:oso/9780199213900.001.0001">https://doi.org/10.1093/acprof:oso/9780199213900.001.0001</a> |
| 42 | Spohn, H. (1978). <i>J. Math. Phys.</i> 19, 1227.                      | <a href="https://doi.org/10.1063/1.523789">https://doi.org/10.1063/1.523789</a>                                                   |

### A.9 Quantum Information

| #  | Citation                                                                   | DOI / URL                                                                                             |
|----|----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| 43 | Holevo, A. S. (1973). <i>Problems Inform. Transmission</i> 9, 177.         | <a href="http://mi.mathnet.ru/eng/ppi903">http://mi.mathnet.ru/eng/ppi903</a>                         |
| 44 | Holevo, A. S. (1998). <i>IEEE Trans. Inf. Theory</i> 44, 269.              | <a href="https://doi.org/10.1109/18.651037">https://doi.org/10.1109/18.651037</a>                     |
| 45 | Schumacher, B., & Westmoreland, M. D. (1997). <i>Phys. Rev. A</i> 56, 131. | <a href="https://doi.org/10.1103/PhysRevA.56.131">https://doi.org/10.1103/PhysRevA.56.131</a>         |
| 46 | Bennett, C. H., et al. (1999). <i>PRL</i> 83, 3081.                        | <a href="https://doi.org/10.1103/PhysRevLett.83.3081">https://doi.org/10.1103/PhysRevLett.83.3081</a> |
| 47 | Bennett, C. H., & Wiesner, S. J. (1992). <i>PRL</i> 69, 2881.              | <a href="https://doi.org/10.1103/PhysRevLett.69.2881">https://doi.org/10.1103/PhysRevLett.69.2881</a> |

### A.10 Berry Phase and Topology

| #  | Citation                                                                   | DOI / URL                                                                                             |
|----|----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| 48 | Berry, M. V. (1984). <i>Proc. R. Soc. A</i> 392, 45.                       | <a href="https://doi.org/10.1098/rspa.1984.0023">https://doi.org/10.1098/rspa.1984.0023</a>           |
| 49 | Simon, B. (1983). <i>PRL</i> 51, 2167.                                     | <a href="https://doi.org/10.1103/PhysRevLett.51.2167">https://doi.org/10.1103/PhysRevLett.51.2167</a> |
| 50 | Thouless, D. J., et al. (1982). <i>PRL</i> 49, 405.                        | <a href="https://doi.org/10.1103/PhysRevLett.49.405">https://doi.org/10.1103/PhysRevLett.49.405</a>   |
| 51 | Wilczek, F., & Zee, A. (1984). <i>PRL</i> 52, 2111.                        | <a href="https://doi.org/10.1103/PhysRevLett.52.2111">https://doi.org/10.1103/PhysRevLett.52.2111</a> |
| 52 | Xiao, D., Chang, M.-C., & Niu, Q. (2010). <i>Rev. Mod. Phys.</i> 82, 1959. | <a href="https://doi.org/10.1103/RevModPhys.82.1959">https://doi.org/10.1103/RevModPhys.82.1959</a>   |

### A.11 Quantum Phase Transitions and Tensor Networks

| #  | Citation                                                                        | DOI / URL                                                                                                       |
|----|---------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| 53 | Sachdev, S. (2011). <i>Quantum Phase Transitions</i> (2nd ed.).                 | <a href="https://doi.org/10.1017/CBO9780511973765">https://doi.org/10.1017/CBO9780511973765</a>                 |
| 54 | Calabrese, P., & Cardy, J. (2004). <i>J. Stat. Mech.</i> P06002.                | <a href="https://doi.org/10.1088/1742-5468/2004/06/P06002">https://doi.org/10.1088/1742-5468/2004/06/P06002</a> |
| 55 | Eisert, J., Cramer, M., & Plenio, M. B. (2010). <i>Rev. Mod. Phys.</i> 82, 277. | <a href="https://doi.org/10.1103/RevModPhys.82.277">https://doi.org/10.1103/RevModPhys.82.277</a>               |
| 56 | White, S. R. (1992). <i>PRL</i> 69, 2863.                                       | <a href="https://doi.org/10.1103/PhysRevLett.69.2863">https://doi.org/10.1103/PhysRevLett.69.2863</a>           |
| 57 | Schollwöck, U. (2011). <i>Ann. Phys.</i> 326, 96.                               | <a href="https://doi.org/10.1016/j.aop.2010.09.012">https://doi.org/10.1016/j.aop.2010.09.012</a>               |
| 58 | Verstraete, F., Murg, V., & Cirac, J. I. (2008). <i>Adv. Phys.</i> 57, 143.     | <a href="https://doi.org/10.1080/14789940801912366">https://doi.org/10.1080/14789940801912366</a>               |
| 59 | Orús, R. (2014). <i>Ann. Phys.</i> 349, 117.                                    | <a href="https://doi.org/10.1016/j.aop.2014.06.013">https://doi.org/10.1016/j.aop.2014.06.013</a>               |

### A.12 Quantum Computing

| #  | Citation                                                      | DOI / URL                                                                                               |
|----|---------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| 60 | Preskill, J. (2018). <i>Quantum</i> 2, 79.                    | <a href="https://doi.org/10.22331/q-2018-08-06-79">https://doi.org/10.22331/q-2018-08-06-79</a>         |
| 61 | Cerezo, M., et al. (2021). <i>Nat. Rev. Phys.</i> 3, 625.     | <a href="https://doi.org/10.1038/s42254-021-00348-9">https://doi.org/10.1038/s42254-021-00348-9</a>     |
| 62 | Bharti, K., et al. (2022). <i>Rev. Mod. Phys.</i> 94, 015004. | <a href="https://doi.org/10.1103/RevModPhys.94.015004">https://doi.org/10.1103/RevModPhys.94.015004</a> |
| 63 | Reina, J. H., et al. (2002). <i>Phys. Rev. A</i> 65, 032326.  | <a href="https://doi.org/10.1103/PhysRevA.65.032326">https://doi.org/10.1103/PhysRevA.65.032326</a>     |

### A.13 Information Geometry and General Relativity

| #  | Citation                                                                  | DOI / URL                                                                                                                   |
|----|---------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| 64 | Amari, S. (1985). <i>Differential-Geometrical Methods in Statistics</i> . | <a href="https://doi.org/10.1007/978-1-4612-5056-2">https://doi.org/10.1007/978-1-4612-5056-2</a>                           |
| 65 | Amari, S. (2016). <i>Information Geometry and Its Applications</i> .      | <a href="https://doi.org/10.1007/978-4-431-55978-8">https://doi.org/10.1007/978-4-431-55978-8</a>                           |
| 66 | Amari, S. (1998). <i>Neural Computation</i> 10, 251.                      | <a href="https://doi.org/10.1162/089976698300017746">https://doi.org/10.1162/089976698300017746</a>                         |
| 67 | Wald, R. M. (1984). <i>General Relativity</i> .                           | <a href="https://doi.org/10.7208/chicago/9780226870373.001.0001">https://doi.org/10.7208/chicago/9780226870373.001.0001</a> |
| 68 | Maldacena, J. (1999). <i>Int. J. Theor. Phys.</i> 38, 1113.               | <a href="https://doi.org/10.1023/A:1026654312961">https://doi.org/10.1023/A:1026654312961</a>                               |
| 69 | Ryu, S., & Takayanagi, T. (2006). <i>PRL</i> 96, 181602.                  | <a href="https://doi.org/10.1103/PhysRevLett.96.181602">https://doi.org/10.1103/PhysRevLett.96.181602</a>                   |

|    |                                                             |                                                                                                           |
|----|-------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| 70 | Van Raamsdonk, M. (2010). <i>Gen. Rel. Grav.</i> 42, 2323.  | <a href="https://doi.org/10.1007/s10714-010-1034-0">https://doi.org/10.1007/s10714-010-1034-0</a>         |
| 71 | Jacobson, T. (1995). <i>PRL</i> 75, 1260.                   | <a href="https://doi.org/10.1103/PhysRevLett.75.1260">https://doi.org/10.1103/PhysRevLett.75.1260</a>     |
| 72 | Verlinde, E. (2011). <i>JHEP</i> 2011(4), 29.               | <a href="https://doi.org/10.1007/JHEP04(2011)029">https://doi.org/10.1007/JHEP04(2011)029</a>             |
| 73 | Padmanabhan, T. (2010). <i>Rep. Prog. Phys.</i> 73, 046901. | <a href="https://doi.org/10.1088/0034-4885/73/4/046901">https://doi.org/10.1088/0034-4885/73/4/046901</a> |
| 74 | Bekenstein, J. D. (1973). <i>Phys. Rev. D</i> 7, 2333.      | <a href="https://doi.org/10.1103/PhysRevD.7.2333">https://doi.org/10.1103/PhysRevD.7.2333</a>             |
| 75 | Hawking, S. W. (1975). <i>Comm. Math. Phys.</i> 43, 199.    | <a href="https://doi.org/10.1007/BF02345020">https://doi.org/10.1007/BF02345020</a>                       |

#### A.14 Cosmology and the Anthropic Principle

| #  | Citation                                                                              | DOI / URL                                                                                         |
|----|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| 76 | Carter, B. (1974). In <i>Confrontation of Cosmological Theories</i> . Reidel.         | —                                                                                                 |
| 77 | Barrow, J. D., & Tipler, F. J. (1986). <i>The Anthropic Cosmological Principle</i> .  | —                                                                                                 |
| 78 | Smolin, L. (1992). <i>Class. Quantum Grav.</i> 9, 173.                                | <a href="https://doi.org/10.1088/0264-9381/9/1/016">https://doi.org/10.1088/0264-9381/9/1/016</a> |
| 79 | Eigen, M., & Schuster, P. (1979). <i>The Hypercycle</i> .                             | <a href="https://doi.org/10.1007/978-3-642-67247-7">https://doi.org/10.1007/978-3-642-67247-7</a> |
| 80 | Wheeler, J. A. (1990). In <i>Complexity, Entropy and the Physics of Information</i> . | —                                                                                                 |

#### A.15 Modern AI Public Releases (as of May 2026)

| #  | Citation                                                 | URL                                                                                                                         |
|----|----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| 81 | OpenAI (2024). <i>Learning to Reason with LLMs</i> (o1). | <a href="https://openai.com/index/learning-to-reason-with-llms/">https://openai.com/index/learning-to-reason-with-llms/</a> |
| 82 | Anthropic (2025). <i>Claude Opus 4</i> .                 | <a href="https://www.anthropic.com/claude">https://www.anthropic.com/claude</a>                                             |
| 83 | Google DeepMind (2024). <i>Gemini 2.0 Deep Think</i> .   | <a href="https://deepmind.google/technologies/gemini/">https://deepmind.google/technologies/gemini/</a>                     |

#### A.16 Controversial / Speculative References

| #  | Citation                                                           | DOI / URL                                                                                             |
|----|--------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| 84 | Penrose, R., & Hameroff, S. (2014). <i>Phys. Life Rev.</i> 11, 39. | <a href="https://doi.org/10.1016/j.plrev.2013.08.002">https://doi.org/10.1016/j.plrev.2013.08.002</a> |
| 85 | Tegmark, M. (2000). <i>Phys. Rev. E</i> 61, 4194. (opposing view)  | <a href="https://doi.org/10.1103/PhysRevE.61.4194">https://doi.org/10.1103/PhysRevE.61.4194</a>       |

## Appendix B: Symbol List

### B.1 Classical CID Symbols

| Symbol                     | Meaning                                            |
|----------------------------|----------------------------------------------------|
| $\varphi(\mathbf{x}, t)$   | System field (neuronal activations, hidden states) |
| $\mu(\varphi)$             | Drift term                                         |
| $\gamma(t)$                | Damping memory kernel                              |
| $\xi(t)$                   | Fluctuation term                                   |
| $U(\varphi)$               | Potential function                                 |
| $\mathbf{v}(\varphi)$      | Curl field                                         |
| $T_1, T_2$                 | Two bath temperatures                              |
| $A^{(1)}, A^{(2)}$         | Coupling matrices                                  |
| $s$                        | Sub-Ohmic exponent                                 |
| $\beta$                    | Noise spectral exponent ( $\beta = 1 - s$ )        |
| $H$                        | Hurst exponent ( $H = 1 - \beta/2$ )               |
| $\tau$                     | Avalanche-size exponent                            |
| $\xi_{\text{correlation}}$ | Correlation length                                 |
|                            | Predictive mutual information (intelligence)       |
| $S_{\text{prod\_rate}}$    | Entropy-production rate (energy cost)              |
| $k_B$                      | Boltzmann constant                                 |
| $Q, K, V$                  | Query / Key / Value in attention                   |
| $d_k$                      | Attention-key dimension                            |

### B.2 Quantum QID Symbols

| Symbol                               | Meaning                                   |
|--------------------------------------|-------------------------------------------|
| $\hat{\rho}$                         | Density operator                          |
| $\hat{H}_S, \hat{H}_B, \hat{H}_{SB}$ | System / bath / coupling Hamiltonians     |
| $\hat{L}_k$                          | Lindblad jump operator                    |
| $\hat{a}_k, \hat{a}_k^\dagger$       | Bosonic annihilation / creation operators |
| $\tilde{\xi}(t)$                     | Quantum fluctuation operator              |
| $J(\omega)$                          | Spectral density                          |
| $\omega_k$                           | Bath-mode frequency                       |
| $g_k$                                | Coupling constant                         |
| $\mathcal{L}$                        | Lindblad generator                        |
|                                      | Projection super-operator                 |
| $\hbar$                              | Reduced Planck constant                   |
| $T_1, T_2^*$                         | Relaxation time, decoherence time         |
| $\chi$                               | Holevo information                        |
| $A(R), F(R)$                         | Berry connection, Berry curvature         |
| $C$                                  | Chern number                              |
| $c$                                  | CFT central charge                        |
| $\lambda$                            | Coherence-to-dissipation ratio            |
| $Q$                                  | Quantum conditional mutual information    |

### B.3 Field-Theoretic FID Symbols

| Symbol        | Meaning                  |
|---------------|--------------------------|
| $\mathcal{M}$ | Information manifold     |
| $g_{\mu\nu}$  | Manifold metric          |
| $g^{\mu\nu}$  | Inverse metric           |
| $R_{ijkl}$    | Riemann curvature tensor |

|                                    |                                    |
|------------------------------------|------------------------------------|
| $R_{ij}$                           | Ricci tensor                       |
| $R$                                | Scalar curvature                   |
| $G_{\mu\nu}$                       | Einstein tensor                    |
| $T_{\mu\nu}^{\text{(info)}}$       | Predictive stress–energy tensor    |
| $\Lambda$                          | Intelligence cosmological constant |
| $\kappa_I$                         | Intelligence coupling constant     |
| $c_I$                              | Information speed of light         |
| $\Gamma^k_{ij}$                    | Christoffel connection             |
| $\mathcal{L}_{\text{data}}$        | Data Lagrangian                    |
| $V(\phi; J)$                       | External-data potential            |
| $\mathcal{R}_{\text{color}}[\phi]$ | Colored-noise functional           |
| $h_{\mu\nu}$                       | Weak-field metric perturbation     |
| $\square$                          | d'Alembert operator                |

#### B.4 UID Overall Symbols

| Symbol | Meaning                                                        |
|--------|----------------------------------------------------------------|
| C1–C5  | The five physical conditions for the emergence of intelligence |
| SOC    | Self-organised criticality                                     |
| GLE    | Generalised Langevin equation                                  |
| QLE    | Quantum Langevin equation                                      |
| MZ     | Mori–Zwanzig projection                                        |
| FT     | Fokker–Planck                                                  |
| (A)    | Empirically verified                                           |
| (B)    | Theoretically estimated                                        |
| (C)    | Falsifiable target                                             |
| (D)    | Philosophical conjecture                                       |

#### Appendix C: Glossary

| Term                                    | Explanation                                                                                                                                               |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Generalised Langevin equation</b>    | A Langevin equation with a memory kernel; can be derived from a full Hamiltonian system via Mori–Zwanzig projection                                       |
| <b>Mori–Zwanzig projection</b>          | An operator method that projects a full phase space onto a slow-variable subspace, producing the generalised Langevin equation                            |
| <b>Detailed balance</b>                 | A symmetry condition of equilibrium states: forward and backward transition rates between any two states, weighted by occupation probabilities, are equal |
| <b>Fluctuation–dissipation theorem</b>  | Fluctuations (noise) and dissipation (friction) are linked by the same temperature                                                                        |
| <b>Helmholtz–Hodge decomposition</b>    | Unique decomposition of a vector field into a gradient part and a divergence-free part                                                                    |
| <b>Self-organised criticality (SOC)</b> | A system that, through intrinsic feedback, automatically tunes itself to the vicinity of a critical point                                                 |
| <b>Correlation length</b>               | The distance over which the statistical correlation between two points decays to $1/e$                                                                    |
| <b>Hurst exponent</b>                   | An exponent describing long-range dependence in a time series; $H > 0.5$ indicates persistence                                                            |



|                                  |                                                                                                                                           |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1/f noise</b>                 | Noise with a power spectrum $S(\omega) \sim 1/\omega$ , closely associated with critical systems                                          |
| <b>Avalanche exponent</b>        | The power-law exponent of avalanche-size distributions in critical systems                                                                |
| <b>Modern Hopfield network</b>   | An associative-memory network using a softmax potential with exponential storage capacity                                                 |
| <b>Attention mechanism</b>       | The core of a Transformer; can be derived from Hopfield dynamics                                                                          |
| <b>Caldeira–Leggett model</b>    | The standard open-quantum-system model in which a system coordinate is linearly coupled to a bath of infinitely many harmonic oscillators |
| <b>Lindblad equation</b>         | The standard equation describing the evolution of Markovian open quantum systems                                                          |
| <b>Holevo bound</b>              | An upper bound on the classically extractable information from a quantum state                                                            |
| <b>Berry phase</b>               | A geometric phase acquired by a quantum state under adiabatic evolution                                                                   |
| <b>Chern number</b>              | The topological quantisation of the Berry curvature; an integer                                                                           |
| <b>Tensor network / MPS</b>      | A mathematical structure that efficiently represents weakly entangled quantum states                                                      |
| <b>NISQ</b>                      | Noisy Intermediate-Scale Quantum, referring to noisy mid-scale quantum devices                                                            |
| <b>Information manifold</b>      | A geometric manifold formed by a family of probability distributions, with the Fisher information as the metric                           |
| <b>Fisher information metric</b> | The natural metric on an information manifold                                                                                             |
| <b>Einstein tensor</b>           | $G_{\mu\nu} = R_{\mu\nu} - (1/2) R g_{\mu\nu}$ , the central geometric object of general relativity                                       |
| <b>AdS/CFT duality</b>           | A duality between (n+1)-dimensional anti-de Sitter gravity and n-dimensional conformal field theory                                       |
| <b>Anthropic principle</b>       | The interpretive principle that cosmic parameters are observed precisely because they permit observers                                    |
| <b>First principles</b>          | Derivation from basic physics that does not rely on empirical fitting                                                                     |
| <b>Falsifiability</b>            | Core notion in Popper's philosophy of science: a theory must be refutable by experiment                                                   |

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**Corresponding author contact:** guilichina@163.com

**Data-availability statement:** All cited references provide public DOI or open-access URLs and are directly clickable. The companion code repository provides an engineering reference implementation of CID and a falsifiable validation suite.

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Author: Gui LI [guilichina@163.com](mailto:guilichina@163.com), Dangyang JIE [jiedy@jodell.cn](mailto:jiedy@jodell.cn), Haitao KANG [kanght@jodell.cn](mailto:kanght@jodell.cn) Date: 2026-05-24

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